# ARCH 2001 – A Space Odyssey

The Architecture of Microgravity Environments

By

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A thesis submitted to the Faculty of Graduate and Post Doctoral Affairs in partial fulfillment of the requirements for the degree of

Master of Architecture

in

2-year stream with Collaborative Specialization in Climate Change

Carleton University

Ottawa, Ontario, Canada

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# I Abstract

Space Architecture represents a new frontier for architectural thought, theory, and practice. As increasing emphasis is placed on space exploration, the discipline of architecture must expand to include the design of humane and sustainable habitation in realms where gravity, logistics, and environmental factors operate differently. In addition to their use for telecommunications and military purposes, spacecraft and microgravity environments have a vital role to play in medical research and in the study of climate change.

This thesis aspires to be a primer for students of architecture in the design of orbital environments, covering a range of topics beginning with a brief history of space exploration. Embracing design as a form of research, the thesis includes the design of cancer and climate research lab modules within a space station concept in low-Earth-orbit (LEO) as a case study to better understand the range of challenges that architects face in this milieu.

# II. Acknowledgements

**Personal Acknowledgements**

In loving memory of Roger Soto, my grandfather. When I was a child, he frequently saw me gazing at the sky and began taking me to the airport every weekend, where we admired the wonderful flying machines that disappeared into the clouds.

To my loving wife - Shereen Munshi - whose positive light has always encouraged me to persevere in the pursuit of my dreams,

Thank you, we did it!

To my parents - Ramon Renderos Sr. and Sandra Soto - who taught me the value of integrity, hard work, and dedication, and to my little brother - Alejandro Renderos Soto - who inspires me to stretch further every day,

Thank you!

To the innovative engineers and life science professionals who have developed today's cancer screening and treatment technologies, and to the health care professionals who guided me through my cancer treatments.

Thank you!

**Academic Acknowledgements**

To Professor Benjamin Gianni - for your guidance and support in this challenging and final chapter of the Master of Architecture Degree at Carleton University.

Thank you!

In addition, I extend a warm appreciation to the professors and instructors throughout the BAS and M.Arch process who provided me a foundational education, encouraged curiosity, and expanded my horizons over my 7-year academic journey within the Azrieli School of Architecture & Urbanism.

**Professional Acknowledgements**

Thank you to my mentors at Merrick & Company who encouraged me to pursue a post-secondary education in architecture, and enhanced my capabilities as a design professional, exposing me early in my career to the impact that architectural thought and design have on the world.

And finally, to the members of the space industry - for your engagement, your inclusive perspectives, and your inspirational work.

Thank you!

# III. Table of Contents


**I ABSTRACT ..... 2**

**II. ACKNOWLEDGEMENTS ..... 3**

**III. TABLE OF CONTENTS ..... 4**

**IV. GLOSSARY OF TERMS / ABBREVIATIONS ..... 5**

**V. LIST OF TABLES ..... 6**

**VI. LIST OF ILLUSTRATIONS ..... 6**

**CHAPTER 1: EXECUTIVE SUMMARY ..... 9**

1.1 Introduction ..... 9
1.2 Objectives ..... 10
1.3 Collaborative Specialization in Climate Change ..... 11
1.4 Thesis Content Summary ..... 12

**CHAPTER 2: HISTORICAL CONTEXT AND CURRENT CONDITIONS ..... 13**

2.1 The Rocket ..... 13
2.2 Early Human Space Flight ..... 15
2.3 Space Stations: Conception ..... 23

**CHAPTER 3: SCIENCE IN MICROGRAVITY ..... 41**

3.1 What is Microgravity? ..... 41
3.2 Science in Microgravity ..... 42

**CHAPTER 4: DESIGN METHODOLOGY ..... 44**

4.1 Research Process ..... 45
4.2 Project Mission(s) ..... 48
4.3 Orbital Selection Criteria ..... 53
4.4 Design Parameters ..... 58

**CHAPTER 5: SPACE STATION DESIGN ..... 65**

**CHAPTER 6: ARCHITECTURAL PERSPECTIVES ..... 66**

6.1 Inspiring Architectural Projects ..... 68
6.2 Architectural Design Process ..... 71
6.3 The Millenium Charter ..... 72
6.4 Mythology, Architecture and Outer Space ..... 72
6.5 Space Architecture Resources ..... 73

**CHAPTER 7: CONCLUSION ..... 74**

7.1 Personal Experience ..... 75
7.2 Key Takeaways ..... 76
7.3 Reflections on Outcome and Method ..... 76
7.4 Reflections on Engagement with Industry Professionals ..... 77
7.5 Next Steps and Conclusion ..... 78

**CHAPTER 8: BIBLIOGRAPHY ..... 79**

**APPENDICES** ..... **83**

Appendix A - Science In Microgravity ..... 84
Appendix B - Research Trip: SpaceCom24 and Space Coast ..... 88
Appendix C - Space Radiation ..... 99
Appendix D – AC$^3$ROSS Project: Conceptual Design ..... 101

## IV. Glossary of Terms / Abbreviations

| Terms / abbreviations | Description(s) |
|---|---|
| AO | Atomic Oxygen |
| CDO | Circulation Driven Orientation |
| CSS | Chinese Space Station |
| E-EBM | Extendable Engineering Bay Module |
| E-HM | Extendable Habitat Module |
| E-LM | Extendable Laboratory Module |
| ESA | European Space Agency |
| EVA | Extravehicular Activities |
| GCR | Galactic Cosmic Radiation |
| GEO | Geosynchronous Orbit |
| HLVs | Heavy Lift Vehicles (i.e., Falcon Heavy) |
| LINC | Logistics Integration Nexus Connector |
| ISS | International Space Station |
| LEO | Low Earth Orbit |
| MEO | Medium Earth Orbit |
| MM/OD | Micrometeoroid Orbital Debris |
| MSG | Microgravity Science Glovebox |
| NASA | National Aeronautics and Space Administration |
| PAD | Personal Adaptive Dormitory |
| REM / RADs / Grays (GY) | Radiation units |
| RUD | Rapid Unscheduled Disassembly |
| SCP | Station Construction Payload |
| SS | Space Shuttle, Space Station |
| STC | Sample Transfer Core |

## V. List of Tables

Table 1 - Space Station Critical Assembly Sequence ..... 32
Table 2 - Gravitational Force Comparisons ..... 41
Table 3 - Research through Design, Design through Research ..... 46
Table 4 - Examples of satellite use cases to foster green economy – Credit: SGAC (2022) ..... 50
Table 5 - Earth - Space Load Comparisons – Credit: SICSA (2024) ..... 59
Table 6 - Radiation type quality factors for REM calculation – Cr. NASA Radiation Math (2011) . 100
Table 7 - Radiation Effects on Humans – Cr. Spacecraft Systems Design & Operations (2004) .... 100

## VI. List of Illustrations

Figure 1 - Hurricane Idalia – Observed from the ISS, August 29-30, 2023. Cr. NASA (2023) ..... 11
Figure 2 - First Flying Device. Adobe Stock – Cr. Archivist (2024) ..... 13
Figure 3 - Chinese fire arrow, 13th Century. Cr. NASA (2021) ..... 14
*Figure 4 - View of Earth - Freedom 7, Cr. NASA (1961)* ..... 16
Figure 5 - Gemini IV - First American Spacewalk , Cr. NASA. (1965) ..... 17
Figure 6 - Apollo 11 lunar module, Credit: NASA (1969) ..... 18
Figure 7 - Saturn V, Apollo 11, Cr. NASA (1969) ..... 19
Figure 8 - The historic handshake in space. Cr. NASA (1975) ..... 20
Figure 9 - Robotic Canadian Handshake in space. Credit: CSA (2001) ..... 21
Figure 10 - Skylab Detail - News Reference. Credit: NASA (1973) ..... 23
Figure 11 - SS Discovery, launchpad 39a, KSC, Cape Canaveral, FL. Cr: NASA / Bill Ingalls (2011) . 24
Figure 12 - STS-97. Endeavour, with payload bay doors open. Cr. NASA (2000) ..... 25
Figure 13 - STS-51L Crew Back Row: Ellison S. Onizuka, Christa McAuliffe and Gregory B. Jarvis, Judith A. Resnik, Front Row: Pilot Michael J. Smith, Commander Francis R. Scobee, Ronald E McNair. Cr. NASA (1986) ..... 26
Figure 14 - Left to Right: STS-107 crew David M. Brown, Rick D. Husband, Laurel B. Clark, Kalpana Chawla, Michael P. Anderson, William C. “Willie” McCool, and Ilan Ramon. Cr. NASA (2003) ..... 26
Figure 15 - Atlantis docked to MIR Station - July 4, 1995. Photo by Mir-19 crew. Cr. NASA (2023) 27
Figure 16 - Zarya Module. NASA. Cr. Roscosmos, (1998) ..... 28
Figure 17 - Unity node being attached by Canadarm robotic arm. STS-88. Cr. NASA (1998) ..... 29
Figure 18 - Left to Right: Zvezda Service Module, Zarya Module, Unity Module - Above Earth's Horizon. Cr. NASA (2000) ..... 30
Figure 19 - International Space Station (ISS) elements by space agency. Credit: NASA (2023) ..... 32
Figure 20 - Destiny Laboratory Module Interior. Credit: ESA/NASA (2016) ..... 33
Figure 21 - Columbus Laboratory Module. Credit: ESA/NASA (2021) ..... 34
Figure 22 - Kibo Laboratory Module, comprising four components. Credit: NASA (2024) ..... 35
Figure 23 - Nauka Multi-purpose Laboratory Module - Roscosmos. Credit: NASA (2021) ..... 36
Figure 24 - Space Stations Comparisons. Cr. Center for Strategic & International Studies (2024) . 37
Figure 25 - Axiom Space Station. Credit: Axiom Space (2024) ..... 38
Figure 26 - Architect Philippe Starck's design for Axiom crew quarters. Cr. Starck.com (2024) ..... 38
Figure 27 - Orbital Reef space station, rigid and inflatable modules. Cr. Blue Origin (2021) ..... 39
Figure 28 - Gateway elements designed to support the Artemis campaign. Cr. NASA (2024) ..... 40
Figure 29 - Crew Dragon Launch . SpaceX, Cr. NASA (2020) ..... 44
Figure 30 - Global Spaceport Alliance Summit Attendees. Credit: GSA (2024) ..... 47
Figure 31 - Hurricane Irma, 2017. Captured by Cosmonaut Sergey Ryazansky. Cr. NASA (2017) ... 49

Figure 32- Concept image of Canadarm3. Credit: CSA (2024) ..... 52
Figure 33 - Reaching Orbit ..... 53
Figure 34 - Orbiting Earth, Ramon RS. Base image Credit: wall.alphacoders.com (2024) ..... 53
Figure 35 - Orbital Inclination Diagram. ..... 55
Figure 36 - ISS in LEO, from SpaceX Crew Dragon. Credit: NASA JSC (2021) ..... 56
Figure 37 - ECLSS functions and facts. Cr. NASA MSC (2014) ..... 58
Figure 38 - Concentrations of objects in L/M/GEO regions. Cr. NASA ODPO (2019) ..... 60
Figure 39 - Comparison of US and Russian MM/OD Systems – Cr. SSD&O (2004) ..... 61
Figure 40 - The UK and Ireland from the shaded side – Cr: André Kuipers NASA/ESA (2012)..... 62
Figure 41 - Solar Flashes captured by NASA's Solar Dynamics Observatory. Cr. NASA (2014) ..... 63
Figure 42 - Pre- and post-flight Long Duration Exposure experiment, indicating atomic oxygen and ultraviolet degradation. Cr. NASA JSC. (2020) ..... 64
Figure 43 - Aerial view of Moon Village plan. Credit: © SOM | Slashcube GmbH (2024) ..... 68
Figure 44 - Lunar Habitat. Credit: Foster + Partners (2012) ..... 69
Figure 45 - Project Olympus, Lunar Habitat. Credit: BIG (2020) ..... 70
Figure 46 - Ramon RS, General Hospital Ottawa, after final radiation therapy treatment (2018).. 75
Figure 47 - Rendering - SS Asclepius with approaching spacecraft ..... 103
Figure 48 - Conceptual Adjacency Diagram ..... 104
Figure 49 - Early Program Concept - Habitation Areas ..... 105
Figure 50 - Early Program Concept - Science Laboratories ..... 105
Figure 51 - Early Program Concept - Logistics ..... 106
Figure 52 - Early Program Concept - Engineering Bays ..... 106
Figure 53 - Early Program Concept – Command ..... 106
Figure 54 - Space Station Modular Massing Diagram ..... 107
Figure 55 - Exo-Skeleton Structural Concepts - rigidity and substrate for exterior shell ..... 108
Figure 56 - Space Station, Conceptual Visualization - Exterior Space Station Cladding ..... 109
Figure 57 - SS Asclepius - Forward side, Command Center viewing window. ..... 110
Figure 58 - SS Asclepius Program ..... 111
Figure 59 - Module Types, Count, and Launches ..... 112
Figure 60 - Module Design Concepts - Sketches by Ramon Renderos Soto ..... 113
Figure 61 - Comparison of Conventional Space Construction Materials – Credit: SSD&O (2004) 114
Figure 62 - Life support and emergency systems – Credit: SSD&O (2004) ..... 114
Figure 63 - Global Spaceports & Launch Sites. Credit: go-astronomy.com (2024) ..... 115
Figure 64 - Starship IFT-3 Screenshot from orbit. Credit: SpaceX ..... 117
Figure 65 - SpaceX Starship Space Shuttle Comparison. Credit: quora.com ..... 118
Figure 66 - E-LM within Starship payload. Base images Cr. X, Fael097 (2021), SpaceX (2020) ..... 118
Figure 67 - Concept SS Asclepius Module within Starship. Base image Cr. ERCX, X (2022) ..... 119
Figure 68 - LEO Market Projections. Credit: Deloitte (2022) ..... 120
Figure 69 - Extendable Module Concept Design ..... 121
Figure 70 - E-LM Concept, un-extended. ..... 122
Figure 71 - Left to Right -"Sleep among the stars", Cr. CNN (2017). JAXA Astronaut Soichi Noguchi, CSA Astronaut Josh Cassada. "Sleeping in Space". Cr. CNN (2023) ..... 123
Figure 72 - Left to Right - Attic bedrooms. Alex Geerts, Jonathan Borba, Cr. Unsplash (2024), Attic Bedroom, Cr. thenorrdroom.com (2024) ..... 123
Figure 73 - PAD Module Concept Design. ..... 124
Figure 74 - PAD Module Concept ..... 124
Figure 75 - E-EBM Concept, un-extended ..... 125

Figure 76 - E-HM Concept, un-extended. ..... 126
Figure 77 - LINC Module Concept. ..... 127
Figure 78 - Terrestrial Laboratory – Credit: labnetinternational.com (2019) ..... 128
Figure 79 - Lab with Large Equipment – Credit: medium.com (2023) ..... 129
Figure 80 - Lab with Large Benchtop Equipment. Credit: labnetinternational.com (2019) ..... 129
Figure 81 - Terrestrial Laboratory Plan and Section. ..... 130
Figure 82 - Christina Koch, Nasa Astronaut. Credit: NASA (2019) ..... 131
Figure 83 - Astronaut at Glovebox – Credit: electronicsweekly.com (2020) ..... 131
Figure 84 - ISS National Laboratory – Credit: plasticstoday.com (2024) ..... 132
Figure 85 - Tianhe Module, CSS. Taikonaut Wang Yaping. Cr. global.chinadaily.com.cn (2021) ... 132
Figure 86 - Capsule rotated upright. ..... 133
Figure 87 - E-LM Lab Levels and circulation ..... 134
Figure 88 - E-LM Sample Transfer Core and Wall Space Utilization ..... 135
Figure 89 - E-LM Concept Plan Diagrams: Standard and Circulation ..... 136
Figure 90 - E-LM Concept Plan Diagrams: Reporting and Bench Space ..... 137
Figure 91 - E-LM Concept Plan Diagrams: Sample Transfer Core and Equipment Racks ..... 138
Figure 92 - E-LM Concept Plan Diagrams: Equipment and Standard Labs ..... 139
Figure 93 - SS Asclepius, Launch and Assembly Sequence ..... 140
Figure 94 - Launch Phase 1 ..... 141
Figure 95 - Launch Phase 2 ..... 141
Figure 96 - Launch Phase 2, extended ..... 142
Figure 97 - Launch Phase 3 - Hangers / Cargo Bays ..... 142
Figure 98 - Launch Phase 4 - LINC Module + E-LM ..... 143
Figure 99 - Launch Phase 5 - PAD Modules ..... 143
Figure 100 - Launch Phase 6 - LINC Module + E-LM ..... 144
Figure 101 - Launch Phase 7 - LINC Modules ..... 144
Figure 102 - Launch Phase 8 - Final E-EBMs ..... 145
Figure 103 - Full Program ..... 146
Figure 104 - Rendering, space eye view of SS Asclepius ..... 147
Figure 105 - Rendering - Zenith View of SS Asclepius ..... 148

# Chapter 1: Executive Summary

## Architecture is the will of an epoch translated into space…

- Le Corbusier, *Vers une architecture* (1923)

## 1.1 Introduction

A dream dares us to venture out of our comfort zones and journey beyond our current perceived limits. Achieving our dreams requires us to make decisions – to shed old identities and commit ourselves to a new trajectory, leaving behind what we currently belief to be true. Since the laying of the first stone and the breaking of the first branch, architects have been motivated to build upwards. A further leap takes us beyond terrestrial environments, bound as they are by gravity and the cardinal directions, into **astrotectonics**, defined by the Sasakawa International Center for Space Architecture (SICSA) as "the science of constructing space structures and facilities for use in orbit, on crewed voyages beyond Earth's magnetosphere, and to create lunar / planetary habitats"[1].

Traditional architecture challenges us with its complexity and demands rigorous study, technical knowledge, mindful planning, and execution of skill, while also requiring us to grapple with varying societal needs, to exercise creativity, and to contend with abstraction. While Astrotectonics requires this multifaceted thinking, the transition from traditional architecture to architecture in outer space requires an immense leap into the void. In addition to a leap of faith, however, a more objective approach is required. Canadian astronaut Chris Hadfield made clear that, "No astronaut launches to space with their fingers crossed. That is not how we deal with risk."[2] So, the path from traditional architecture to a basic understanding of astrotectonics follows a more calculated direction as this field brims with aerospace engineering concepts, is rife with design constraints, and exhibits its own vivid history.

[1] Sasakawa International Center for Space Architecture, "Astrotectonics", pg.1
[2] MasterClass, "Chris Hadfield, Teaches Space Exploration"

## 1.2 Objectives

In his book, *Ultralearning*, Scott H. Young describes several strategies for acquiring knowledge in an accelerated but effective manner. He suggests dividing the given field of interest into three categories -- concepts, facts, and procedures – otherwise stated, things to understand, memorize and practice. This strategy informed my approach to the research of space architecture and informed the way I have organized the content of this thesis. . The thesis is split into two sections. Chapters 1 through 4 are intended as a primer on space architecture for architects who may be considering expanding their horizons. It is structured as summaries of a series of topics, supported by links to references for further study. By contrast, Chapters 5 through 7 chronicle the design of a hypothetical laboratory, and offer perspectives on the use of design as a form of research. These two sections work harmoniously, having the following objectives:

**Primary Objective**

The primarily objective of this thesis was to document and summarize my research journey into an emerging realm of architecture, contributing to the bridge already being built by the academics, design professionals, and visionaries of the last century, namely the bridge between traditional (terrestrial) architecture and a future architecture in space. A tremendous body of research exists on this subject, an introduction to which the chapters of this thesis are intended to provide, as well as a repository of key references and resources.

**Secondary Objective**

To engage in research through design, i.e., to augment and test my understanding of key concepts in the field of astrotectonics through the design of orbital research space station modules within a functional space station framework.

**Tertiary Objective**

To identify and articulate the roles and responsibilities of architects with respect to the design of non-terrestrial environments – sharing my own viewpoint and those of subject matter experts.

Questions to be addressed…

- What can architects contribute to the design of environments that accommodate human habitation in micro-gravity environments such as the Earth's orbit and the surface of the moon?
- Do our current definitions of architecture and our understanding of ourselves as professionals hinder or contribute to our participation in the design of extraterrestrial environments?
- To what degree is gravity implicit in the concept of architecture? Is there architecture beyond the constraints of gravity?
- What can the history of architecture and the built environment teach us about creating healthy and sustainable environments in space?

**The creation of a new architecture is not merely a question of rearranging forms, but of thoroughly revolutionizing our entire conception of architecture.**

➢ Le *Corbusier, Vers une architecture* (1923)

## 1.3 Collaborative Specialization in Climate Change

"The climate crisis is the most pressing issue of our time, and our ability to address it will be a defining moment for society. The problem is complex and multifaceted - and can only be addressed through collaboration between groups spanning the humanities, sciences, engineering, business, and public policy." – Carleton University, Collaborative Specialization in Climate Change.

As part of a Collaborative Specialization in Climate Change at Carleton University I have had a unique opportunity to engage with internationally recognized climate change researchers. The specialization has enabled me to collaborate with students and professors from the schools of Business, Engineering, Science and Economics as well as with community partners, notably the Sustainable Laboratories Canada -- a network created to facilitate interaction among professionals on the topic of sustainable laboratory design and operation. Together we have explored the interconnected nature of climate change and debated solutions for mitigation and adaptation to the climate crisis at the level of communities, institutions, and entire ecosystems.

Data from the International Space Station (ISS) and various satellite have provided us with enormous insights into activity on our planet's surface as well as oceanic and atmospheric conditions. It is safe to say we would know much less about our current climate conditions if it were not for our efforts in outer space. In this respect – and however tangentially --this thesis also addresses climate change by highlighting historic, current, and future contributions to Earth and solar observation and to earth science research aboard an orbital research station. As an aside, questions often arise about the environmental impact of the fuel-intensive rockets that send payloads and crews into outer space. To this end, sub-chapter 4.5: *Mission Architecture*, addresses launch-related pollution and discusses green fuel technologies designed to reduce and eventually eliminating fossil fuel consumption for aerospace applications.

*Figure 1 - Hurricane Idalia – Observed from the ISS, August 29-30, 2023. Cr. NASA (2023)*



## 1.4 Thesis Content Summary

### Historical Context and Current Conditions

Chapter 2 briefly chronicles the history of human space flight. Aspiring space architects must understand the strategies and paths that have led to the current state of design. It is important to recognize the human effort, technological pioneering, and, in a few tragic instances, the loss of life that the dawn of human space flight has brought with it. The chapter describes early concepts for space stations including NASA's Space Shuttle Program and the ongoing contributions of Canada. Given the planned decommissioning of the International Space Station (ISS) in 2030, interest in commercial aerospace is growing rapidly -- as is interest on the part of architects. The chapter includes concepts for the next generation of space stations.

**As an architect you design for the present, with an awareness of the past, for a future which is essentially unknown.**

- Norman Foster

### Science In Microgravity

As one of the primary functions of space stations has been to conduct scientific research in microgravity environments, it was important to identify the nature and value of such research. To this end, Chapter 3 describes how microgravity relates to gravity on Earth and outlines some of the many benefits that scientific research in microgravity has provided over the decades.

### Design Method

Building on the above, Chapter 4 describes the research process and outlines the key concepts in astrotectonics, namely the issues and challenges associated with constructing habitable facilities for use in orbits beyond Earth's magnetosphere. Sub-chapters describe:

- What? The **project missions** – in this case a habitable space station that supports cancer-related research and research on climate change.
- Where? **Orbital "Site" Selection Criteria** -- the location, characteristics, challenges, and benefits of various orbits.
- How? **Design Logistics** and **Mission Architecture** -- an introduction to the structural and environmental design parameters associated with orbiting structures, including assembly sequences and launch-related constraints.

### Space Station Design

With a focus on research through design, Chapter 5 applies the basic principles and constraints outlined above to the design of a space station intended to support research in microgravity. In such environments overall "volume" supersedes "floor area" when considering the allocation and organization of space.

### Architectural Perspective and Conclusion

Chapters 6 considers historic and current views on space architecture from architectural professionals and academics and explores questions about the future of architecture and its role in the design of astrotectonics. The concluding and 7$^{th}$ Chapter reflects on the value of research through design and summarizes the lessons learned over the course of the exploration.

# Chapter 2: Historical Context and Current Conditions

This section provides a summary of humanity's history in space. It highlights the component-based and modular approach to the construction of space stations and outlines the major functional components that comprise these stations.

## 2.1 The Rocket

Great technologies evolve in their capabilities and their use by humanity. The rocket has deep origins in ancient history, scientific understanding in the modern age, and in recent advancements that have given us the opportunity to embark on deep space human exploration. Traditional architects understand the logistics of transport here on Earth, i.e., the constraints, widths, and docking requirements involved with shipping by air, sea, trains, and trucks. Similarly, space architects must understand the historic development of the vessels that have enabled materials to be transported into and beyond Earth's orbit.

For a history of rockets, visit:

- https://www.grc.nasa.gov/www/k-12/TRC/Rockets/history_of_rockets.html

### Ancient Origins[3]

A Roman named Aulus Gellius authored a telling story about an ancient Greek philosopher, mathematician, and military general named Archytas. Archytas lived in Tarentum, Magna Graecia (modern day Southern Italy) around 400 B.C and was believed to be the first to invent a self-flying device – a flying pigeon made of wood – in order to understand avian flight. The flying pigeon had an aerodynamic design, wings and an internal bladder containing a small steam boiler[4], providing the internal pressure and propulsion that enabled it to fly for several hundred meters. The flying pigeon employed the action-reaction principle of physics, not formally articulated as scientific law for another 2000 years (17th century). The Archytas crater on the moon pays homage to the achievements of this ancient influencer




*Figure 2 - First Flying Device. Adobe Stock – Cr. Archivist (2024)*

[3] NASA, "Brief History of Rockets".
[4] Reese, "Pigeon of Archytas".

Around the first century A.D., the Chinese experimented with gunpowder-filled bamboo tubes used as festive explosives. After this, the tubes were added to the ends of arrows, and shot from bows. When it became apparent that the bow was unnecessary and the fuel-filled tubes could launch themselves, the age of rocketry was born.

The reported date for the first use of a rocket was 1232, during the battle of Kai-Keng against the Mongols. Following this battle, the Mongols began development of rockets and are thought to be responsible for this knowledge spreading into Europe in the ensuing centuries. Experiments and advancements in England, France, and Italy led to the development of longer-range, higher-accuracy weapons.



*Figure 3 - Chinese fire arrow, 13th Century. Cr. NASA (2021)*

## Rocket Science

Modern rocket science advancements are underpinned by the scientific laws discovered by Sir Isaac Newton (1642-1727). Newton's laws of physics explained how rockets worked within our atmosphere and in the vacuum of space. Since conception, rockets have oscillated as weapons of war, and tools for scientific exploration. A brief timeline of rocket development is listed below:

- 1720: Dutch professor William Gravesande built model vehicles with steam powered jets.
- 1700-1800s: Rockets are used once again as military weapons.
- Indian rocket barrages against the British during battles (1792, 1799) inspire British artillery.
- 1812: Colonel William Congreve, artillery expert, uses Congreve rockets on British Ships to attack Fort McHenry in the war of 1812. The "rockets' red glare" of Francis Scott Key's poem, written about this battle, eventually became the lyrics of the Star-Spangled Banner.
- 1898: Russian schoolteacher Konstantin Tsiolkovsky (1857-1935) published a report suggesting the use of liquid propellants to enable rockets to achieve higher altitudes and be used for space exploration. Tsiolkovsky is considered the father of modern astronautics.
- Robert H. Goddard (1882-1945) has been called the father of modern rocketry. His achievements include:
    - 1915: Goddard experiments with solid-propellant rockets
    - 1919: Goddard published "A Method of Reaching Extreme Altitudes," a mathematical analysis of the meteorological sounding rocket. A sounding rocket is a research rocket that carries instruments to conduct experiments during its sub-orbital flights.
    - March 16, 1926 – Goddard launches a rocket using liquid oxygen and gasoline as propellants. The rocket flew 12.5m high, 56m away, in 2.5 seconds, inaugurating a new era of rocket flight.
    - Goddard developed gyroscope systems for flight control, payload compartments, and parachute recovery methods.
- Hermann Oberth (1894-1989)
    - was born in Hermannstadt, Transylvania and died in Nuremberg, Germany.
    - Published a book in 1923 about rocket travel into outer space.

    - This book inspired the formation of rocket societies around the world, notably, the German *Verein fur Raumschiffahrt* (society for space travel) which led to the development of the V-2 rocket (called the A-4 in Germany). Development was eventually directed by Wernher von Braun, who used hundreds of these rockets against London, England during World War II.
- Post World War II, 1945:
    - Allies captured V-2 rockets, studies, and components.
    - Rocket scientists from Germany migrated to the United States and the Soviet Union.
    - Von Braun and other rocket scientists were recruited by the U.S. under "Project Paperclip" for rocket development.
    - Rocket weaponry programs develop in the U.S. and the Soviet Union.
    - U.S. program develops high-altitude atmospheric sounding rockets, giving birth to the U.S. space program.
- V-2 rocket evolution: Both the U.S. and Soviet Union developed rockets based on V-2 designs:
    - U.S Army: Redstone rocket
    - U.S. Navy: Vanguard missile
    - U.S. Air Force: Atlas rocket
    - Soviet Union: R-Series rockets, including the first intercontinental ballistic missile (ICBM), the R-7.
- 1957: The space race begins.

## 2.2 Early Human Space Flight

The space race was a one component of the decades-long competition between the US and Soviet Russia known as the Cold War. Tensions escalated with the development of nuclear arms and as the range and accuracy of rockets advanced. This sub-chapter summarizes early influences and sub-orbital / orbital milestones of these competing nations.

For a more detailed timeline of the space race, visit:

- https://www.rmg.co.uk/stories/topics/space-race-timeline [5]
- https://www.history.com/topics/cold-war/space-race [6]

### 1957: Sputnik 1 and 2

On October 4th, 1957, Russia launched the world's first human-made satellite into earth's orbit using an R-7 ICBM. It was named "Sputnik" -- the Russian word for "traveler." Not only did the launch of Sputnik beat the U.S. to the punch, but it also showcased the power of R-7 rockets and their potential as a military threat. To add insult to injury, the Soviet Union launched Sputnik 2 into orbit on November 3rd of the same year, this time carrying a dog named Laika – successfully sending a living organism into orbit. Alas, Laika did not return.

[5] Royal Museums Greenwich, "Space Race Timeline"
[6] History.com, "The Space Race"

## 1958: Explorer I and NASA

Designed by the U.S. Army and under the direction of Wernher von Braun, the U.S. launched Explorer I, its own orbital satellite. Explorer, I carried equipment that led to the discovery of the Van Allen radiation belts.

## 1960: Sputnik 5

Two dogs, Belka and Strelka, along with some plants, are launched, orbited, and returned alive aboard a Sputnik 5 vessel. Sputnik 5 was launched on August 19th, 1960.

## 1961: Soviet Triumph

Although Ham, a U.S. chimpanzee became the first hominid in space to survive a landing on January 31st, 1961, the Soviets triumphed over the U.S. yet again on April 12th of the same year by successfully orbiting the first man around the Earth. Yuri Gagarin made a single orbit, aboard the Vostok 3KA.

## 1961-1963: Project Mercury

Freedom 7 flight: This mission on May 5th, 1961, carried astronaut Alan Shepard into space on a sub-orbital flight, becoming the first American to do so. He was launched on a Redstone rocket and the flight lasted 15 minutes and 28 seconds. Shepard named his Mercury spacecraft “Freedom 7” paying homage to the seven Mercury astronauts. Shepard later went on to land on the moon on Apollo 14.

Project Mercury was the first U.S. program to put humans in orbit. Between 1961 and 1963, the program made 25 total flights, including six crewed flights. The single-person Mercury capsule was designed by NASA for water landing, with a parachute that deployed at 7,500 meters.

The six human flights included two sub-orbital missions, starting with Freedom 7, and four orbital missions. These missions investigated spacecraft orbit, biological functions in space, and full-mission planning, including safe crew and spacecraft recovery. Less than a year after Yuri Gagarin, John Glenn become the first American to orbit the Earth on February 20th, 1962. He travelled aboard the Friendship 7 Mercury spacecraft on the 3rd human mission of the Mercury missions. The Atlas ICBM rocket that launched him into space used for the following three Mercury flights.



*Figure 4 - View of Earth - Freedom 7, Cr. NASA (1961)*

## 1964-65: The Gemini Program

While project Mercury focused on getting humans into space, the goal of the Gemini program was to understand how humans might stay and work in space, in preparation for the later Apollo missions. Missions included testing of long-duration astronaut missions, rendezvous and docking procedures between multiple vehicles in orbit, and orbital re-entry methods. At this point, long-duration flights amounted to 14 days and their effects on astronauts were thoroughly researched.

The first human flight, Gemini III, was launched on March 23, 1965. On March 16th, 1966, astronauts Neil Armstrong and David Scott, aboard the 2-passenger Gemini VIII spacecraft, conducted a rendezvous and docking procedure with the Agena spacecraft, a target vehicle launched the same day. This historic procedure set the precedent for many docking procedures to come.

For more details on the Gemini Program, visit:

- https://www.nasa.gov/gemini/ [7]
- https://airandspace.si.edu/stories/editorial/what-was-gemini-program [8]



*Figure 5 - Gemini IV - First American Spacewalk , Cr. NASA. (1965)*

[7] NASA, “Project Gemini”
[8] National Air and Space Museum, ”What was the Gemini Program?”

## 1962-1972: The Apollo Program

The Apollo Program was focused on landing humans on the moon and bringing them safely back on Earth. The program included both uncrewed and crewed earth- and lunar-orbit flights, six of which were successful, full-crewed missions to the lunar surface and back. These were Apollos 11, 12, 14, 15, 16 and 17. These missions returned with immense scientific data and approximately 400 kilograms of physical lunar samples, which continue to be used for research.

On July 20th, 1969, Apollo 11 marked the historic moment when Neil Armstrong became the first man to set foot on the moon, along with his partner, astronaut Buzz Aldrin. The Apollo spacecraft was launched by a three-stage 363-foot Saturn V rocket at Launch Complex 39A at the Kennedy Space Center.

For more details on the spacecraft, missions and chronology of the Apollo Program, visit:

- https://www.nasa.gov/the-apollo-program/ [9]
- https://nssdc.gsfc.nasa.gov/planetary/lunar/apollo.html [10]

*Figure 6 - Apollo 11 lunar module, Credit: NASA (1969)*



[9] NASA, "The Apollo Program"
[10] NASA, GSFC, "The Apollo Program (1963-1972)



*Figure 7 - Saturn V, Apollo 11, Cr. NASA (1969)*

## Historic Handshake



*Figure 8 - The historic handshake in space. Cr. NASA (1975)*

While space exploration had benefitted from competition between rivals, it was thought that cooperation would drive evolution. On July 17, 1975[11], the Apollo-Soyuz Test Project (ASTP) successfully docked two spacecraft – the U.S. Apollo and the Soviet Soyuz – each with astronauts/cosmonauts onboard. The collaboration between nations to achieve mission success in the ASTP led to years of subsequent achievements in the Shuttle-Mir and International Space Station (ISS) programs. For the full story, visit: https://www.nasa.gov/history/45-years-ago-historic-handshake-in-space/

## Canadian History in Space

Canada has an extensive history in space studies and exploration, dating back to the 1800s when the first magnetic observatory was established at the University of Toronto.[12] Contributions continued from there, leading up to 1957 where John Chapman, recognized as the father of the Canadian Space Program, and fellow scientists became the first to record Sputnik 1's beeps. Canada has continued to contribute to international space programs through the development of satellite, sensor, and robotic technologies, astronaut training and missions. It now looks to the future as the NASA Artemis Program once again prepares to land humans on the moon. Below is a short list of notable milestones in Canadian space history. For a comprehensive chronological list of Canada's numerous accomplishments, refer to:

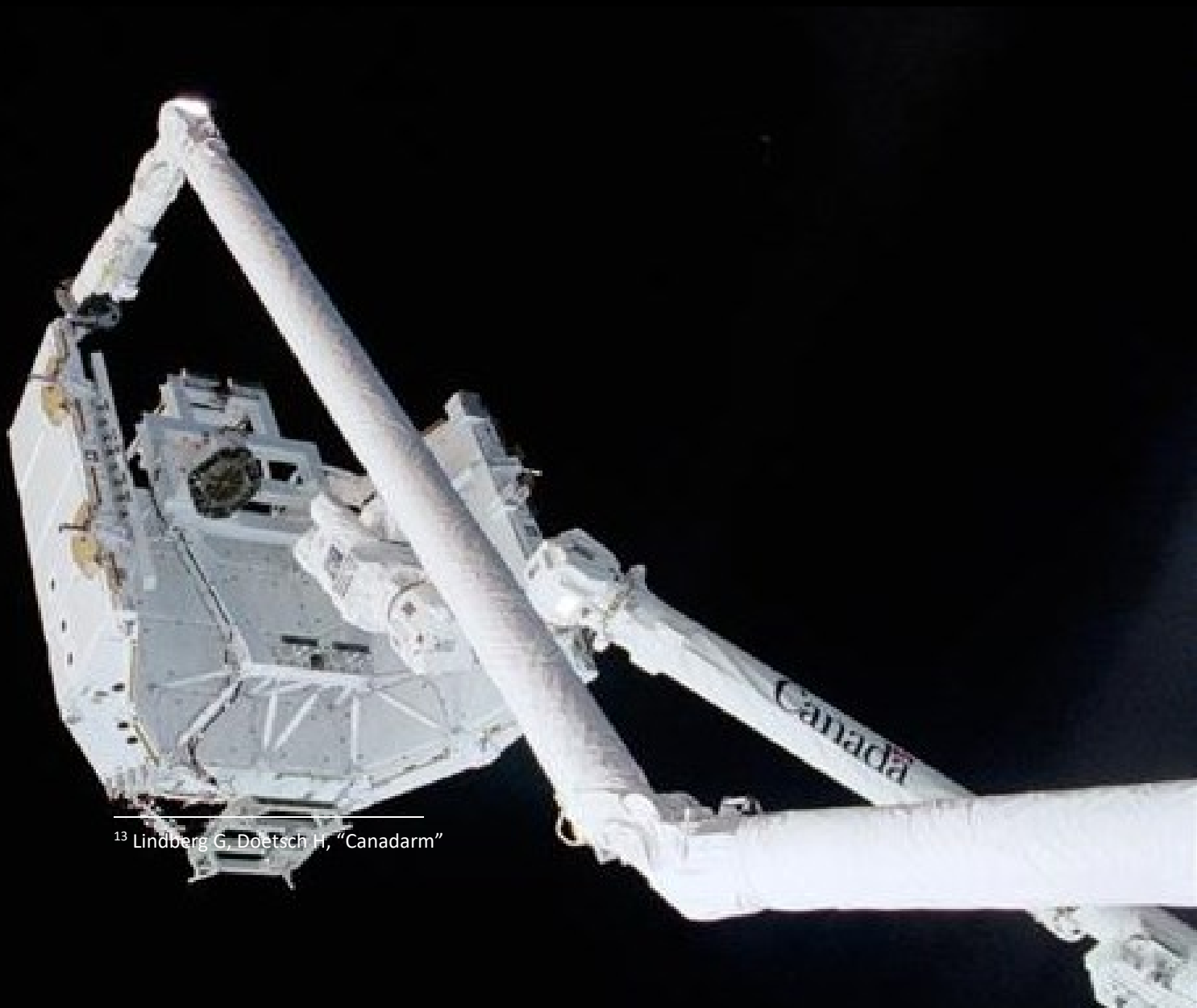
https://www.asc-csa.gc.ca/eng/about/milestones.asp

- **1974** – NASA awarded Canada the responsibility of designing, developing, and building the Shuttle Remote Manipulator System (SRMS) known as the Canadarm.
- **1981** – Launch of Canadarm aboard Space Shuttle Columbia (STS-2). Canadarm assisted the Space Shuttle Program by deploying satellites and cargo, positioning astronauts and various maintenance activities.
- **1982** – Canadarm performs so well, NASA invites a Canadian astronaut to fly in space. This marks the beginning of the Canadian astronaut program
- **1984** – astronaut Marc Garneau becomes first Canadian in space aboard Space Shuttle Challenger (STS-41G).
- **1985** – Canada and the U.S. sign a Memorandum of Understanding for the space station project.
- **1986** – Canada signs international agreement to become a full partner in the ISS program.
- **1989** – Creation of the Canadian Space Agency (CSA).

[11] John Uri, NASA, "Historic Handshake"
[12] Canadian Space Agency, "Canadian Space Milestones"

- **1992** – Roberta Bondar becomes the second Canadian, and fist Canadian woman, in space. She participated in the first international Microgravity Laboratory Mission.
- **1995** – Air Force Major, Chris Hadfield becomes the first Canadian Mission Specialist and the first Canadian aboard space station Mir during the STS-74 Atlantis-Mir Docking Mission.
- **1998** - Canadarm is used for the first International Space Station Assembly mission where the Orbiter Docking System is built onto the U.S. Unity Node aboard Space Shuttle Atlantis and later connected to the Soviet Zarya module.[13]
- **2001** – The Space Shuttle Endeavor (STS-100), including astronaut Chris Hadfield, delivers Canadarm2 to the ISS. Canadarm2, also known as the Space Station Remote Manipulator System (SSRMS), served a similar function for the ISS as Canadarm1 did for the Space Shuttles. On this same mission, astronaut Chris Hadfield becomes the first Canadian to conduct an Extra Vehicular activity (EVA), or spacewalk. A historic moment for Canadian contribution to human development in space was when Canadarm2 (on the ISS) transferred its launch cradle to Canadarm (on the shuttle). This event was named the "Canadian Handshake."

*Figure 9 - Robotic Canadian Handshake in space. Credit: CSA (2001)*




[13] Lindberg G, Doetsch H, "Canadarm"

- **2007** – NASA's Phoenix Mars lander is launched. Canadian meteorological instruments onboard the spacecraft track the weather and climate on Mars. This technology officially becomes the first Canadian technology to land on another planet on May 25th, 2008.
- **2009** – Canadarm captures a Japanese free-flying vehicle from space and successfully docks it to the ISS, under the control of CSA astronaut Robert Thirsk. The first "Cosmic Catch" for the robotic arm on the ISS.
- **2013** – Chris Hadfield completes a 5-month mission aboard the ISS, conducting a record 130 scientific experiments and becoming the first Canadian to command the Station.
- **2019** – Prime Minister Justin Trudeau announces Canada's commitment to partner with NASA on the Lunar Gateway – a component of the Artemis program aimed at assembling a space station in lunar orbit – and its contribution of the Canadarm3, the next generation of autonomous robotics.
- **2023** – CSA astronaut **Jeremy Hansen** was announced as the first Canadian ever to be part of a lunar mission as part of Artemis II. Astronauts on this mission will travel around the moon, ensuring that the Orion spacecraft's systems work as designed. This will make Canada the second country in history to go around the moon.
- **2023 –** OSIRIS-Rex is a spacecraft sent to an asteroid in our solar system named Bennu. The spacecraft touched down on Bennu, collected samples and safely returned to Earth on September 24, 2023. Canada contributed the OSIRIS-Rex Laser Altimeter (OLA), which provided information about shape and surface features. Canadian technology helped select the best site to collect the sample and will obtain a portion of the sample to be studied for generations.

## 2.3 Space Stations: Conception

This sub-chapter summarizes the early evolution of space stations in low earth orbit (LEO). As of November 2nd, 2020, humans have occupied LEO for 20 years. These are the roots of human space habitation, which is still in its infancy.

### 1973-1974: Skylab – America's First Experimental Space Station

After more than a decade of space research, exploration, and short-duration human space flight, the world started working toward a more permanent future for humans in space. The Skylab missions laid the foundations for the International Space Station (ISS), long-duration space habitation, and scientific research, medical experiments, and earth and solar observation. Skylab missions included three crewed missions (3-person crews) and were occupied for a total of 171 days and 13 hours by astronauts. In total, approximately 300 experiments were conducted on Skylab, including some of the first medical experiments in micro-gravity.

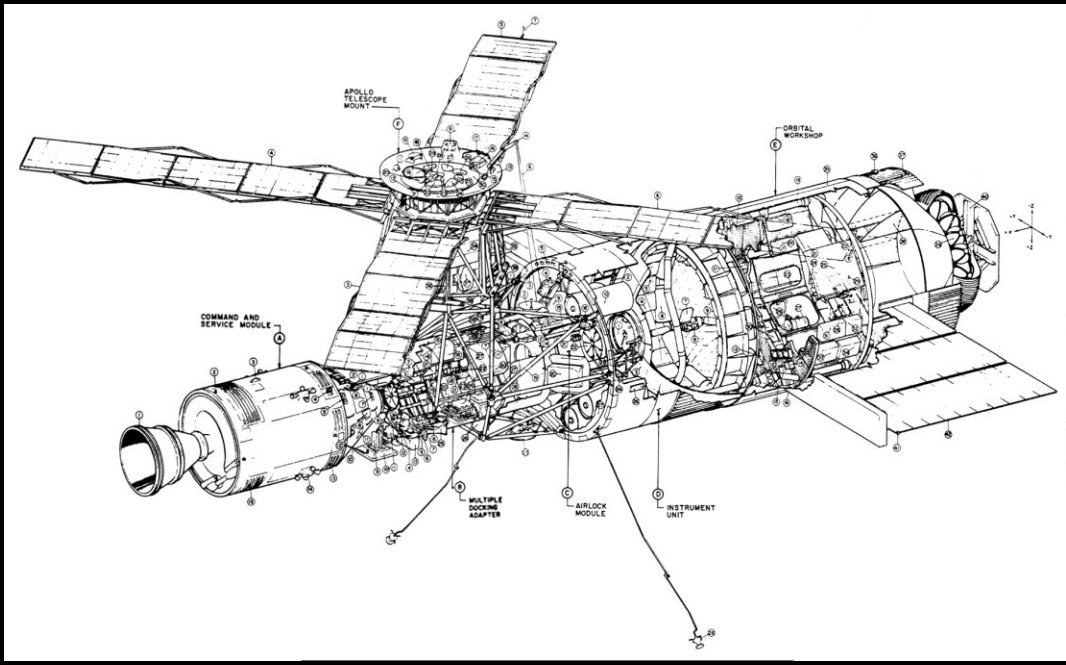



*Figure 10 - Skylab Detail - News Reference. Credit: NASA (1973)*

## 1981-2011: The Space Shuttle Program

The 30-year schedule for the space shuttle program was a foundational step forward for human spaceflight and orbital science. It was also a key to the in-orbit assembly and construction of the International Space Station (ISS) – the largest human structure in space. Shuttles were also key to transporting crews to and from the ISS.

The first shuttle to launch was the Columbia on April 12, 1981. Unlike the Apollo capsules that descended via parachute over the ocean, the space shuttle used wings to glide down to Earth onto a runway like an airplane, making it re-usable. The Space Shuttle Program flew a total of 135 missions and accommodated some 852 shuttle fliers. The orbiter vehicle (OV), commonly known as the space shuttle, was the spacecraft component of the overall shuttle assembly.

*Figure 11 - SS Discovery, launchpad 39a, KSC, Cape Canaveral, FL. Cr: NASA / Bill Ingalls (2011)*



The Space Shuttle included a crew cockpit and a 60-foot-long payload bay large enough to carry satellites and laboratories as well as ISS modules and assembly-related components. It made use of a robotic arm (Canadarm) to assist with maneuvering payload and with other operational tasks.

Below are some notable facts about the Space Shuttle Program:

- The shuttles were built in California by Rockwell International.
- Each space shuttle was named after an historical science or exploration vessel. They were as follows: Enterprise (sub-orbital), Columbia (OV-102), Challenger (OV-099), Discovery (OV-103), Endeavour (OV-105) and Atlantis (OV-104).
- The average cost to prepare a shuttle mission was approximately $775 million USD
- Space Shuttles themselves costed approximately $1.7 billon USD to build
- The overall cost of the program was $113.7 billion USD
- The overall length of the shuttle was 122 feet, 184 with booster stack.
- Wingspan: 78-feet, Height: 57-feet.
- The payload bay: 60-feet (18.3m) x 15-feet (4.6m) diameter. This set the design constraints for equipment and ISS modules being sent up into orbit.
- The heaviest spacecraft deployed was an X-Ray observatory, weighing 25 tons.

Getting things into space and maintaining them there is expensive. Current commercial endeavors seek to increase the size of payload bays and reduce the cost per kilogram ($/kg) of sending things into space via rockets.

*Figure 12 - STS-97. Endeavour, with payload bay doors open. Cr. NASA (2000)*



## Tragedy: The Life Cost of Space Flight

Space flight, engineering, architecture, research, and exploration are remarkable feats of engineering and human ingenuity. But they have come with the loss of life -- reminding us of the dangers of pushing the boundaries of our existence. Below are two tragedies of the space shuttle program – it must be noted that these were not the only incidents where astronauts or cosmonauts (Russian astronauts) lost their lives in pursuit of space flight and exploration.

- On January 28, 1986, the space shuttle Challenger and its seven astronauts were lost on flight STS-51L due to an explosion caused by a booster seal failure.
- On February 1, 2003, the space shuttle Columbia and its seven astronauts were lost when it broke apart during re-entry on flight STS-107.



*Figure 13 - STS-51L Crew Back Row: Ellison S. Onizuka, Christa McAuliffe and Gregory B. Jarvis, Judith A. Resnik, Front Row: Pilot Michael J. Smith, Commander Francis R. Scobee, Ronald E McNair. Cr. NASA (1986)*



*Figure 14 - Left to Right: STS-107 crew David M. Brown, Rick D. Husband, Laurel B. Clark, Kalpana Chawla, Michael P. Anderson, William C. "Willie" McCool, and Ilan Ramon. Cr. NASA (2003)*

## 1986-2001: MIR, Russian Space Station

The MIR space station orbited the Earth between 1986-2001. At the time it was the largest space structure ever created. Occupied by humans for 13 years, MIR station hosted 125 cosmonauts and astronauts from 12 countries. A critical partnership was founded during the Shuttle-Mir program, where American astronauts rendezvoused ten times with MIR station, aboard the Space Shuttle. This partnership began an era of collaboration, paving the way for the International Space Station. [14]

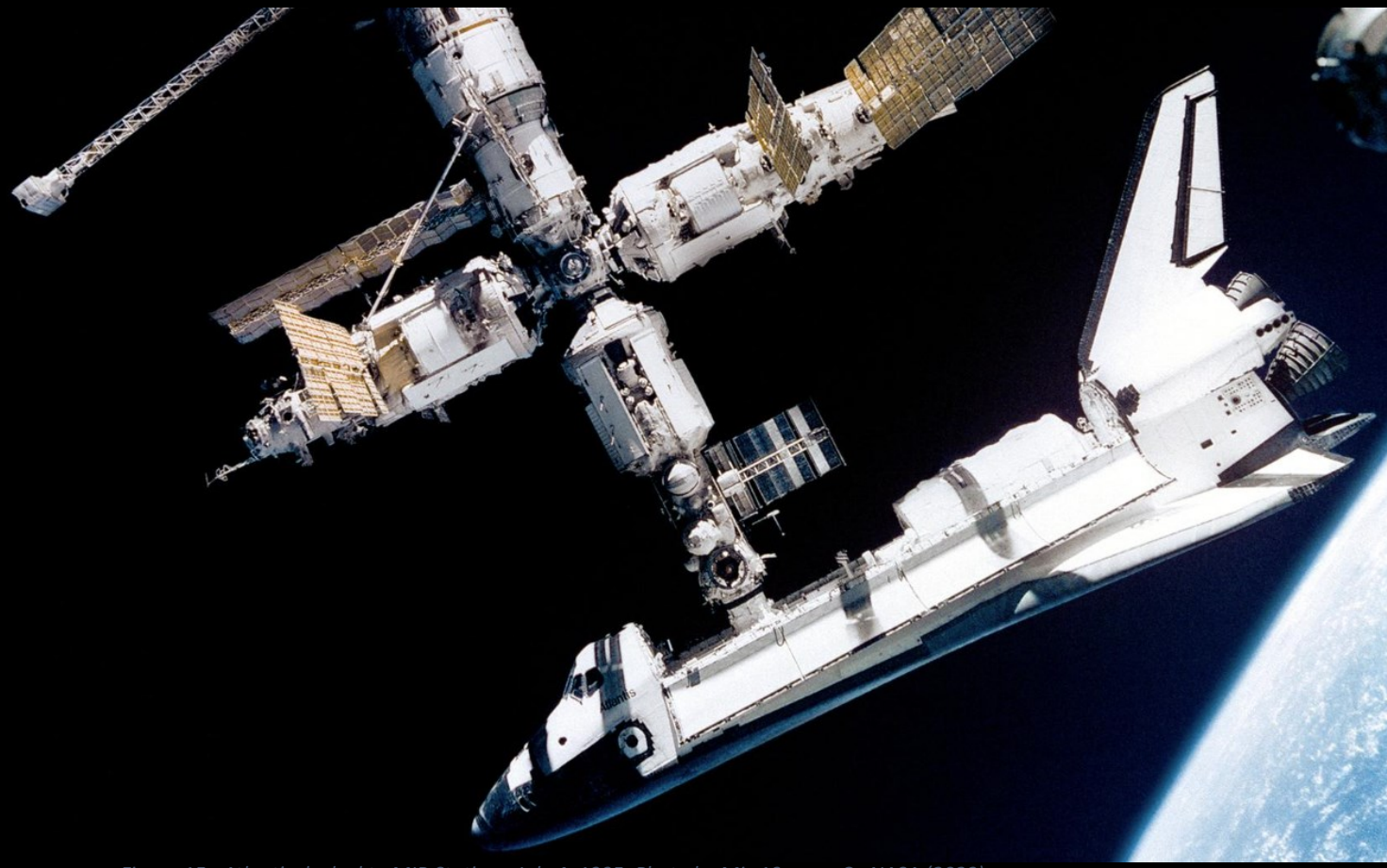


*Figure 15 - Atlantis docked to MIR Station - July 4, 1995. Photo by Mir-19 crew. Cr. NASA (2023)*

[14] Johnson, S. L., NASA "Shuttle-Mir".

## The international Space Station (ISS)

This section outlines the construction and transformation of the ISS over time and highlights some of the scientific, environmental, and technological contributions made by scientists who worked in space. It is intended to help aspiring space architects understand the additive, modular structure of the space station, which is relevant to the construction of future research facilities in space. These include both scientific laboratory modules and the critical modules required for space station operation and habitability.

Detailed information on the ISS's orbital location is described along with orbital concepts in Chapter 4.3 *Orbital Selection Criteria.*

**ISS Assembly and Construction[15]**

The "first brick in the wall" of the ISS was laid on November 20th, 1998, when the Zarya module – or Functional Cargo Block -- was launched into orbit. Its purpose was to provide altitude control and to generate power through its solar arrays. While funded by the U.S, the module was built and launched by Russia's Roscosmos Agency as the first experiment in a decades-long collaboration between the two countries. Altitude control and power generation will be the starting point for the construction of any new facility orbiting around the earth.



*Figure 16 - Zarya Module. NASA. Cr. Roscosmos, (1998)*

Zarya orbited the earth for sixteen days awaiting the Space Shuttle Endeavour crew (STS-88) to deliver the second module of the ISS, namely the Unity Module. Since then, seven international and commercial partners have contributed some 43 additional modules to the ISS.

[15] Garcia, NASA, "ISS Assembly Elements".

Listed and described below are the notable ISS assembly phases.

- **1998 - Unity Module (Node 1)**
  On December 6, 1998, the STS-88 ("Space Transportation System '88") crew mated the Zarya and Unity modules within the Space Shuttle's payload bay, using the **Canadarm** robotic arm and one of two mating adapters brought and installed during the assembly mission. Unity provided living and working space for future crews, supported by fluid and gas lines and electrical cabling that stretched six miles (9.65 km). As it included a total of six berthing ports, Unity – also referred to as Node 1 – also served as a node into which to connect future modules.



*Figure 17 - Unity node being attached by Canadarm robotic arm. STS-88. Cr. NASA (1998)*

- **2000 – Zvezda Service Module**
  This third module, contributed by Russia, included additional living quarters, enhanced life support and service systems on the station, and provided propulsion and additional berthing capabilities to the overall assembly. The Zvezda Service Module also expanded the power generation and storage capacity by adding an additional solar array with a span of 97.5 feet (~29.7m).

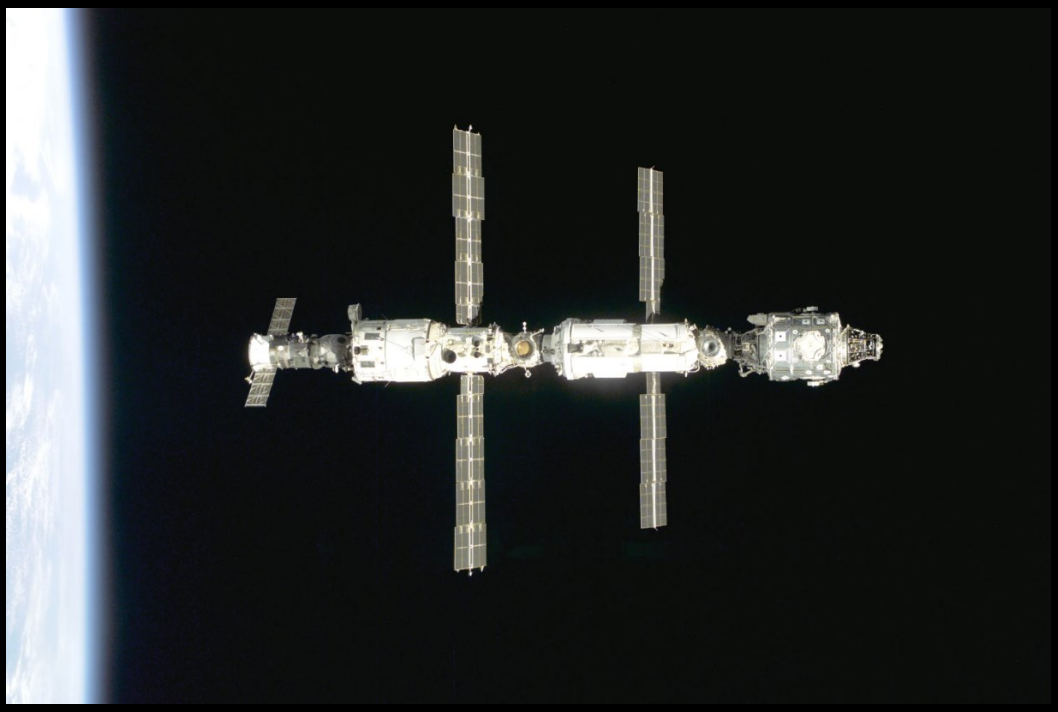


*Figure 18 - Left to Right: Zvezda Service Module, Zarya Module, Unity Module - Above Earth's Horizon. Cr. NASA (2000)*

- **2000 x 2** – Integrated Truss Structures (ITS) Solar Arrays, Thermal Control Radiators, External Payloads
- **2001** – Destiny Laboratory Module.
- **2001** (2005) – External Stowage Platform (ESP) – unpressurized ESPs required electricity to power the heaters of stored equipment
- **2001** – Robotic arm upgrade – Canadarm 2
- **2001** – Quest Airlock – spacesuit maintenance, safe and enhanced EVA capabilities
- **2002** – Mobile Base System (MBS) – movable work platform for Canadarm 2 and future Dextre robot.
- **2007** – Harmony Module (ESA) – living + working space – passageway for international modules + docking ports.
- **2008** – Columbus Laboratory (ESA) – laboratory for microgravity experiments + external payloads for space science and Earth observation
- **2008** – DEXTRE is the most sophisticated space robot ever built and does routine jobs in the harsh space environment.[16]

[16] Canadian Space Agency, “About Dextre”.

- **2008** – Kibo Laboratory Module (JAXA) – Space research facility – biology, physics, technology. Kibo research racks.
- **2009** – Japanese Exposed Facility – Earth observations. Experiments in the vacuum of space
- **2009** – EXPRESS Logistics Carriers 1 and 2 (ELC-1, ELC-2) – support external payloads.
- **2010** – Tranquility Module – life support systems, bathroom, exercise facilities.
- **2010** – Cupola – added for astronauts to have a "window to the world".
- **2016** – Bigelow Expandable Activity Module (BEAM).
- **2020** – NanoRacks Bishop Airlock – research capacity on the outside of the ISS. Allows deployment of larger satellite and transfer of tools from the interior to exterior of the ISS
- **2021**-**23** – Roll-Out Solar Arrays (iROSA) 2B/4B, 3A/4A, 1A/1B: 20 kw. The array provides a 30% increase over current power generation from the 8 existing arrays.

Summarizing the above, the table below was created to categorize the components and capabilities required to build a habitable space station. For future space stations, some of these functions may be combined within the same module.

*Table 1 - Space Station Critical Assembly Sequence*

| Space Station Critical Assembly Sequence | |
|---|---|
| **Assembly Order** | **Capability Provided** |
| 1 | Altitude Control and Power Generation (to regulate orbit and power future modules) |
| 2 | Crew Living and Workspace + Comms (Habitat + Initial Operations) |
| 3 | Additional Crew Quarters + Enhanced Power Generation / Storage + Propulsion |
| 4 | Exterior capabilities – experimental platforms, robotics. Transfer airlocks for interior to exterior interfacing |

For detailed station and module information, visit these NASA resources:

- https://www.nasa.gov/international-space-station/international-space-station-assembly-elements/ [17]
- https://www.nasa.gov/international-space-station/space-station-facts-and-figures/ [18]
- https://www.nasa.gov/international-space-station/space-station-overview/ [19]

For an animated sequence of ISS assembly, see the NASA Johnson video online:

- https://www.youtube.com/watch?v=yRqUPjl3tTQ&ab_channel=NASAJohnson [20]

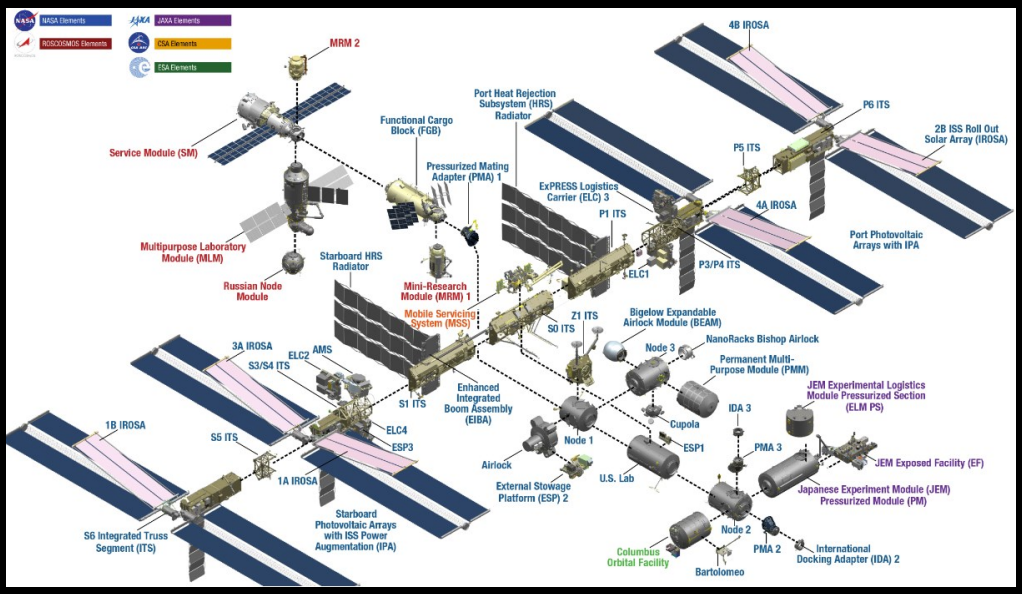



*Figure 19 - International Space Station (ISS) elements by space agency. Credit: NASA (2023)*

[17] Garcia, NASA, "Station Assembly Elements".
[18] Hall, NASA, "ISS Facts".
[19] Graf, NASA, "Station Overview".
[20] NASA, JSC, "ISS Assembly".

**Lab Modules**

For this thesis, I began to focus on the scientific functionality of the ISS. This sub-section highlights a few of the science specific modules. The first decade of the ISS's lifespan was dedicated to its modular assembly, with much of the astronauts' time dedicated to assembly procedures and facility testing and maintenance. Crew time available for scientific activities during this initial assembly phase was limited to 10-20 hours per week. That number now reaches highs of 125 hours per week. It took 10 years for the first 1000 experiments and half the time for the same amount, during the second decade.[21]

**2001 - Destiny Laboratory Module[22]**

- This was the fourth module and first laboratory module to be added to the ISS. It was brought up by SS Atlantis.
- Destiny's Microgravity Science Glovebox[23] (MSG) enables physical and biological research.
- Destiny accommodates 24 equipment racks to house different system and scientific instruments.

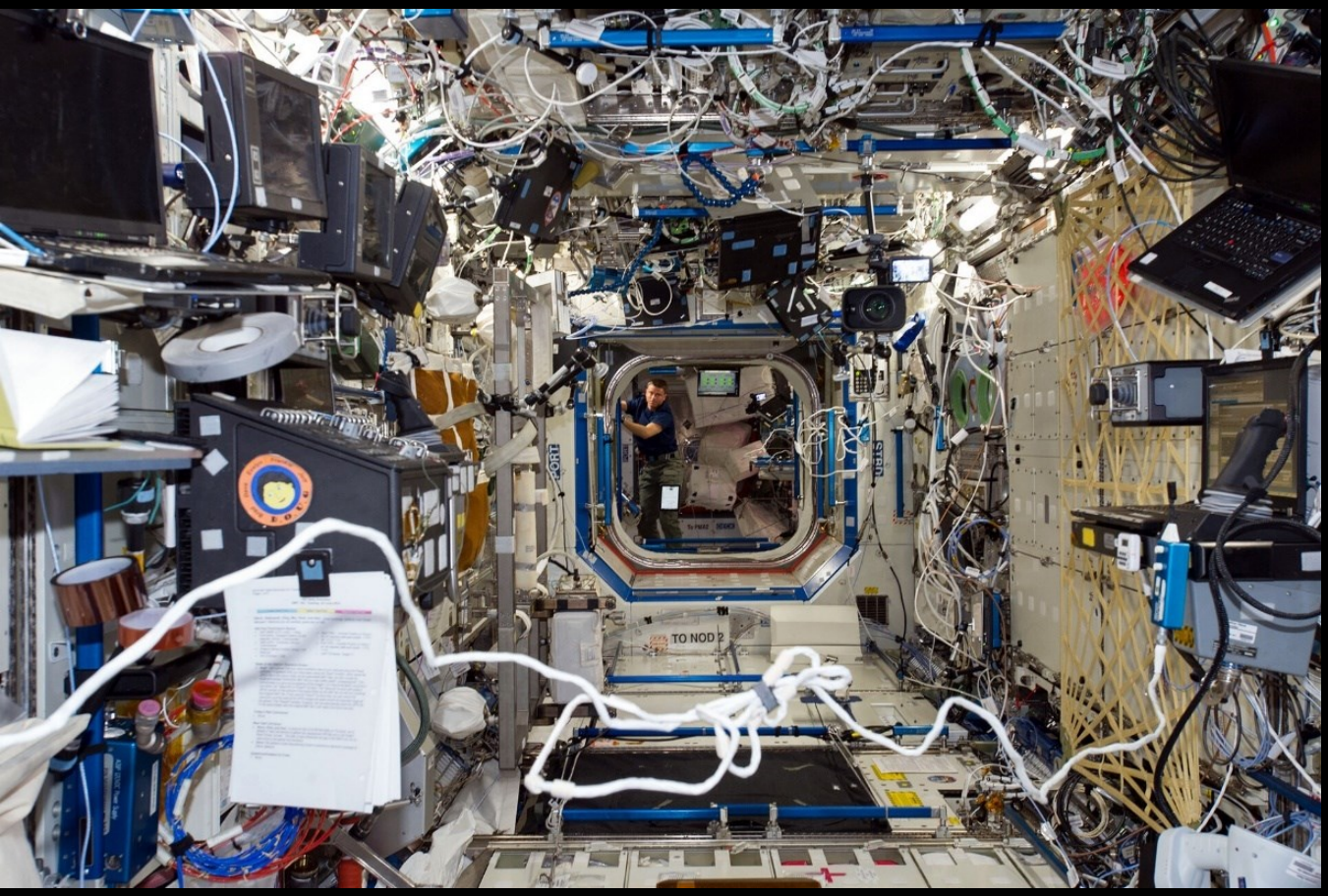



*Figure 20 - Destiny Laboratory Module Interior. Credit: ESA/NASA (2016)*

[21] ISS Program Science Forum, NASA, "ISS Benefits for Humanity".
[22] Hall, NASA, "Destiny Laboratory Module".
[23] Enclosed equipment where scientists manipulate contained experiments using integrated gloves.

**2008 - Columbus Laboratory Module**[24]

- The Columbus module is Europe's largest contribution to the ISS assembly through the European Space Agency (ESA)
- The four external platforms accommodate external payloads for space science, technology, and Earth observation.
- Internal laboratory facilities support a range of research, with a focus on the effects of microgravity on the human body. They include the Advanced Plant Habitat for botany research, supporting deep-space crew sustenance goals.

*Figure 21 - Columbus Laboratory Module. Credit: ESA/NASA (2021)*




[24] Hall, NASA, "Columbus Laboratory Module".

**2008/9 – Kibo Laboratory Module[25]**

- Launched and assembled over three space shuttle missions the Kibo Modules belong to the Japanese Space Exploration Agency (JAXA)
- The Kibo Laboratory Module includes four components: a pressurized lab module, an Exposed Facility (EF) for experiments in the vacuum of space and for Earth observation, the Logistics Module for experiments and materials storage, and a robotic arm.



*Figure 22 - Kibo Laboratory Module, comprising four components. Credit: NASA (2024)*

[25] Garcia, NASA, "Japanese Experiment Module Kibo".

**2021 - Nauka Multipurpose Laboratory Module (MLM)[26]**

- Nauka is a MLM contributed by Roscosmos – the Russian Space Agency.
- The Russian word for "science" Nauka includes a research facility, a docking port, and a spacewalk airlock for future EVAs.



*Figure 23 - Nauka Multi-purpose Laboratory Module - Roscosmos. Credit: NASA (2021)*

[26] Dunbar, NASA Blogs, "Nauka".

## New Space Station Precedents

In anticipation of material and technological deterioration, the ISS is scheduled to be decommissioned by the year 2030. This has led to increased interest, investment, and partnering between governmental organization and private-sector players on the planning, design, and execution of commercial space stations for the twenty-first century. "NASA is already laying the foundation of this future. By extending the operation of ISS through 2030, we are giving U.S. private industry time to develop the capabilities and operational experience to operate in LEO and to deploy the platforms that will meet the needs of NASA and other users there."[27]

This section describes a few of the other space stations in orbit, as well as those being planned and developed.

**Tiangong, Chinese Space Station (CSS)[28]**

- Tiangong means "Heavenly Palace." "The station orbits the Earth at an altitude between 217 and 280 miles (340 to 450 km), approximately the same height as the ISS"[29]
- Tiangong was built, launched, and is operated by the Chinese Manned Space Agency (CMSA). Its three modules were launched separately, and the station was assembled between 2021-2022.
- It is significantly smaller than the ISS, closer in size to the historic Mir space station.
- For more information on the CSS, visit: https://www.space.com/tiangong-space-station

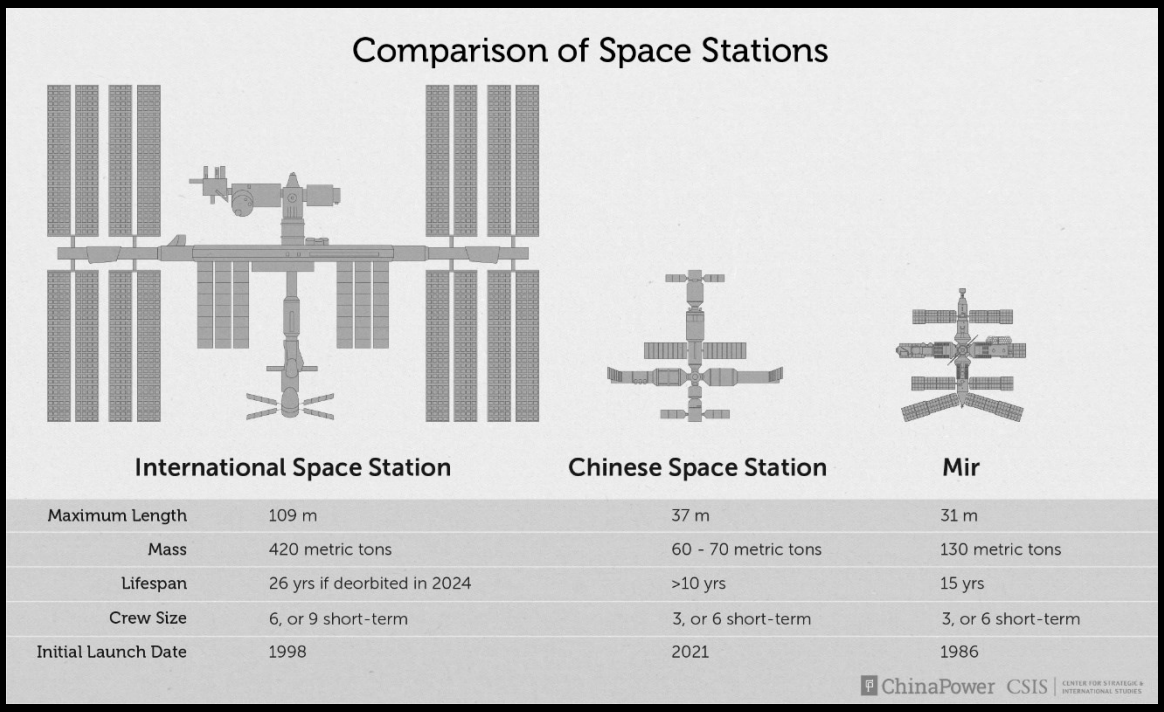



*Figure 24 - Space Stations Comparisons. Cr. Center for Strategic & International Studies (2024)*

[27] Bleacher, NASA, "International Space Station Transition Plan".
[28] China Power Team, "What's Driving China's Race to Build a Space Station?"
[29] Jones, Dobrijevic, "China's Space Station, Tiangong: A complete guide".

**Axiom Station**

In January 2020, NASA signed a contract with Axiom Space allowing it to use the forward Node 2 port on the ISS for a commercial module they are planning to deploy in the mid-2020s. This module will support private activities in LEO that could otherwise not be accommodated on ISS, testing new technologies and new forms of collaboration between public- and private-sector interests. Axiom plans to add more modules to this complex, with the eventual aim of detaching from the ISS and becoming a free-flying facility in LEO.[30]




*Figure 25 - Axiom Space Station. Credit: Axiom Space (2024)*

The Axiom station is the world's first commercial space station, and the module shells are being manufactured by the contractor Thales Alenia Space, a global space manufacturer that was responsible for approximately 40% of the ISS's shell manufacturing.[31] An architectural contribution to the Axiom station is architect Phillipe Starck's design of the crew quarters within the habitation module. Starck's vision was to create a nest, a quilted egg of soft materials. The walls are "sprinkled with hundreds of nano-LEDs that change color with the mood and biorhythms of its osmotic inhabitant while a window provides continuous views of the universe."[32]



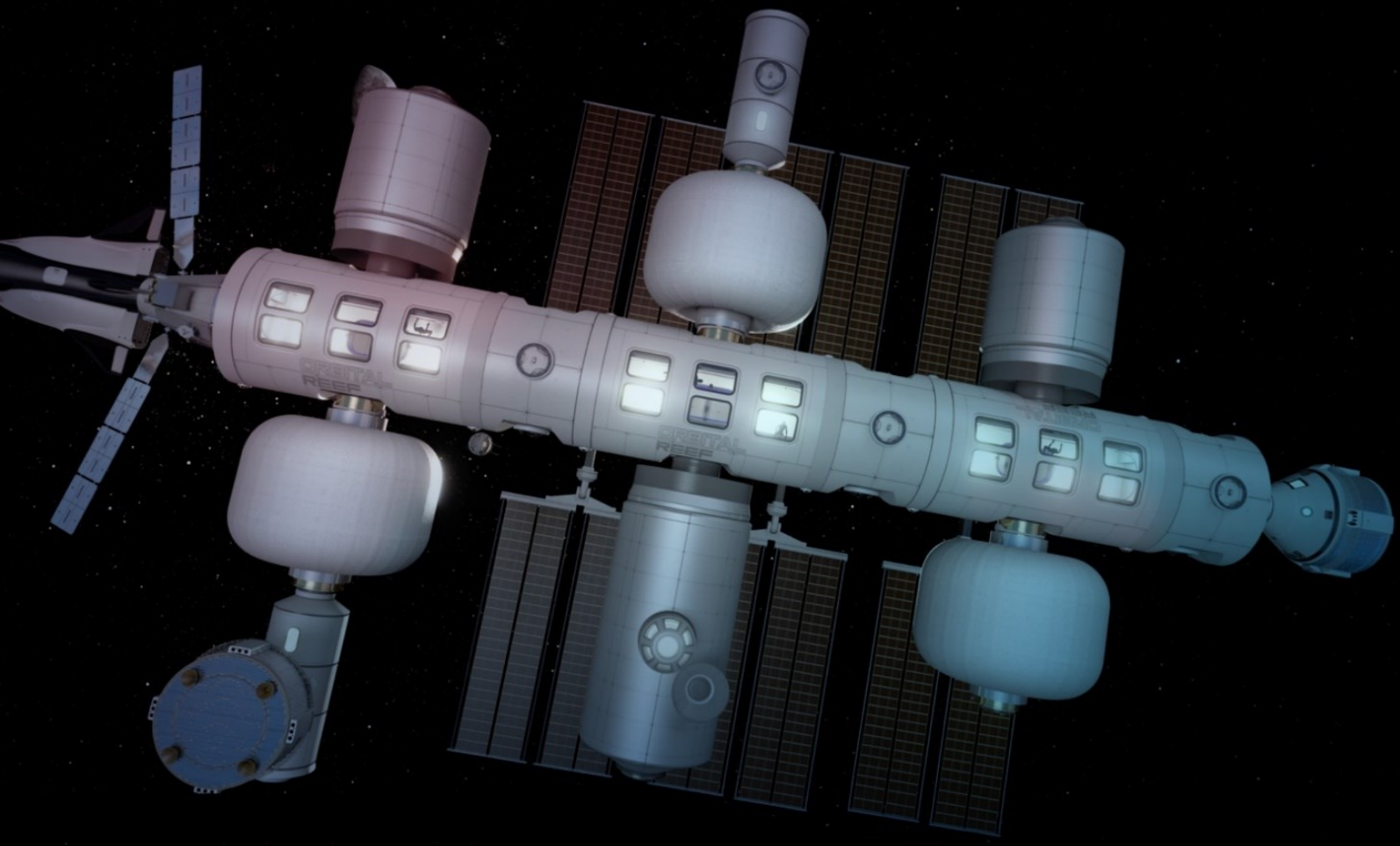
*Figure 26 - Architect Philippe Starck's design for Axiom crew quarters. Cr. Starck.com (2024)*

[30] NASA, "International Space Station Transition Report"
[31] Gebhardt, "Axiom module additions to ISS"
[32] Starck, "Axiom Space Station's crew quarter habitation module".

**Orbital Reef**

Blue Origin and Sierra Space, two other private-sector space companies, announced plans for an "Orbital Reef," a commercially developed, owned, and operated space station to be built in low earth orbit. The station will usher in the next chapter of human space exploration and development by supporting the growth of a vibrant ecosystem, like a coral reef, and business model for the future. Orbital Reef is backed by space industry leaders and other partners including Boeing, Redwire Space, Genesis Engineering Solutions, and Arizona State University.[33]

*Figure 27 - Orbital Reef space station, rigid and inflatable modules. Cr. Blue Origin (2021)*



[33] Blue Origin, "Developing commercial space station".

**Project Gateway**

The NASA-led Gateway Program is an international collaboration to build the first space station that orbits around the Moon. It is a vital component of the larger Artemis campaign that is once again aiming to put humans on the moon. Gateway will help NASA and its partners test the technologies and capabilities required for a sustained human presence in deep space and chart a path for the first human missions to Mars.[34] The Gateway program is an international partnership with contributions by commercial partners and organizations such as the CSA, ESA, JAXA and the Mohammad Bin Rashid Space Center of the United Arab Emirates.

For detailed station assembly, capabilities and updates, visit:

- https://www.nasa.gov/mission/gateway/ [35]
- https://www.nasa.gov/gateway-capabilities/ [36]

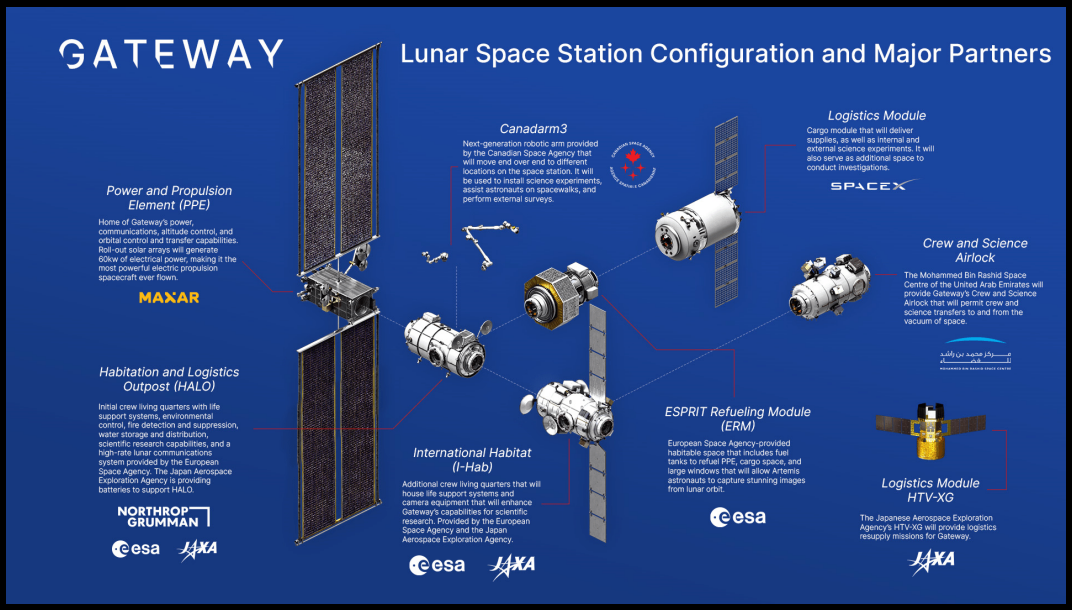



*Figure 28 - Gateway elements designed to support the Artemis campaign. Cr. NASA (2024)*

[34] Zamora, NASA, "Gateway Space Station".
[35] Zamora, NASA, "Gateway".
[36] Zamora, NASA, "Snapshot: Gateway Capabilities".

# Chapter 3: Science in Microgravity

This section describes the fundamentals of microgravity – the gravitational forces active in low-earth orbit (LEO) – and how it compares to gravity on Earth. It describes the benefits of scientific research in microgravity environments. Scientific facts will be summarized and expanded upon in Chapter 3.1: *Project Missions* as they relate to cancer and climate change research for this thesis.

## 3.1 What is Microgravity?

The phenomenon of gravity was described over 300 years ago by Sir Isaac Newton. Through a series of experiments, he was able to demonstrate that falling objects accelerate towards Earth's surface at 9.8 meters per second per second (32.2 ft/sec/sec). Newton categorized this as "normal gravity," or "1g." He developed a thought experiment in which a cannonball is fired from a cannon on the top of a mountain. The distance the cannonball travels before hitting the ground is relative to the speed at which it is fired from the cannon. Given the curvature of the earth, Newton postulated that, if fired fast enough, the cannonball would reach a state of continuous freefall around Earth, a state we have defined as orbit.[37] Objects in this state – whether humans, space stations, satellites, or spacecraft – appear to float. This occurs when objects are moving at a speed of 28,000 km per hour (17,500 mph), achieving the state we call "zero gravity," "0g," or microgravity (1x10-6g). The distance above the gravitational mass, in this case Earth, affects the orbital speed. The different orbits are summarized in chapter 4.2 *Orbital Selection Criteria*.

It is a common misconception that there is no gravity in outer space. Gravity exists around every object in the universe in proportion to its mass. In outer space the gravitational pull of other celestial bodies, including the sun, are ever-present and act on objects simultaneously. The table below compares gravitational forces on earth with other celestial bodies:

*Table 2 - Gravitational Force Comparisons*

| Location | Gravitational Force (Earth=1g) |
|---|---|
| Earth | 1g |
| Orbital Environment* | $10^{-5}$ – $10^{-6}$ (0g) |
| Moon | .16g (approx. 1/6g) |
| Mars | .38g (approx. 1/3g) |
| *Includes LEO, MEO, GEO | |

**Architects spend an entire life with the unreasonable idea that you can fight against gravity.**

➢ Renzo Piano

[37] Sands, Bowman, NASA, "What is Microgravity?".

## 3.2 Science in Microgravity

Conducting scientific experiments in a microgravity environment enables methods and breakthroughs that would not be possible in normal gravity environments, i.e., in laboratories on Earth. Phenomena are observed and manipulated in different ways in this "floating" environment, as physical and chemical processes behave differently in 0g. Basic concepts that are true in 1g are starkly different in 0g; hot air, for example, does not rise, and roots do not grow "downwards." Microgravity environments facilitate innovative experiments spanning every major scientific discipline – physics, microbiology, botany, etc. – limited only by the space available to conduct them. Below are some facts and highlights about general scientific research conducted on the ISS and in microgravity environments. An extensive list of contributions to life science, environmental science, and technological advancements in microgravity, see Appendix A: *Science In Microgravity.*

- More than 180 scientific instruments and facilities have been used on the ISS[38]
- Science on the ISS has expanded to include research on the impacts of microgravity on organisms such as plants, animals, and microbes.
- Human bodies include 100,000+ protein types that provide critical information about our biology, when analyzed. Protein crystal growth (PCG) experiments have demonstrated that higher quality crystals can be produced in microgravity, enabling scientists to crystalize such proteins for closer study – learning more about the human body, disease, and potential treatments.
- Everyday products such as shampoo, toothpaste, and pharmaceuticals have all been improved through liquid suspension research conducted on the ISS.
- As of September 2021: 2000 peer reviewed publications from 4000+ researchers have been supported by research conducted on the ISS.

### Life Science Contributions in Microgravity

Hundreds of life-science and physical-science experiments in microgravity have directly or indirectly contributed to improved healthcare on Earth. The list below highlights some amazing contributions[39] to give aspiring space architects an idea of positive impacts of science in 0g:

- Protein crystal growth (PCG) experiments have helped scientists further understand disease and treatments for cancer, muscular dystrophy, heart disease and more.
- AngieX studies cultured endothelial cells in microgravity – these cells line the walls of blood vessels. AngieX has the goal of developing safer and more cost-effective treatments for cancer by finding ways to deprive tumor cells of oxygen and nutrients through blood flow.
- Axiom Space missions (Ax-1,2 and 3) between 2022 and 2024 have all conducted cancer research on the ISS. The latest mission returned with a potential breakthrough in cancer research.[40]
- "Tumors in Space" is an ESA and Norwegian Space Agency funded project conducting innovative cancer organoid research, planned for launch to the CSS in 2025.[41]

[38] ISS Program Science Forum, NASA, "ISS Benefits for Humanity".
[39] ISS Program Science Forum, NASA, "ISS Benefits for Humanity".
[40] Whittington, "Axiom 3's return to Earth comes with a potential cancer breakthrough"
[41] Parisi, HUB Organoids, "Tumors in Space"

## Environmental Science Contributions[42]

- Space stations have become places for the manufacturing and deployment of satellites:
  - Roscosmos deployed Nanosputnik in 2005 – a satellite that flew a 4-month mission conducting experiments.
  - The Kibo module (JAXA) allowed for even more deployment of small satellites and facilities installed in 2013 (JAXA / Nanoracks) and provided automation capabilities for further developments.
  - As of November 2020 – 250+ CubeSats (small satellites) for Earth observation activities have been deployed in LEO.
- The Total and Spectral Solar Irradiance Sensor (TSIS-1) is an external instrument measuring total energy flow from the sun into Earth, a critical measurement for climate modeling.
- Launched in 2017 - Stratospheric Aerosol and Gas Experiment III (SAGE-III) is an externally mounted instrument whose data informs assessments of ozone depletion and continued understanding of the Earth's atmosphere.
- Earth observing instruments including cameras and other tools hang off of the JAXA Experiment Module, which set a precedent for exterior experiment platforms.

## Scientific Technological Advancements[43]

- Experiments and related instruments have been developed to run autonomously, greatly decreasing the need for crew interaction, and greatly increasing experiment success and result fidelity.
- Canadian Space Agency's Mobile Base, Canadarm2 and Dextre support spacewalks, vehicle docking, maintenance and other external activities.
  - These technologies led to medical advancements in surgeries such as the NeuroArm[44]
  - Image-Guided Autonomous Robot (IGAR) – this advanced surgical robot is derived from advanced ISS robot technology and is able to conduct highly accurate and minimally invasive procedures. It is entering clinical trials for use in the early diagnosis and treatment of breast cancer. [45]
  - Canadarm3 is being developed and will be included in NASA's Gateway project.
- Italian Space Agency / NASA's collaboration for portable on-board printing has allowed for many studies into the efficiency and feasibility of in-space manufacturing for continuous LEO habitation and long-term deep space missions.
- Fiber optics manufacturing in space can lead to the next evolution in communication speeds and technologies.
- ISS Roll-out Solar Array (iROSA- 2021) allows more efficient roll-out solar arrays providing 20-30% increase in power generation.

[42] ISS Program Science Forum, NASA, "ISS Benefits for Humanity".
[43] ISS Program Science Forum, NASA, "ISS Benefits for Humanity".
[44] neuroArm – for more information visit, https://neuroarm.org/
[45] Canadian Space Agency, "Robots from space"

# Chapter 4: Design Methodology

A childhood passion for space exploration, spacecraft, and all things NASA was nurtured by the Space Shuttle Program. Having begun well before I was born, the program ended on July 21st, 2011, when I was 19 years old and commencing my studies in architecture. This passion was re-ignited during my second year of undergraduate studies (May 2020), when SpaceX's Crew Dragon mission transported humans to the ISS, launched by a Falcon 9 rocket aboard a Dragon space craft. This took place nearly a decade after the Space Shuttle Atlantis returned to Earth on its final mission, and during the global COVID-19 lockdown. With the sense of wonder returning, I questioned whether and how, as an aspiring architect, I might be involved with space exploration. To this end, I began exploring the few but rich resources available.

This initial research into space architecture laid the groundwork for this thesis. This chapter describes the process by which I conducted my research and how I approached the symbiotic relationship between research and design. The chapter will clarify the relationships between the written and design portions of the thesis, following the hierarchy of the three (3) objectives listed in Chapter 1: *Executive Summary*.

**"You want to wake up in the morning and think the future is going to be great – and that is what being a spacefaring civilization is all about. It is about believing in the future and thinking that the future will be better than the past. And I cannot think of anything more exciting than going out there and being among the stars."**

- Elon Musk

*Figure 29 - Crew Dragon Launch . SpaceX, Cr. NASA (2020)*



## 4.1 Research Process

Research for this thesis began with the exciting realization of a cross-over between the fields of aerospace engineering and architectural design, although this relationship is in its infancy. Evidence of the connection appeared through resources produced by **The Space Architecture Technical Committee** (SATC), one of 70 Technical Committees of the **American Institute of Aeronautics and Astronautics** (AIAA).[46] The resources, precedents and educational opportunities referenced in SATC documents outlined the steps required to become a space architect. Several of these are mentioned in Chapter 6: *Architectural Perspectives.*

Upon reviewing these resources, I became aware of the impacts that the ISS has had on humanity as a research laboratory. This further sparked my interest as it suggested an opportunity to apply my experience and interest in laboratory design to research in space.

It quickly became apparent that, to design a research laboratory in outer space and justify this potential career trajectory, I would need to familiarize myself with the history of space exploration, acquire a basic understanding of issues involved in transporting equipment and supporting human life in space, while maintaining my perspective and applying my expertise as an architect. The first order of business, then, was to learn what I needed to learn.

Knowing **what** to research was critical. **The 10 per cent rule** described by Scott H. Young in *Ultralearning*,[47] suggests that approximately ten per-cent of one's expected learning time should be invested into research prior to starting. It's helpful to identify key concepts and resources ahead of time in order to spend less time – preferably none – on content that may be interesting, but not immediately applicable to the task at hand. I see this as investing time in building a concise learning curriculum. In the case of this thesis, I needed to understand the elements and logistics associated with orbital space stations in order to focus in on the design a laboratory module and conceptual space station framework.

After subtracting the time required for other work and academic commitment, I was left with a roughly 24h-per-week, 7-month (~28 weeks) schedule to focus on thesis research.

(28 weeks x 10%) = 2.8 weeks, (2.8 weeks x 24h/week) = 67.2 hours of research.

Based on this, approximately 2.8 weeks (67.2 hours) of research was allocated to determining **what** to research, before diving into deeper research and design thinking.

Chapters 2 and 3, *Historical Context and Current Conditions*, and *Science in Microgravity* summarize the initial research, i.e., what I needed to know to move forward. The goal was to learn enough to start designing. Once having sketched out basic concepts and configurations, however, I was confronted with decisions that required additional research. It was a challenging but fulfilling process that allowed me to drill deeper into a field that inspires me. Every question, led to a plethora of answers, debates and further questions. This was like a game of tennis between research and design – each challenging the other with every swing. I also began to engage with people in the industry, who provided valuable insights and, eventually, a real-world case study for my program. The table below illustrates the iterative nature of the research and development process.

[46] For more information on the AIAA – visit, www.aiaa.org
[47] Young, *Ultralearning*, 61-69

*Table 3 - Research through Design, Design through Research*

| Research / Design Milestone | Resulting Findings and Direction |
|---|---|
| 10% Rule[48] | ➢ Invested time in learning about aerospace curriculum, learning objectives and available resources.<br>➢ Archived papers, articles, and websites for future focused research (set myself up for success to learn key subjects). |
| Is there a place for me, for Architects? | ➢ Discovered AIAA-SATC.[49]<br>➢ This led to new resources and suggested next steps.<br>➢ This was an inspirational, "diamond in the rough" find – and give me confidence that I was on the right path. |
| What to design? | ➢ I studied the ISS and other precedents – stations and laboratory modules. |
| What knowledge is required to design a laboratory as part of a space station?<br>(Who, what, where, when and why?) | ➢ This phase involved research of key concepts of the space environment following architectural thinking – Precedent, Program, Materials, Environments, Systems.<br>➢ Brought myself up to date with current initiatives – and selected SpaceX *Starship* as the proposed launch vehicle for my space station. This provided key logistic constraints for the design |
| Designing a Program + Concepts | ➢ Highlighted more gaps in research such as orbit selection, vessel envelope construction, and launch logistics. |
| Deeper Research and Engagement. Why is this needed? How is it different from previous work? | ➢ Drilled deeper into historical research, learning additional concepts and facts.<br>➢ Attended a space conference - SPACECOM24[50] in Orlando Florida (Jan.29-Feb.3, 2024), Inclusive of a Global Spaceport Alliance[51] (GSA) Summit) where I engaged with industry professionals. This highlighted current and **real** needs.<br>➢ Toured space laboratories on display at the Kennedy Space Center and watched rocket launch in person – a cargo resupply (CRS) SpaceX launch from Cape Canaveral, Florida.<br>➢ Additional context and photographs from my trip to the space coast is included in Appendix B - *Research Trip: SpaceCom24 and Space Coast*<br>➢ Identified a specific need. Found case studies and people to support my design goals (designing an orbital space laboratory) |
| Research / Design Hybrid decision. Who is it for? | ➢ Decided to structure the thesis as a kind of primer or manual for aspiring space architects, i.e., architects who aspired to play a role in the design of facilities in space. Recommended that a course on space |

[48] Young, *Ultralearning*, 61-69
[49] SpaceArchitect main website, https://spacearchitect.org
[50] SPACECOM website, https://www.spacecomexpo.com/
[51] GSA website, https://www.globalspaceportalliance.com/

| | |
|---|---|
| | architecture as a course elective option at Carleton University. (“ARCH 2001”) – offered jointly by the School of Architecture and the Department of Mechanical and Aerospace Engineering.<br>➢ Applied the research to date to the design of a space station. It was intended as a case study both to test what I had learned and to determine what I needed to research further. |
| Recalibration | ➢ Knowledge gained and direction having been decided, I continued researching. Summarizing my learning process became primary objective.<br>➢ Began to focus in on the design of a laboratory module focused on research on cancer and climate-change.<br>➢ Continued to identify and engage with knowledge experts and key stakeholders. Further refined the design requirements for the laboratory module. |
| Execution | ➢ The final 6 weeks of the thesis process were focused on design and writing, with some room for exploring and evolving in a way that did not distract from the core undertaking. |




*Figure 30 - Global Spaceport Alliance Summit Attendees. Credit: GSA (2024)*

## 4.2 Project Mission(s)

The goal of the design portion of this thesis, namely the design of a hypothetical scientific research module as part of a hypothetical space station, was to meet specific programmatic requirements through design and technological solutions – and to learn about requirements and test out solutions in the process. The term "project mission" refers to the design parameters as defined by the programmatic needs of researchers, research methods, anticipated outcomes, and the larger issues the research was intended to address. This required an understanding of the spaces in which researchers worked, and the equipment and environmental conditions necessary to conduct research. It was important to consider other aspects of living and working in space, including habitats, life safety, and support systems.

### Life Science Mission

There were an estimated 18.1 million cancer cases around the world in 2020[52] In 2022, the world fact sheet from the Global Cancer Observatory, owned by the World Health Organization estimated a global number of new cancer cases at 19,976,499[53] (nearly 20 million) resulting in 9.7 million deaths.[54] Cancer rates are expected to rise at an alarming rate, costing millions of lives and billions of dollars for global economies.

The life science mission of the conceptual design in this thesis hopes to use an ongoing research project called "Tumors in Space" as case-study and design program. The principal investigator for the *Tumors in Space* project is Dr. Tricia Larose at the University of Oslo, Norway. This innovative project, which sits at the intersection of space technology and stem-cell biology, is funded by the European Space Agency (ESA) with support from the Norwegian Space Agency.

**"Over 35 million new cancer cases are predicted in 2050, a 77% increase from the estimated 20 million cases in 2022"**

- World Health Organization, 2024

Tumors in Space uses tumor biopsies donated by cancer patients in collaboration with Hubrecht Organoid Technology in the Netherlands. These 3-dimensional cancer organoids hold the same structure and function of the cancer patient's tumor tissue, making Tumors in Space the most advanced and physiologically relevant cancer research conducted in outer space. Dr. Larose is hoping to identify a unique mutational signature of cosmic radiation for early detection of cancer and other chronic diseases in astronauts and jet-fighter pilots. She also hopes to slow or stop the growth of tumor organoids via gene expression changes after exposure to microgravity in outer space.[55] Having been underway for several years[56], Tumors in Space will culminate in a spaceflight experiment scheduled for 2025 aboard the Chinese Space Station (CSS). I am grateful to have connected with Dr. Larose and learned about her current and future needs for research in microgravity. Our conversations have informed program, workflows, and included discussion of areas of potential innovation.

[52] WHO, "Worldwide cancer data".
[53] WHO, "World Fact Sheet" 2022.
[54] WHO, "Global cancer burden growing, amidst mounting need for services".
[55] University of Oslo, "Tricia Lynn Lois Larose".
[56] Larose, "Tumors in Space: Preparation for Spaceflight".1-14

## Environmental Science Mission

As a part of Carleton University's Collaborative Specialization in Climate Change, this thesis also attempts to address climate change by highlighting connections between space research and climate action. The orbital laboratory being designed as part of this thesis will include a module for conducting environmental science research. It will include sensors, instrumentation for monitoring and mapping, and will support research on innovative energy production and storage. The scientific laboratory module is designed to allow reconfiguration of flexible space, equipment, and workflows to support different scientific processes. For the purposes of this thesis, the one lab module design aims to support cancer research, with a secondary module (of same size and shape) for environmental science research.



*Figure 31 - Hurricane Irma, 2017. Captured by Cosmonaut Sergey Ryazansky. Cr. NASA (2017)*

A few case studies shared by NASA describe the positive impact that sensors, monitoring, and research on the ISS has had on Earth:[57]

- **Monitoring and documenting hurricanes:** Cosmonaut Sergey Ryazansky photographed Hurricane Irma in 2017. Photographs were then georeferenced for use by hazard teams on Earth. Similar imagery is used for wildfire events – passing information from space to Earth such as the direction of smoke plume travel.
- **Orinoco River between Colombia and Venezuela:** Flooding in July of 2020 affected 1,200 families and destroyed 1,600 hectares of crops (4,000 acres). The Colombia Institute of Hydrology, Meteorology, and Environmental Studies asked NASA's Earth Applied Sciences Disasters Program to help analyze the impacts of the flooding. NASA responded as follows:
    - Crew Earth Observations provided astronaut photographs from the ISS.
    - NASA's Earth Science and Remote Sensing Unit at Johnson Space Center (JSC) matched photos with geo-locations.
    - This helped identify crucial sites for immediate emergency response.

[57] ISS Program Science Forum, NASA, "ISS Benefits for Humanity".

    - This displayed the purpose and benefits of the International Disasters Charter, which works with international experts to acquire information to support disaster relief – saving lives, property, infrastructure, and the environment.[58] For news on past and current Charter activities, visit: https://disasterscharter.org/
- **Volcano spotting**: Astronaut Jeff Williams first spotted and photographed activity from the Cleveland Volcano in Alaska (2006). He alerted the Alaska Volcano Observatory, providing them advanced notice for any response required on the ground.
- **Nighttime imagery:** Studies provide data and graphics on tracking light pollution and effects of urbanization on climate change.

The following table – from a Space Generation Advisory Council (SGAC) report entitled "Saving Our Future on Earth Through Our Presence in Space"[59] – includes examples of how satellites have been instrumental in fostering a green economy. As defined by the United Nations Environment Program, green economies are "low carbon, resource efficient, and socially inclusive."[60] The examples below highlight aspects of climate crisis discussed in Carleton's specialization in climate change, namely that climate action must be multi-faceted – there are no one-size-fits-all solutions – and requires study from all disciplines and by levels of government as well as by industry to ensure that mitigation and adaptation strategies are equitable and just.

*Table 4 - Examples of satellite use cases to foster green economy – Credit: SGAC (2022)*

| Sector | Satellite Use Case |
|---|---|
| Urban | • Assessing the extent of urban footprint and its growth over time<br>• Warnings about potential infrastructure instability |
| Agriculture | • Analyzing climate impact on crops and fields' productivity<br>• Identifying conditions for disease outbreak, famine, or disasters |
| Transport | • Supporting real-time traffic management systems<br>• Detailed mapping of roads and their features |
| Waste | • Identifying the optimal waste landfill sites<br>• Detecting and early warning of landfill subsurface fires |
| Energy | • Optimizing of operations to minimize costs and emissions<br>• Supporting authorities to define, manage, and monitor compliance with the law |
| Forestry | • Detecting forest carbon stocks/ emissions<br>• Monitoring illegal deforestation |
| Blue Economy | • Monitoring of coastal and ocean environments/ecosystems<br>• Assessing ocean conditions for workers/fishers at sea |

[58] The International Charter Space and Major Disasters, "About the Charter".
[59] SCAG, "Saving Our Future on Earth Through Our Presence in Space", 51.
[60] United Nations Environment Programme, "Towards a Green Economy: Pathways to Sustainable Development and Poverty Eradication," 2011.

Earth observation technologies contribute significantly to climate action, and spin-off technologies support climate action here on Earth. The following list from the SGAC report includes examples of spin-off technologies, with additional references from NASA and ESA.

- Carbon capture, energy efficiency, insulation, clean energy production and solar technologies[61]
- Space sensors reducing emissions from heating systems[62]
- Mars spectrometer to detect methane gas leaks[63]
- Detoxifying soil and groundwater, and space-inspired farming and closed-loop systems[64]
- Green building and smart monitoring systems[65]
- Durable wind turbines using Mars technologies[66]

For more Information on ESA's Technology Transfer Programme Office (TTPO) visit:
https://www.esa.int/Applications/Technology_Transfer
For more information on NASA's Technology Transfer Program, NASA's spin-off technologies and specifically how it helps in the fight against climate change, visit:
https://technology.nasa.gov/ , https://spinoff.nasa.gov/ , https://spinoff.nasa.gov/climate-change

For more information on the United Nations Office for Outer Space Affairs (UNOOSA) and the relationship between outer space and the UN Sustainable Development Goals (SDGs), visit:

https://www.unoosa.org/

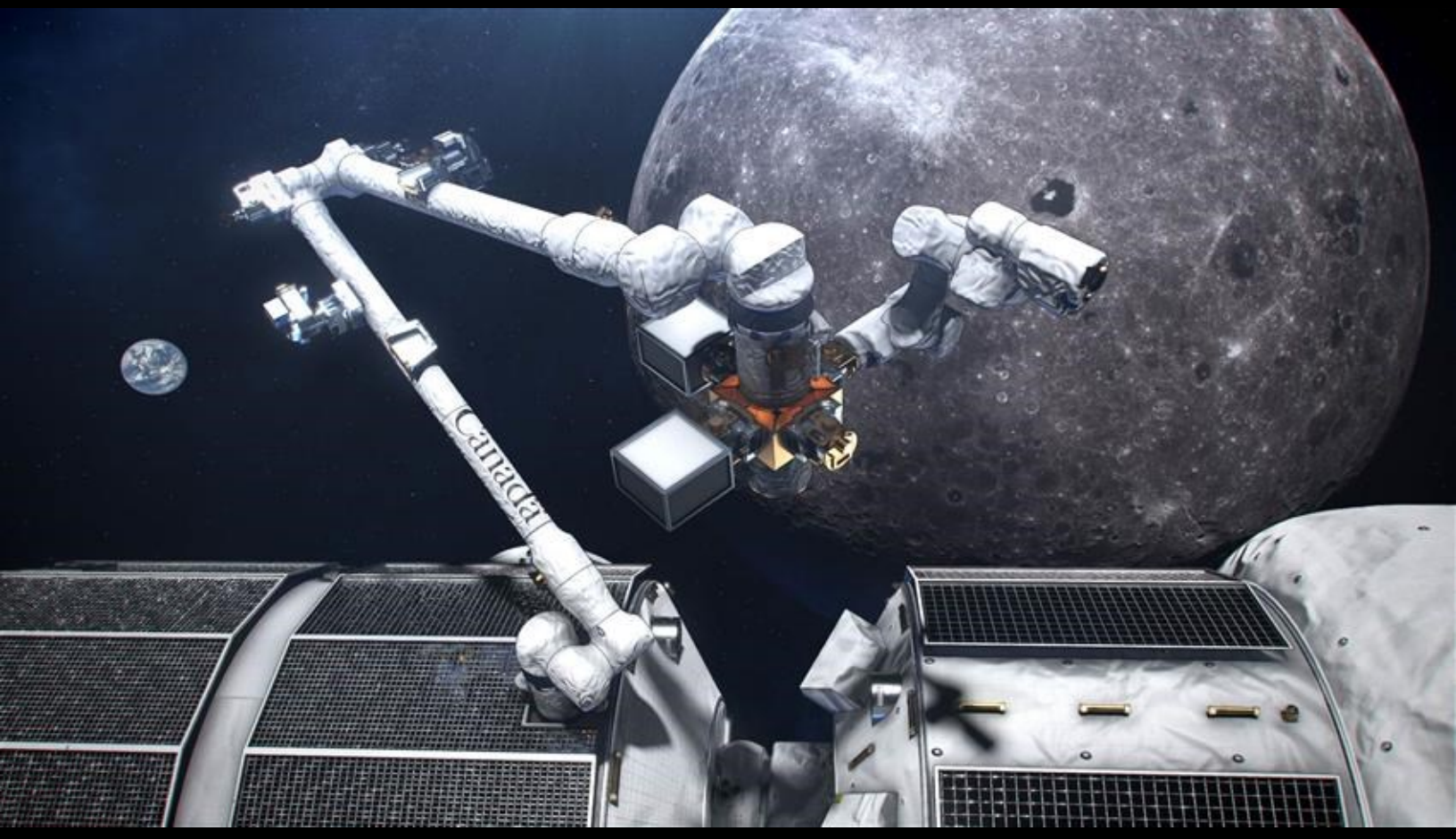
[61] NASA, "NASA Technologies Spin off to Fight Climate Change".
[62] ESA, "Space technology helps mitigate climate change".
[63] NASA Spinoff, "Methane Detector Sniffs Out Leaks".
[64] NASA Spinoff, "SPINOFF 2024 Report", 47.
[65] NASA Spinoff, "Building-Monitoring System Provides Insights for Sustainability".
[66] NASA Spinoff, "Mars Technology Spawn Durable Wind Turbines".

## Technological Mission

In the course of my research, I have come across innovative ideas about how to address future needs and support additional activities on orbital space stations. I hope to engage with some of these in the design of the research module and the space station as a whole.

Several of the technological innovations listed in Chapter 3.2 *Science in Microgravity* should become baselines for design. These include advances in super-heavy payload capabilities, robotic technologies, in-place manufacturing, in-orbit servicing, bio-manufacturing, and solar arrays. The research module I am designing as a proof-of-concept project will include customized equipment and will accommodate new research workflows in response to a call for international standards for cancer research in microgravity laboratories.[67]

*Figure 32- Concept image of Canadarm3. Credit: CSA (2024)*




[67] Discussions with Dr. Tricia Larose highlighted the reality that equipment rack standards are different on the ISS and the CSS, requiring her and her team do design customized research equipment to conduct the same types of experiments on both stations. In a future where more space stations exist, including stations dedicated to cancer research, a global standard for cancer research equipment, workflows and processes in space would provide a benefit to scientists.

## 4.3 Orbital Selection Criteria

Site selection is a crucial consideration in architectural design – ideally something that is done early in the design process. Similarly, selection of the orbit in which a laboratory will operate will have an important impact on the kinds of research it can support. This section describes different orbit types, their respective characteristics, and the rationale of siting the conceptual design in this thesis in low-Earth orbit (LEO).

**Orbit**: The curved path of a celestial object or spacecraft around a star, planet, or moon, especially a periodic elliptical revolution

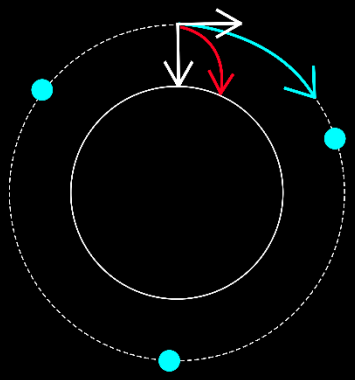


*Figure 33 - Reaching Orbit*

To reach orbit, we use rockets to launch payloads (crews, cargo, satellites) into outer space, following Isaac Newton's thought experiment described in Chapter 3 *Science in Microgravity*. Once rockets reach their target orbit (speed and altitude) the payloads are deployed (e.g., satellites), or in the case of spacecraft docking with a space station, rendezvous and docking maneuvers are conducted. Chapter 4.5 *Mission Architecture* discusses rocket and payload concepts and characteristics in greater detail.

For the purpose of the research in microgravity environments, the assumption is that modules, equipment, and/or crew are being introduced to circular or nearly circular orbits in LEO. Objects in these orbits spend equal amounts of time in opposite hemispheres. Alternative orbital paths exist, such as a highly elliptical orbit (HEO).

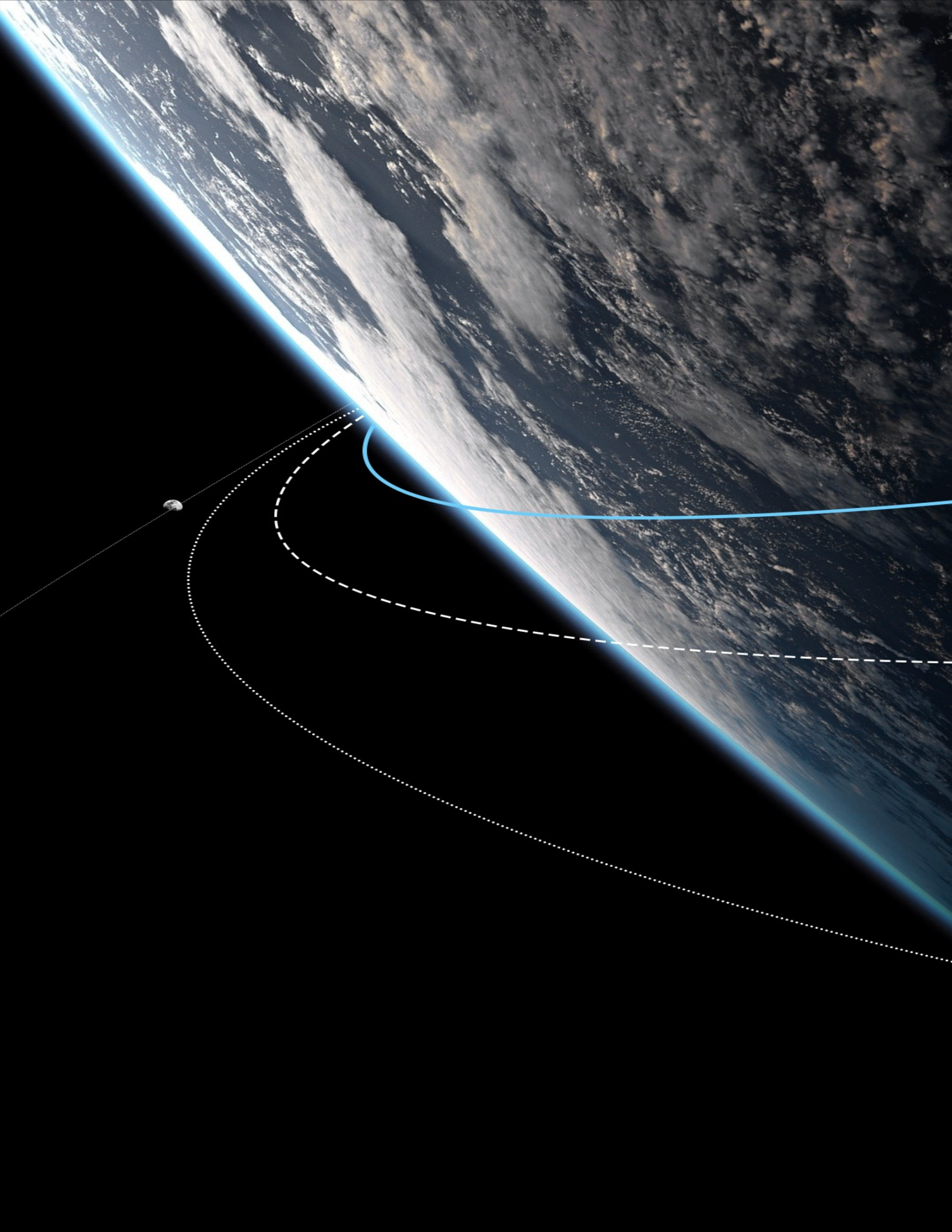
For more learning on both circular and non-circular orbits visit:

- https://aerospace.csis.org/aerospace101/earth-orbit-101/ [68]
- ESA - Types of orbits [69]
- https://science.nasa.gov/learn/basics-of-space-flight/chapter5-1/[70]

Following page:

*Figure 34 - Orbiting Earth, Ramon RS. Base image Credit: wall.alphacoders.com (2024)*

[68] Roberts, "Popular Orbits 101".
[69] ESA, "Types of Orbits".
[70] NASA, "Basics of Spaceflight, Chapter 5:Planetary Orbits".



## Low Earth Orbit (LEO) – 160 km to 2,000 km Above the Earth

Objects in Low Earth Orbit (LEO) follow an orbital path at altitudes between 160 km and 2,000 km above Earth's surface. A notable example is the International Space Station (ISS), which orbits at an altitude of between 370-460 km (200-250 miles) and travels at a velocity of 28,000 km/h (17,900 mph, ~ 8 km per second). Rocket thrusters provide periodic altitude adjustments to keep the station in its orbit.

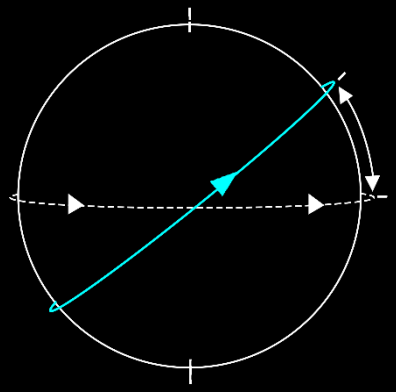


*Figure 35 - Orbital Inclination Diagram.*

Orbital inclination is an important consideration for objects in LEO. Think of orbital selection as a determinant of **how much** you can see, e.g., as a function of your distance from earth and field of vision, while orbital inclination will determine **what** you see along any given path of travel. Figure 35 illustrates this basic yet critical concept for orbital selection.

The orbital inclination of the ISS, for example, is 51.6° (measured from the equator) and covers 90% of the Earth's population (strategic inclination). The station's 90-minute orbital periods allow crews and instruments to observe 16 sunrises and sunsets in each 24-hour period.[71] This is a function of the speed and altitude associated with a low-earth orbit.

The *Space Station Orbit Tutorial* by NASA JSC provides key facts and visual explanations of the ISS orbit: https://eol.jsc.nasa.gov/Tools/orbitTutorial.htm [72]

Notably:

- 55% of all satellites are located in LEO.
- With the exception of the Apollo missions to the moon, all human space activity has occurred in LEO. [73]
- Commercial airplanes fly at an average altitude of 8-11 km (5-7 miles)[74]. Comparatively, even the lowest object in LEO is more than 10x further from the earth than your average flight.
- LEO is ideal for habitation – due to our current research record and the relative ease of access due to its proximity to Earth's surface.
- With our current technology, LEO provides us the highest resolution imaging, sensing, and monitoring capabilities of the Earth's surface in support of climate change research.

[71] Garcia, NASA "ISS".
[72] Evans, Robinson, "Space Station Orbit Tutorial".
[73] Roberts, "Popular Orbits 101".
[74] Flight Deck Friend, "How High Do Planes Fly?".

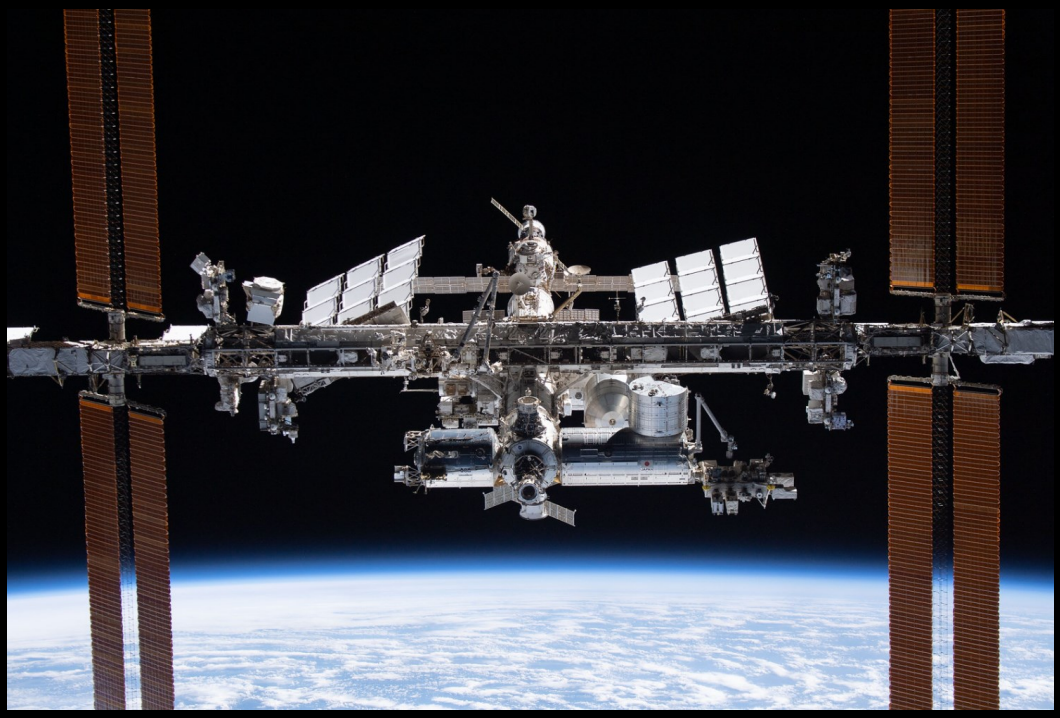


*Figure 36 - ISS in LEO, from SpaceX Crew Dragon. Credit: NASA JSC (2021)*

## Medium Earth Orbit (MEO) - 2,000 km to 35,785 km Above the Earth

Orbital periods and velocities vary in MEO due to a relatively large altitude range , i.e., 2,000 to 35,785 km above the Earth. In MEO, objects can be placed in semi-synchronous orbit at an altitude of 20,200km, where the orbital period is 12 hours (2 times/day) at velocities of approximately 13,900 km/h.

- MEO is used for many different satellite applications across many orbital inclinations.
- Satellites in MEO can see more of the Earth at one time compared to LEO.
- Transmission/signal delays in MEO are slightly larger than LEO.
- Van Allen Belts are located in MEO. These are two doughnut-shaped magnetic fields, centered on Earth's polar axis, that trap charged particles from cosmic rays and solar winds. This charged environment causes damage to electronics, solar arrays and poses a threat to biological life. For these reasons, this is not a popular orbit for satellites and certainly not for human habitation due to the biological threat of high radiation.[75]
- Satellites within these higher radiation environments require additional shielding to lengthen their operational lifespans.
- The Global Positioning System (GPS) operates in MEO. 24 satellites have been positioned in near semi-synchronous orbit such that 4 satellites can be accessed from anywhere on planet Earth at any given time.

[75] By comparison, it is faster for humans to travel back and forth from LEO vs. MEO. However, the high radiation environment in MEO does present an opportunity for advanced radiation research by well shielded and autonomous vessels using advanced robotics, minimizing the need for human presence to conduct science experiments – This autonomous future is true for all outer space environments.

- European Galileo system satellites operating in MEO provide significant coverage at all times.

## Geosynchronous / Geostationary Orbit (GEO) - 35,786 km + Above the Earth

Objects in GEO are approximately 35,786km from Earth's surface and require an orbital speed of 11,110 km/h (3 km per second). This orbital speed precisely matches Earth's rotational velocity such that objects in GEO have an orbital period of 1 day (24 hours).

- Approximately 35% of operational satellites are in GEO[76]
- Satellites in GEO can see approximately 33% (1/3) of the Earth's surface at all times.
- Most satellites in GEO follow the equator (have no inclination). These are categorized as geostationary as they appear in the same location in the sky at all times, making them beneficial for broadcasting, communications, and weather monitoring.
- At this altitude, transmission delays are approximately 0.25 seconds (83 times the delays associated with LEO).
- GEO is significantly further away than current and proposed space stations. Due to transmission delays and required travel distance, it is less than ideal for human habitation – with our current capabilities. Radiation, however, is less of an issue in GEO.

## Axes in Space

Due to the presence of gravity on earth architects make a strong distinction between *area* (X,Y) and *volume* (X,Y,Z). And where the Z axis is concerned, *up* is significantly different than *down*. Moreover , on earth, distinctions between X and Y are important for the purposes of direction and orientation, e.g., front vs. back, North, East, South, West, etc. In outer space, specifically on a LEO spacecraft, we must use different terminology to communicate. Some terms reference traditional shipping language, using the Earth and the direction of travel as reference points.

NASA' uses the following terminology[77]:

- **Zenith** – directly above (away from Earth), referred to as "overhead" while inside the spacecraft.
- **Nadir** – directly below (towards Earth), referred to as "deck" while inside the spacecraft.
- **Port** – direction to the left side, relative to space station / craft direction of travel.
- **Starboard** – direction to the right side, relative to space station / craft direction of travel.
- **Forward / Bow** – direction of travel
- **Aft** – or wake, opposite direction of travel.

[76] Roberts, "Popular Orbits 101".
[77] NASA, "Guide to the International Space Station – Assembly Complete Edition".

## 4.4 Design Parameters

Design parameters for astrotectonics involve similar issues as those that apply to architecture on Earth. These include concepts like human health and safety, designing for specific environmental conditions and basic structural principles that underlie the design of buildings. This section discusses some of these topics and includes references for further study.

### Space Station Base Requirements – Human Health and Safety

Life safety, the highest priority for architects on Earth, is an ever-present challenge involving ethical considerations, design knowledge, and technical solutions. While this topic includes fire and smoke prevention and suppression as well as emergency egress, it could extend to physical accessibility, indoor air quality, and access to potable water – all of which are essential for human health and well-being. When designing for space, these considerations must be applied to environments that have no access to oxygen, limited egress options, limited life-support systems, and no access to external emergency response (for now). The United States Space Force (USSF) has made progress in the development and deployment of tactical emergency response capabilities in space that could be applied to space station emergencies.[78]

The Environmental Control and Life Support System (ECLSS) in operation on the ISS is a useful precedent.[79]

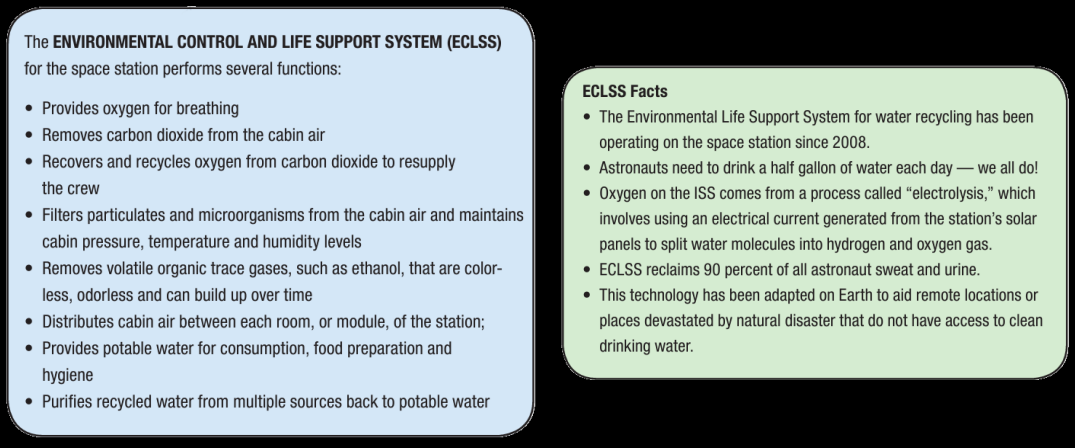



*Figure 37 - ECLSS functions and facts. Cr. NASA MSC (2014)*

[78] USSF, "successfully concludes VICTUS NOX Tactically Responsive Space mission".
[79] NASA, MSC, "Environmental Control and Life Support System (ECLSS) Fact Sheet".

## Structural Requirements

As with buildings on earth, structural loads must be addressed in the design of space architecture. The table below compares typical loads for Earth to those in Space to help space architects understand similarities and differences.

| Earth Structures | Space Structures |
|---|---|
| **Live Loads:** | **Acceleration Loads:** |
| People / Activities | Launch |
| Equipment Operations | Reentry / Landing |
| **Static Loads:** | **Static Loads:** |
| Equipment / Structures | Pressurization (tensile) |
| Snow and Ice | Artificial / Low Gravity |
| **Environmental Loads:** | **Environmental Loads:** |
| Wind | Thermal stresses and Structure Deformation |
| Earthquakes | |
| **Vibration / Impact Loads:** | **Vibration / Impact Loads:** |
| Equipment Systems | Equipment / Structures |
| Machinery Operations | Flight Operations |

*Table 5 - Earth - Space Load Comparisons – Credit: SICSA (2024)*

Although orbiting space stations may appear to be floating in space, they are actually moving rapidly at speeds that vary across orbits. As noted above, space stations in LEO travel at approximately 17,700 miles per hour (28,485 km/h). Since these stations are subject to the Earth's gravitational pull, they require periodic thruster adjustments to maintain their orbital height. In addition, stations must accommodate impacts associated with the docking of spacecraft. Both the movement of the station and spacecraft docking cause vibrations and stresses that must be considered in the structural design.

## Environmental Conditions

In order to contend with environmental factors on Earth, buildings must be designed to address daylighting, heat retention or protection, noise mitigation, wind, air and particle infiltration, precipitation, and seismic activity. Space architects must also consider the environments for which their buildings are designed. Environmental factors in outer space include micrometeoroids, thermal loads, material degradation, and various types of radiation. To endure the unique and extreme conditions and forces at play in outer space, astrotectonics must also consider structure, systems, and envelopes.

To understand the environmental conditions with which spacecraft must deal, it may be helpful to consider what the "weather" is like in space:

- What are the equivalents to wind and precipitation?
- What are effects of the diurnal cycles on heat gain, access to daylight, and quality of life?
- How do these conditions change across different orbital selections?
- What are the assembly design considerations in a realm with no or a vastly different atmosphere?

Space architects must understand these concepts, given their impact on structure, materials, program adjacencies and systems design.

For a comprehensive understanding of the environmental factors at play in space, refer to NASA's Johnson Space Center: *Researcher's Guide to: International Space Station, Space Environmental Effects.*[80] This document includes information on tests and experiments conducted over the years.

**Hazardous Debris – Micrometeoroids and Human Debris**

Although space may appear empty to the naked eye, millions of floating particles of varying sizes travel through it in every direction and at different speeds. While space debris is primarily natural, much of what is found closer to Earth consists of particles from old rockets, satellites, and human operations in space. Where natural particles are concerned, over 7,000 asteroids have been identified in our solar system alone. Asteroids are 1,000 km or less in diameter and are typically found in our solar system's asteroid belt. By and large they pose little threat to space stations in Earth's orbit as they tend to follow their own orbit around the sun and are documented and monitored. For more resources on orbital debris visit NASA's Orbital Debris Program Office (ODPO) Website: https://orbitaldebris.jsc.nasa.gov/ [81]

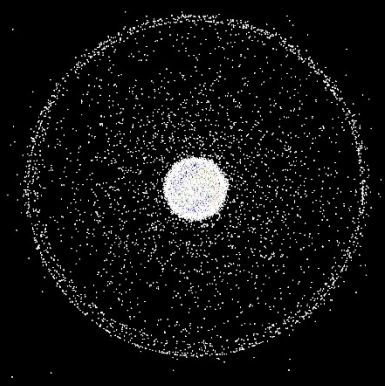


*Figure 38 - Concentrations of objects in L/M/GEO regions. Cr. NASA ODPO (2019)*

[80] Finckenor, de Groh, "A Researcher's Guide to: ISS, Space Environmental Effects".
[81] Visit: NASA Orbital Debris Program Office

The real threat for space stations are **meteoroids –** defined as small rock or metallic bodies ranging from the size of grains to a meter wide – and **micrometeoroid orbital debris** (MM/OD) – defined as microscopic particles in space, usually weighing less than a gram. MM/OD can travel as fast as 60 km per second[82] and is much harder to track than meteoroids.

Design strategies that space architects must consider to protect spacecraft from hazardous debris include:

- Shielding strategies that prevent penetration of debris traveling at hypervelocity.
- Strategic placement of main structures and ancillary structures. Forward facing elements are more likely to be struck by debris than those located on the aft.
- Assembly material should prevent penetration into core levels of the station.

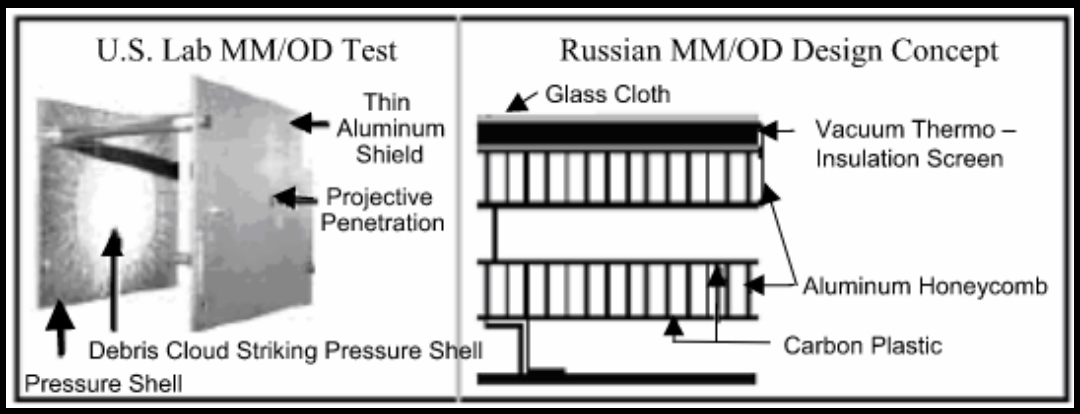



*Figure 39 - Comparison of US and Russian MM/OD Systems – Cr. SSD&O (2004)*

[82] Finckenor, de Groh, "A Researcher's Guide to: ISS, Space Environmental Effects".

**Thermal Loads**

In addition to designing an envelope to withstand space debris, space architects must also address the extreme thermal loads associated with space. With minimal atmospheric protection, temperatures in LEO fluctuate from extreme cold to extreme heat throughout orbital cycles, i.e., extreme heat when the space station is between the sun and the Earth (sunlit side), and extreme cold when the Earth shields the space station from the sun (shaded side). Depending on orbit selection and inclination, radical fluctuations in temperature can occur as many as 16 times a day, or approximately every 90 minutes (ISS orbit). External temperatures on the sunlit side of the space station reach around 121 degrees Celsius, and drop to -157 degrees Celsius on the shaded side – a delta of almost 300 degrees.

Design strategies that space architects must consider to address external thermal loads include:

- Thermal hardening of exposed structural elements and exterior, surface mounted equipment and robotics.
- Heat blankets, coatings, and other shielding to help reflect and/or dissipate heat radiation from the exterior of the station.
- Positioning of window openings to minimize overheating of spaces withing the station. (daylighting strategies)
- Careful selection of structural materials to mitigate thermal degradation and structural fatigue.

*Figure 40 - The UK and Ireland from the shaded side – Cr: André Kuipers NASA/ESA (2012)*



**Radiation**

While there may be no air or wind in outer space, similar, constant, and at times extreme forces exist in the form of **radiation**. Radiation shielding protects personnel and equipment from damage.

Radiation effects come in a few forms in LEO space, including:

- Solar Proton Events (SPE) or Solar Cosmic Rays (SCR) – This is standard solar activity including, solar winds and solar flares that can reach lethal intensities. These occur at different frequencies (sporadically) over 11-year cycles.
- Galactic Cosmic Radiation (GCR) – high energy protons with low flux density that pass through most shields.
- Nuclear radiation – natural and from human activities.

To address user life safety in astrotectonics, space architects must understand radioactivity and the life-threatening characteristics of exposure to radiation. **Ionizing Radiation** refers to subatomic particles or electromagnetic waves that move with sufficient force to ionize atoms and molecules, i.e., to detach electrons from them. They can affect water, air, and living tissues, which are of primary concern for human health aboard space stations.

For an understanding of my research of space conventions and advances in space radiation measurement and application to design considerations, Refer to Appendix C: *Space Radiation*.

For additional learning on space radiation, visit:

https://www.nasa.gov/directorates/esdmd/hhp/space-radiation/

https://www.nasa.gov/missions/analog-field-testing/why-space-radiation-matters/

*Figure 41 - Solar Flashes captured by NASA's Solar Dynamics Observatory. Cr. NASA (2014)*



**Material Degradation - UV and Atomic Oxygen**

The materials, constructional systems, and siting of terrestrial architecture are critical design considerations. Choices are based on a wide range of factors including climate and potential degradation from solar exposure, wind, and precipitation. For astrotectonics, material selection, space station orientation, and internal program adjacencies are influenced by similar factors, including constant and unabated exposure to solar ultra-violet (UV) radiation and to something called *atomic oxygen* (AO), which is present in the Earth's orbital environments.

AO is produced when short-wavelength UV radiation reacts with molecular oxygen in the upper atmosphere.[83] In LEO, AO poses a particular material threat as it oxidizes metals and reacts strongly with carbon, nitrogen, sulfur, and hydrogen bonds. In short, polymers react and erode when exposed to AO.

I recommend that aspiring space architects review NASA's resources, including *A Researcher's Guide to: International Space Station, Space Environmental Effects,* to learn about the many environmental factors in space that lead to material degradation and about the related material research conducted by scientists over the years.

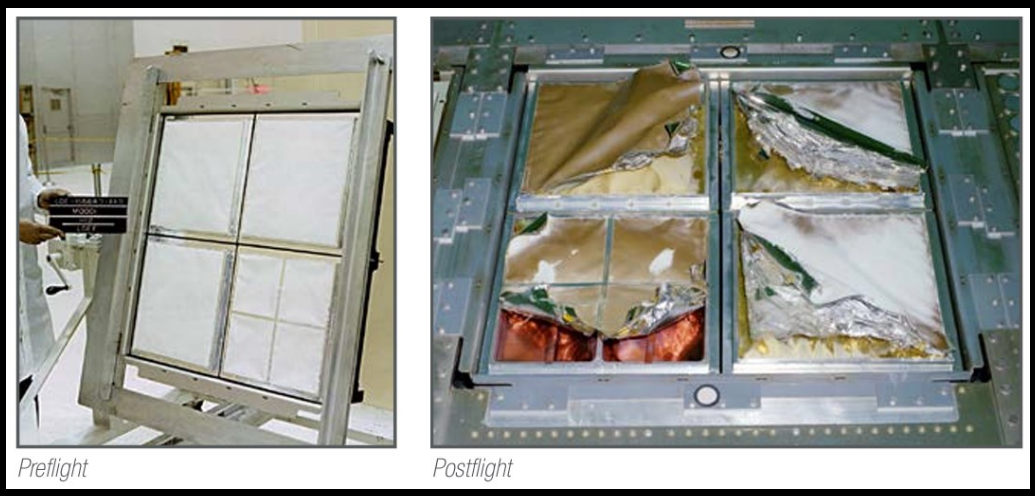



*Figure 42 - Pre- and post-flight Long Duration Exposure experiment, indicating atomic oxygen and ultraviolet degradation. Cr. NASA JSC. (2020)*

[83] NASA, "A Researcher's Guide to: International Space Station, Space Environmental Effects"

# Chapter 5: Space Station Design

Chapter 5 marks the beginning of the second portion of the thesis, where the knowledge gained in assembling the space architecture "primer" (Chapters 2 – 4) was synthesized and applied to the design of a hypothetical research space station. Particular focus was given to laboratory modules to support research on cancer and climate change.

Due to length of this design document, please refer to Appendix D for the AC$^3$ROSS Project – "The Asclepius Climate Change and Cancer Research Orbital Space Station" in which the overall design of Space Station (SS) Asclepius is illustrated.

# Chapter 6: Architectural Perspectives

The questions raised in Chapter 1 concerning the role of the architect in the milieu of outer space have remained in the back of my mind throughout the process of researching and authoring this thesis. That said, my perspective has changed, having been informed by what I have learned from source materials, from individuals working in the field, and through the process of designing for research in microgravity. Insightful conversations with the following industry professionals provided a diverse perspective into need for architects in the field of space exploration.

**NASA Space Architect, Krystofer Dudzinski** works with interdisciplinary teams at the NASA Marshall Space Flight Center focusses on habitat design. He employs architectural thinking, design processes, and his presentation acumen to develop concepts and insights into the human aspects of space habitation and exploration. Krystofer emphasized that the space industry desperately needs architects and holds our professional skillset and human-centric design thinking in high regard. He is excited for the continued engagement and increased interest among architects in this new frontier for architecture.

**Professor Larry Toups** has a rich history of contribution to the space industry. Trained as an architect, he practiced for 14 years prior to obtaining a Master's Degree in *Design for Extreme Environments and Space Architecture* from the University of Houston. He previously worked in NASA's ISS Program Office, serving as the Habitability Systems Lead in the ISS Vehicle Office. Professor Toups currently teaches in the Space Architecture Graduate Program at the Sasakawa International Center for Space Architecture, University of Houston, and supports the evolving Mars campaign in NASA's Exploration Mission Planning Office at Johnson Space Center. He serves as a Professor at Chalmers Institute of Science and Technology, Sweden, involved in research on sustainable homes of the future and the connection of that research with deep space habitation. To add to his experience, Professor Toups is an honorary consultant of the *Master's Program in Architecture and Extreme Environments* at the Royal Danish Academy of Fine Arts School of Architecture, Design and Conservation in Copenhagen, Denmark.

**Dr. Tricia Larose's** work on *Tumors in Space* struck a particular heartstring as it was a crossover of my academic, professional and life experiences with architecture, laboratory design, and cancer. For these reasons – and thanks to her generosity and willingness to connect – my engagement with her was both enriching and inspiring. Dr. Larose currently serves as Senior Advisor / Analyst at the Health Directorate of Norway, while maintaining her responsibilities as principal investigator for *Tumors in Space* at the University of Oslo. She serves as adjunct faculty in Human Performance in Space, Space Masters, and Space Studies Program at the International Space University (ISU), France, and is a Scientist-Astronaut Candidate in the *Polar Suborbital Science in the Upper Mesosphere* project, in Boulder, Colorado.

Dr. Larose holds a Bachelor's Degree in Science and Ethics from Laurentian University, Canada, and studied at the International Space University before acquiring a graduate degree and post graduate diploma in Science, Demography & Health, and Infectious Diseases, respectively, at the University of London's School of Hygiene and Tropical Medicine. She then acquired a PhD in Medicine at the Norwegian University of Science and Technology.

Additional reflection on what I learned through engaging with these individuals can be found in Chapter 7.4: Reflections on Engagement.

The following is an attempt to reflect on – and offer answers to – some key questions:

**Q: What can architects contribute to the design of environments that accommodate human habitation in micro-gravity environments such as the Earth's orbit and the surface of the moon?**

A: Architects design environments to accommodate the human body and facilitate human interaction. The advent of new possibilities for space travel coupled with an increase in the payload capacity for transporting things into space will expand opportunities to design for human habitation and interaction. As agencies plan for more humans in space, for longer periods of time, they will need the support of all disciplines. The Artemis missions, for example, dream of a future return to the moon, opening the potential for lunar habitation and exploration. The development of settlements on the moon, however small their initial scale, is an area exploration in which architects should be involved.

**Q: Do our current definitions of architecture and our understanding of ourselves as professionals hinder or contribute to our participation in the design of extraterrestrial environments?**

A: Change is perhaps the hardest thing for human beings to contend with, which is ironic, considering that we're the most advanced species on Earth due to our exceptional abilities to adapt and to consider different futures. Our current definitions and understanding of ourselves as professionals coupled with the boundaries we construct between the disciplines of architecture and engineering threaten to hinder us from participating in astrotectonic design. As Professor Larry Toups suggested to me at SpaceCom24, "to become a space architect – you must choose," i.e., between traditional architecture and space architecture – the implication being that the latter requires a significant redefinition of our role as architects and the milieux in which we operate. Our contributions as architects begin as soon as we decide to engage, at whatever cost.

**Q: To what degree is gravity implicit in the concept of architecture? Is there architecture beyond the constraints of gravity**?

A: Gravity defines architecture on earth, as both context and constraint. The inherent qualities of the classical order, the pointed cathedrals of the gothic era that reach up to the heavens, our structural rigidity and conceptions of space are all based on the directional ups and downs of our Earthly existence, bound by the almighty gravity. But we know that architecture is more than that – it is the human experience of space, the interactions within and the thresholds between, the beauty and the sublime, the home, and the empire – and as gravity is implicit in almost all of our history as humans, we have now proven that there is a future outside of its forceful grasp. Architecture does not define us, we define it, and as the human being embark on the next stages of exploration in outer space, architecture will follow.

**Q: What can the history of architecture and the built environment teach us about creating healthy and sustainable environments in space?**

A: We know, to a high degree, what needs to be done to build sustainably, and to create healthy and sustainable environments here on Earth – although we stumble through the execution. We know that necessity drives innovation and we have seen that international parties can collaborate in the harshness of outer space. Architecture has a much longer history than that of human existence in outer space, and so there are many lessons to apply to the design of future space stations and, eventually, to settlements on the surface of the moon or Mars.

## 6.1 Inspiring Architectural Projects

The Artemis Campaign, lunar inhabitation, and the exploration of Mars represent incredible opportunities for architects to develop knowledge, to leverage transferable skills in design, planning and project/program management and, more generally, to contribute to the exceptional challenges being explored by the space community. Below are a few contributions to space architecture futures from prominent architecture firms, displaying the fact that this is already being considered by industry leaders.

### Skidmore, Owings & Merrill – SOM

**Moon Village**

"Decades after humans first set foot on the Moon, SOM and the European Space Agency are proposing a bold new initiative to return – this time, on a permanent basis." - SOM

https://www.som.com/research/moon-village/

**New Frontiers Project: https://www.som.com/research/new-frontiers/**

**Graphene-Enhanced Space Habitat:**

https://www.som.com/news/som-and-university-of-manchester-researchers-propose-graphene-enhanced-space-habitat/

*Figure 43 - Aerial view of Moon Village plan. Credit: © SOM | Slashcube GmbH (2024)*



## Foster + Partners

**Lunar Habitation**

“Foster + Partners is part of a consortium set up by the European Space Agency to explore the possibilities of 3D printing to construct lunar habitations. Addressing the challenges of transporting materials to the moon, the study is investigating the use of lunar soil, known as regolith, as building matter.” – Foster + Partners

https://www.fosterandpartners.com/projects/lunar-habitation

**Mars Habitation**

“Foster + Partners have been working on a NASA-backed competition for a 3d-printed modular habitat on Mars. The design for the Mars Habitat outlines plans for a settlement constructed by an array of pre-programmed, semi-autonomous robots prior to the eventual arrival of the astronauts. The habitat – created in collaboration with industrial and academic partners – envisions a robust 3D-printed dwelling for up to four astronauts constructed using regolith – the loose soil and rocks found on the surface of Mars.” – Foster + Partners

https://www.fosterandpartners.com/projects/mars-habitat

**Interplanetary Practice: Learning from Space Analogues**

“On Earth, there is a critical network of advanced, experimental, and environmentally extreme Earth-based architectures that support human space exploration. These simulations are known as 'space analogues'.” – Foster + Partners

https://www.fosterandpartners.com/insights/plus-journal/interplanetary-practice-learning-from-space-analogues

*Figure 44 - Lunar Habitat. Credit: Foster + Partners (2012)*



## Bjarke Ingels Group - BIG Architecture

**Mars Dune Alpha**

"Mars Dune Alpha is a 3D-printed habitat designed to aid in long-duration, exploration-class science missions. Mars Dune Alpha's analog missions – part of NASA's Crew Health and Performance Exploration Analog (CHAPEA) program – will provide valuable insights for NASA's space food system, as well as physical and behavioral health and performance outcomes for future space missions" - BIG

Mars Dune Alpha | BIG | Bjarke Ingels Group

**Mars Science City**

"Mars Science City is a prototypical test site for a sustained city on Mars. Using the same techniques as on the Moon – robotic construction, excavation, 3D printing, and inflatable membranes – BIG aims to build a campus on Earth to host education and engineering, science and agriculture, conferences, and exhibitions related to the interplanetary exploration and settlement." BIG

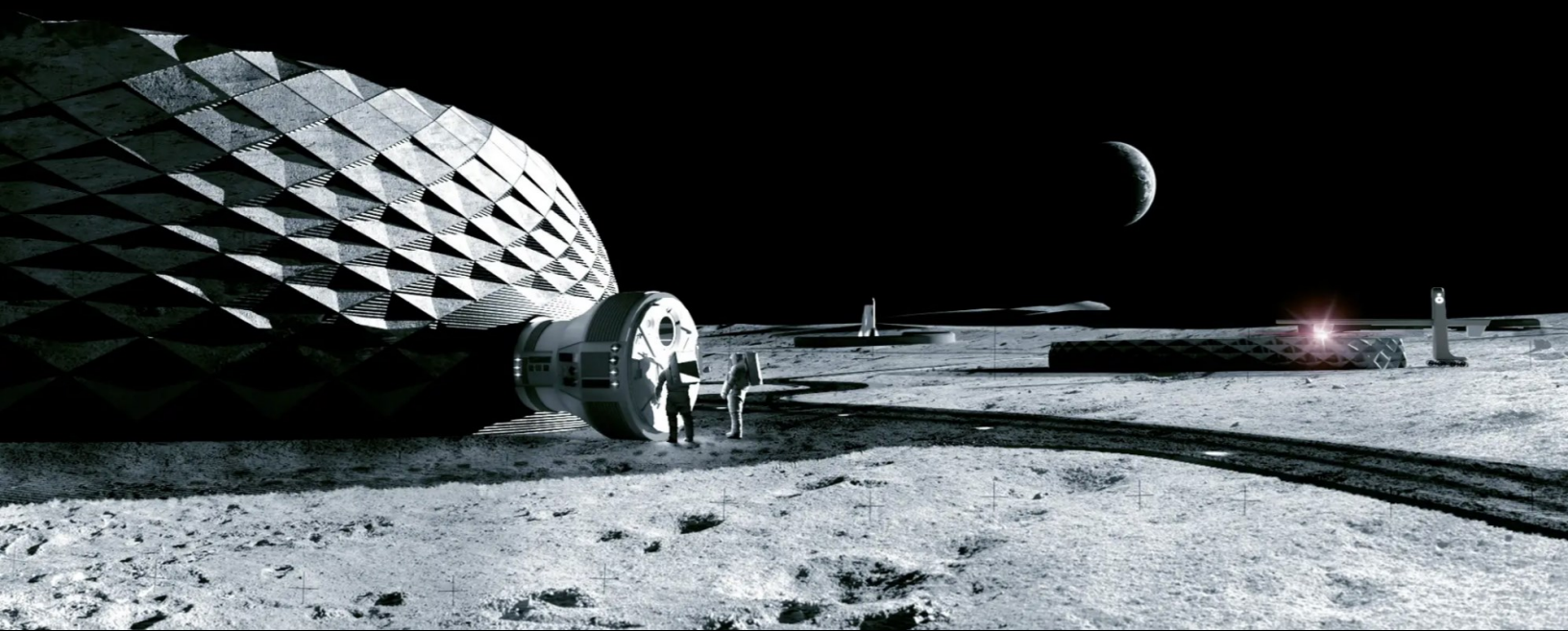
Mars Science City | BIG | Bjarke Ingels Group

**NASA Olympus**

"Big is collaborating with ICON, SEArch+, and NASA to design Project Olympus: a lunar habitat set to be the first human foray into extra-terrestrial construction. Through the Artemis program – a collaboration with commercial and international partners to explore more of the lunar surface than ever before – NASA has signaled that the Moon will be the first off-Earth site for sustainable surface exploration. ICON, a developer of advanced construction technologies such as robotics, software, and building materials, was awarded a government Small Business Innovation Research (SBIR) contract, including funding from NASA, to begin research and development of a space-based construction system that could support future exploration of the Moon. From landing pads to habitats, this collective is driven by the belief that humanity can and should become a spacefaring civilization." – BIG

NASA Olympus | BIG | Bjarke Ingels Group

*Figure 45 - Project Olympus, Lunar Habitat. Credit: BIG (2020)*



## 6.2 Architectural Design Process

While I am uncertain of the degree to which traditional architectural design processes can be applied to astrotectonics, I am confident that elements of it, e.g., the iterative nature of design and the professional, multi-faceted nature of architectural projects, can provide value for the future of astrotectonics. What I have learned about space architecture strongly suggests that there are myriad opportunities for traditional architects to participate, notwithstanding the need for additional education and exposure to the field. The *Tumors in Space* case study was extremely helpful in this regard. It enabled me to apply basic design knowledge and processes, combined with my experience in lab planning, to engage and communicate with someone directly connected to a space-bound scientific exploration. This suggests that, in the right context and with the proper education and connections, our work as architects has value – especially as it relates to the accommodation and interactions of human beings in space.

**Barriers to Entry**

From my perspective, architects face the following barriers to participating in the emerging space economy.

a. There is a steep learning curve to understand the fundamentals of aerospace engineering and astronautics that underpin the industry. Developing a base understanding is the first step to building the vocabulary required to communicate effectively with industry professionals. Being able to speak the language of the industry enables us to translate architectural thought into concepts and reality in outer space. By understanding industry professionals – not unlike the need for traditional architects to understand mechanical, structural, and electrical engineers – architects will be able to help visualize and bring ideas to life.
b. Historically, architectural, and human-centered design has played a secondary role in the engineering-driven design required to assemble safe and habitable spaces in outer space. As the commercial market invests more into the space industry, and as humanity ventures deeper into space for longer durations, human-centered design becomes a higher priority. The industry is beginning to recognize this priority, and architects face a challenge in marketing ourselves as high-value contributors, as these types of contributions may not fit the traditional paradigms of ourselves and our industry.
c. With human-centered design in mind, issues like orientation, ergonomics, and physical and psychological well-being in micro-gravity pose challenges for longer-duration space habitation. Although architects are equipped with human-centered thinking, there is still a significant amount of study, thought and experimentation required for us to develop advanced design standards and best-practices when it comes to human-centered design in micro-gravity. This is especially challenging as so few of us have experienced what it is like to live and work in microgravity environments, nor are we familiar with the constrained spaces in which most astronauts operate.

## 6.3 The Millenium Charter

The *Millennium Charter* – whose name likely references the infamous, modernist *Charter of Athens*[84] -- was signed by 46 signatories from around the world on October 12, 2002, in Houston Texas and included the *Space Architecture Mission Statement*: "Space Architecture is the theory and practice of designing and building inhabited environments in outer space."[85]

The Charter describes architectural motivations, contributions, knowledge. The charter includes philosophical guidelines and the categories for action, listed as the "Team 11 Principles" below:

1. Sustainability
2. Human Interaction
3. The User
4. Human Factors
5. Human Condition
6. Social Aspects
7. Environmental Conditions
8. Education
9. Life Cycle
10. Humility
11. Benefits

Each of these topics is described further in the Charter and could inspire future theses and / or studies. *The Millennium Charter* is attached as an appendix to this thesis.

## 6.4 Mythology, Architecture and Outer Space

The ancient Greeks are known for looking up to the stars and studying the cosmos. In architecture, we are instructed in the Greek order; we marvel at ancient structures like the Parthenon and embrace its paradigms of beauty, symmetry, and order.

In the tradition of naming early NASA missions for Greco-Roman deities such as Mercury and Gemini, NASA manager Abe Silverstein named the Apollo program, noting that the image of "Apollo riding his chariot across the Sun" was appropriate to the grand scale of the proposed program. Apollo is the Greek God of light, music, and the sun. NASA has selected Artemis, the name of his twin sister – and goddess of the moon - for its new lunar program.

Gods have historically been revered by humanity, providing us legends, stories and lessons that evolved into the foundations of our culture and societal beliefs. Architecture on Earth continues to acknowledge these figures, and the space industry evokes them in light of the magnitude of our technological goals and philosophical pursuits for meaning in the universe.

In addition, there is something to be said about the "cool factor" of these mythologies, deities, and namesakes. Having grown up in the Apollo generation and now being considered a part of the Artemis generation, I found it appropriate to name my design case study after Asclepius, son of Apollo and the god of medicine. Our tendency to fall back on mythology in our path forward suggests some inherent need to pay respect to our history as we venture to the future.

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[84] The Charter of Athens envisioned a brave new world and embraced the future, as its signatories understood it.

[85] AIAA, "The Millenium Charter", 1-7

## 6.5 Space Architecture Resources

### AIAA Space Architecture:

The AIAA provides many resources for space architects to immerse themselves into the field of aerospace engineering and architectural topics. The two courses below provide learning opportunities specifically for the design of habitation systems in orbital and lunar environments.

Space Architecture: Designing an Orbital Habitation System: Challenges and Consequences – On-Demand Short Course (aiaa.org)

Space Architecture: Designing a Lunar Habitation System – On-Demand Short Course (aiaa.org)

### University of Houston – Sasakawa International Center for Space Architecture

The Sasakawa International Center for Space Architecture (SICSA) is a research and design organization within the University of Houston. SICSA brings more than 30 years of internationally recognized experience in habitat research, planning and design for space and extreme terrestrial environments.[86] SICSA provides a Master of Science in Space Architecture Degree (Thesis).

Welcome to SICSA | SICSA (uh.edu)

### Space Architect

**Space Architect** is an independent organization and includes space architects from various professional associations. It is a membership organization that provides free access to dozens of projects, many publications and education resources. In addition, they provide a portal for up-to-date industry news, awards and events for members and non-members.

SpaceArchitect.org

### Artificial Gravity

Achieving artificial gravity through advancements in engineering and space architecture is one of the next evolutions of human habitation in outer space, as envisioned by past research, visionaries, and science fictions of the 20th century.

This resource provides links to books, chapters, papers, articles, and presentation on the topic of artificial gravity that could support future architectural studies concepts. In addition, links to software developed to calculate and simulate artificial gravity are shared for future use.

Artificial Gravity (artificial-gravity.com)

### Ground support facilities

The following link leads to a list of global ground support facilities for major space agencies.

https://www.nasa.gov/international-space-station/ground-facilities/

[86] Sasakawa International Center for Space Architecture, “Welcome to SICSA”

# Chapter 7: Conclusion

The initial intent of this thesis was to research topics that would support the design of a space station in low Earth orbit. Naturally, initial thoughts were that a design at a graduate level would be detailed and comprehensive – drawing on all aspects of my academic journey at the Azrieli School of Architecture. Early in the process, the concept of "research through design" was introduced, recasting the endeavor as a kind of exploration more than a demonstration of mastery.

In addition, this thesis sought to answer questions about the architect's role in space habitation and exploration. What could our way of thinking, our way of seeing the world and its built environments, and our design processes do to support and enhance an established, ambitious, and expanding aerospace and astronautical industry? As the thesis progressed, I was increasingly inspired by the past achievements and future aspirations of the space industry. I discovered resources to bridge the gap between traditional terrestrial architecture, bound by gravity, and the space architectures of the future. Architectural precedents and industry engagement enriched the experience, solidifying my conviction that architects must contribute to this future.

Guided by Professor Benjamin Gianni, I decided that this thesis would be a greater contribution if it became a primer on space and astrotectonics, outlining a path for aspiring space architects to follow by providing a summary of my research journey and a repository of resources and recommendations for further study. In addition, the thesis includes a design case-study – intended to test my understanding of the subject matter and as a mechanism through which to learn. The research informed the design, and the design drove further research – a process that promoted a deeper understanding of the complexities and possibilities of researching and designing of astrotectonics.

## 7.1 Personal Experience

My education as an architect has included a Diploma in Architectural Technology from Algonquin College, a Bachelor of Architectural Studies, and a Master of Architecture degree (in progress) from the Azrieli School of Architecture & Urbanism at Carleton University. Supplementing this, I have spent close to a decade in professional practice, specializing in the design of research laboratories. Among other things, this has enabled me to learn about laboratory containment principles, pathogens, the societal impact of scientific research, protocols, and mandates, and, most importantly, the ways in which scientists conduct research as it relates to the design of the environments in which they work. The foray into space architecture, then, is the extension of an ongoing exploration that began at a young age.

I have been long fascinated and inspired by the National Aeronautics and Space Administration (NASA), the Canadian Space Agency (CSA), the Space Shuttle program, Canadarms, and the International Space Station. I have always wanted to "work for NASA" without really understanding what it might mean or in what capacities people worked for the agency -- other than as astronauts, of course. As a leader in space exploration, NASA is more than an agency, it is an idea with a vision for the future. This idea is what inspired me, and this thesis has allowed me to explore the countless ways that thousands of people, from a range of disciplinary backgrounds, contribute to this idea – including architects.

In 2017, during my first year of undergraduate studies, I was diagnosed with cancer – a tumor in my left leg consisting of a soft tissue cancer called a myxoid liposarcoma. Treatments included two surgical removals of tissue and 25 sessions of radiation therapy. This was a life changing experience. When the dust settled and I was cancer-free, I felt extremely grateful for all of the microbiological research and experimental treatment technologies that had been brought to bear to save my life. Thoughts about those who were not as fortunate as I have clouded my mind ever since, and I have dreamt that, through architecture and a knowledge of technology, I might contribute to the global fight against cancer.




*Figure 46 - Ramon RS, General Hospital Ottawa, after final radiation therapy treatment (2018)*

## 7.2 Key Takeaways

Questioning the overlap between architecture and the space industry led me to a vast array of resources. A key takeaway from my research into aerospace engineering, climate action from space, cancer research in microgravity, radiation, and other topics was that each topic constitutes a study of its own, introducing me to organizations, breakthroughs, and networks of dedicated innovators. Constrained by time, it was important to distill and summarize various avenues of exploration, bookmarking them for further study. Among other things, then, this thesis offers a basic introduction to space architecture for students and professionals. It includes links to relevant resources. The hope is that someone in my position embarking on a thesis or design project can use this work to streamline their learning, enabling them to get further faster, dive deeper into an inspiring avenue of exploration, or identify innovative strategies to address challenges – whether global or local.

Another key takeaway from this process was the enriching benefits of approaching a project with a research-through-design mentality. The requirements for thorough research, in order to responsibly design something with strong justifications for decisions and processes, showed me the importance of developing a strong background knowledge on a topic or challenge before suggesting a solution. As designers, sometimes we are quick to pull out our pen and paper to provide an answer – I challenge you to adopt a research-through-design process as I will in all my future architectural endeavors. I am strongly convinced that the knowledge I gained through this iterative and reciprocal focus on research and design gave me an immensely better education on the topic, giving me confidence for future study and engagement with the field of space architecture.

## 7.3 Reflections on Outcome and Method

My initial hypothesis was that I would acquire enough knowledge to be able to design a full space station - with detailed plans, sections, and elevations, i.e., a traditional architectural drawing set. Stemming from an early understanding of what a graduate thesis **should** demonstrate, this expectation evolved after the first few months. Academic advisors offered overlapping guidance about the nature of an architectural thesis, each contributing unique perspectives. Advice ranged in nature; some suggested the thesis should demonstrate a mastery of the craft, some encouraged to take an architectural stance on a topic, while others recommended that the thesis illustrate a speculative architectural future through storytelling. That being said, as my definition of an architectural thesis expanded and I continued to research through design, I spent more time engaging with and learning from industry professionals.

I began leaning away from the design of a hypothetical space station towards a research-focused thesis with a conceptual design component through which to determine what I needed to learn. This approach enabled me to dig deeper into topics of interest, namely, research in microgravity and its benefits to humanity. It enabled me to leverage my experience in laboratory design and focus in on personal and academic areas of interest – namely cancer and climate change research in microgravity. The decision to focus on research – the School often makes a distinction between a ”research” and a “design” thesis – provided the confidence to articulate the thesis objectives (below) more clearly, accurately define design scope and pursue further research and engagement.

I encourage future graduate students embarking on a thesis to be flexible in their objectives, to adopt a learning strategy that includes research through design, to apply the strategies described in *Ultralearning* by Scott H. Young, and to engage with academic and professional advisors to help

develop key definitions and targets. Be patient, take daily steps forward and approach new fields of learning with humility. I.e., hold off on proposing solutions before understanding the history, advancements, and real problems presented by your selected topic. In my journey and in yours, there is nothing new under the sun. Find your personal angle, leverage your passions, and experience and find a way to contribute to future study. This thesis was able to meet the following:

**Primary Objective**

The primarily objective of this thesis was to document and summarize my research journey into an emerging realm of architecture, contributing to the bridge already being built by the academics, design professionals, and visionaries of the last century, namely the bridge between traditional (terrestrial) architecture and a future architecture in space. A tremendous body of research exists on this subject, an introduction to which the chapters of this thesis are intended to provide, as well as a repository of key references and resources.

**Secondary Objective**

To engage in research through design, i.e., to augment and test my understanding of key concepts in the field of astrotectonics through the design of orbital research space station modules within a functional space station framework.

**Tertiary Objective**

To identify and articulate the roles and responsibilities of architects with respect to the design of non-terrestrial environments – sharing my own viewpoint and those of subject matter experts.

## 7.4 Reflections on Engagement with Industry Professionals

The following reflections summarize the outcomes of my direct engagement with NASA Space Architect Krystofer Dudzinski, Professor Larry Toups, and Dr. Tricia Larose – each of whom provided unique and constructive context in support of my research and design goals.

1. The key takeaway from my discussion with Mr. Dudzinski was that the human-centered approach to design that architects prioritize is incredibly valuable, and that NASA and other leaders in the space industry need to integrate it into their design priorities. Secondly, architectural training in visualization techniques and presentation acumen are an asset that architects possess to a greater degree than engineers, who dominate the field.
2. Professor Larry Toups shared that, throughout his career as an architect and design professional in the space industry, **integration** is a key skill that our training as architects provides. Synthetic thinking enables us to build bridges across gaps to connect teams, ideas, and industries. He suggested that architects are able to see the potential of something that is successful in one industry and integrate it in new ways into another more easily than professionals in other disciplines.
3. Dr. Tricia Larose made me aware of the realities of international relationships and their impact on the logistics of conducting research in outer space. She opened my eyes to the network of terrestrial laboratories that support science on space stations and reinforced the value that my experience in laboratory design for science in outer space. She observed that she had never before spoken with a lab architect about human-centered design in outer space – reinforcing, once again, that the current priorities privilege engineering and technical design considerations.

## 7.5 Next Steps and Conclusion

The *Millenium Charter* describes a list of topics that provide architects with clear avenues for further study in the realm of astrotectonics. If I were to continue working on this thesis, I would strengthen my engagement with cancer research in microgravity. There is much more to learn on this topic in order to make a meaningful contribution. I would like to develop an advanced knowledge of scientific workflows, equipment, and automation in outer space. Drawing on my professional experience in containment laboratory design, I would like to contribute to the design of containment laboratories for higher risk science. These types of containment facilities will eventually exist in outer space and are being debated as best practice for sample return missions from outer space.

Other avenues of study with which to engage include:

- Material science and construction process studies for space architecture.
- Human experience and community building in space.
- Psychological effects on human beings engaging with astrotectonics, drawing from our empirical knowledge of our relationships with the spaces we inhabit.
- Adapting architectural design thinking, phases, and processes into the design for outer space architectures – orbital, lunar or Martian based.
- Assist in creating architectural design standards, order and guidelines for astrotectonics.

I find it curious that on both my first day of architectural studies at Algonquin college, and my first day of architectural studies at Carleton University the instructor posed the question….

**“What is architecture?”**

At the time, this re-occurrence put a grin on my face, as I thought I held a strong definition that was about to be updated or elevated. In retrospect, this coincidence compels me to believe that we continuously redefine architecture as we respond to the intensifying and evolving challenges of the world such as cancer and climate change.

Ascribing the human condition to the built environments we inhabit has been our perspective and responsibility for millennia. In the milieu of outer space, we must answer the call from the bold professionals of the space industry and begin asking **what is space architecture?** We must apply ourselves with the same rigor and bravery displayed by those who have gone beyond the limits we once imagined and above all, we must dare to dream.

**Ad Astra**

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End of Bibliography.

# Appendices

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## Appendix A - Science In Microgravity

This list below highlights additional scientific and technological contributions of experiments conducted in the microgravity environment of outer space:

**General**

- In 2019 the Canadian Space Agency Bio-Analyzer was demonstrated to enable astronauts to quickly test bodily fluids – blood, saliva, urine. It provides test results within two to three hours,[87] enabling quicker diagnostics on space stations in support of long-duration deep-space mission.
- Telescope technology behind the "NICER" telescope project has been used to improve CT scanning for patients on Earth, enabling better image quality and lower patient exposure to radiation.[88]
- ISS protein crystal studies associated with Duchenne Muscular Dystrophy (DMD), an incurable genetic disorder, are leading to the development of compounds that could slow the progression of the disease, potentially doubling the lifespan of many patients.[18]
- Robotics developed for space related uses have influenced and robotics development for surgical equipment on Earth.
- Space station air filtration system technologies have been adapted for air purification systems on Earth for the preservation of goods and wine cellars. Companies have used the technology in air purifiers used to eliminate or detect viruses such as SARS-Cov-2.[18]
- Companies have conducted experiments in the manufacturing of artificial retinas on the ISS. These retinas are built-up of 200-layer high quality films, which could help restore meaningful vision for millions of people on Earth suffering from retinal degenerative diseases.[18]
- The ISS houses 12+ commercial research facilities.

**Life Sciences**

- On November 2, 2000, during Expedition 1, Russian cosmonaut Yuri Gidzenko and Sergei Krilkalev and NASA astronaut Bill Shepherd commenced operation of the world's first truly international space station. Experiments included the Plasma Kristall-3 (PK-3) Investigation. PK03 was a physical science experiment collaborated on by The Max Planck Institute for Extraterrestrial Physics + Institute for High Energy Densities part of the Russian Academy of Sciences. Cold plasma can be used to disinfect surfaces and quickly neutralize drug resistant bacteria and has gone on to help with medicine, skin disease treatments and wound care.
- Expedition 1 - Rapid viral detection tests and viral shedding experiments are used today on viruses like mononucleosis, herpes, chicken pox.
- PCG-5 (ISS-National Lab) – Experiments focused on production of high-quality crystalline suspensions, facilitating the injection of cancer related treatments (melanoma, lung cancer). Monoclonal antibodies do not dissolve easily in liquid, this is a challenge that

[87]Government of Canada, CSA. 2019. "Bio-Analyzer: Near-real-time biomedical results from space to Earth." https://www.asc-csa.gc.ca/eng/iss/bio-analyzer.asp

[88] NASA. 2022. "15 Ways the International Space Station Benefits Humanity Back on Earth" https://www.nasa.gov/missions/station/15-ways-the-international-space-station-benefits-humanity-back-on-earth/

PCG-5 aimed to support. Cancer treatment by injection greatly reduces the time required for intravenous treatment as well as cost.

- PCG-5 (2022) – this project aims at experimenting on the creation of even higher quality and uniform crystalline suspensions.
- JAXA – PCG led to development of artificial albumin – the most abundant protein found in blood. Using dog and cat blood, this aims to help with veterinary medicine.
- ESA – Experiment Unit (EU) for cell culture in space – The SPHEROIDS project studies effects of microgravity on endothelial cells (blood vessel wall).
- 2016 -NASA's Kate Rubins conducted the first DNA Sequencing in Space using a MinION sequencing platform, equipment the size of a cell phone.
- NASA's GeneLab contains biological data stemming back to 1995. This data includes both spaceflight and ground experiments.
- JAXA's Integrated Biobank for Space Life Science (ibSLS) is a biorepository that provides similar historical bio data for study between space and earth life science.
- CSA's Vascular series studies the microgravity effects on astronaut blood vessels and hearts.
- Discoveries from the Ring Sheared Drop (RSD) device experiments on protein aggregates are playing a role in Alzheimer's, type-2 diabetes, Parkinson's, and prion diseases.
- Organ.Aut (ROSCOSMOS) – This project includes the culturing of cartilage cells in space.
- REDWIRE also conducted human tissue printing in orbit using their BioFabrication Facility (BFF). They have conducted testing on the manufacturing of cardiac and orthopedic tissues in-orbit, with new programs aimed at manufacturing human vasculature. These kinds of products are aimed at addressing a global issue of organ shortage for critical transplant needs due to a range of diseases.
- IGAR – CSA and Centre for Surgical Innovation and Investigation (CSii) – A surgical robot that can be pre-programmed and operate autonomously to perform minimally invasive / highly accurate surgical procedures. This project is currently seeking final FDA approval and the technology can be used for imaging and treatment of several types of cancers and provide surgical capabilities in extreme or remote environments or during emergency / disaster response scenarios.
- 2021 – The ISS researchers have conducted more than 500 protein crystal growth experiments (PCG) – This is the largest amount of experiments conducted on the ISS per category.
- The Roscomos Kristallizator program is creating single protein crystals for X-ray diffraction analysis.
    - PCG studies that have helped with development of anti-tuberculosis (TB) drugs. TB is responsible for 1 million deaths annually – USA)
- JAXA PCG programs have provided protein structures that have led to discoveries for breast cancer, periodontal / gum disease and muscular dystrophy drugs that are in preclinical trial phases (2019)
- CASIS PCG 19 – studies into the deterioration of monoclonal antibodies (drugs) that aims at increasing drug shelf/refrigerator life – A goal is to greatly reduce the refrigeration needs for drugs as this is a large deterrent for drug access.
- Antimicrobial coatings have been developed in space and have been applied on surface decontaminants used for COVID-19.
- Vaccines in microgravity – research allows for refined composition of the virus that leads to reduced risks for patients with pre-existing conditions (i.e., kidney problems)

- Organ bud research – human stem cells are being used to grow 3-dimensional organ buds (tissue masses that develop into full organs)

**Earth and Climate Science**

- Telescope technology behind the "NICER" telescope project has been used to improve CT scanning for patients on Earth, enabling better image quality and lower patient exposure to radiation.[89]
- Data from NASA's ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS) Earth scanning technology helps people on Earth develop strategies to reduce heat island effect, efficiently water agriculture and reduce the risk of forest fires. [18]
- Orbiting Carbon Observatory 3 (OCO-3 ) launched in 2019 is a space instrument investigating distribution of carbon dioxide on Earth as it relates to urban population and changing patterns of fossil fuel combustion. It is also helping understand plants and their role in the carbon cycle.
- The most accurate map ever made from space was released in 2021 showing human influence on carbon dioxide in Los Angeles. Contributions support effective goal setting and policy creation.
- Carbon storage and removal strategies are explored using the ESA photobioreactor – these experiments study microalgae and their ability to help capture carbon in future life support systems.
- Kuwait's Experiment, E. coli C, studies bacteria modified to consume carbon dioxide as food.
- JAXA – Integrated Standard Imager for Microsatellites (iSIM) takes Earth images at less than 1 meter of resolution.
- The Expedition 1 provided early Earth observation photography. As of 2021, the image capture program has accumulated approximately 4 million photographs of the Earth that have become available to research environmental changes over decades.
- The cupola, installed on February 15th, 2010, greatly increased the opportunity for crew photographs of Earth.
- Launched in 2018 - The Global Ecosystem Dynamics Investigation (GEDI) is a joint mission between NASA and the University of Maryland. Using laser technology, GEDI creates three-dimensional maps of forest canopy height and biomass distribution. GEDI analyzes how forests and biomass change overtime by understanding carbon storage and loss, when they are disturbed (human activity, forest fires)
- CLARREO Pathfinder is a project that will measure reflected sunlight from the Earth and the moon.
- Launched in 2018 - Multi-User system for Earth Sensing (MUSES) is an Earth imaging facility providing Earth observation tools for land surface imagery.

**Scientific Technological Advancements**

- ISS regenerative life support hardware provides clean air and water for crews.

[89] NASA. 2022. "15 Ways the International Space Station Benefits Humanity Back on Earth" https://www.nasa.gov/missions/station/15-ways-the-international-space-station-benefits-humanity-back-on-earth/

- The Universal Waste Management System (UWMS) is a smaller, lighter, and more comfortable space toilet.
- Testing (2020) of the Urine Processor Assembly shows that recent upgrades create service life of 4,300 hours against the 1,400 hours previously anticipated.
- The Bigelow Expandable Activity Module (BEAM - 2016), is proven expandable model and is now a fixture on the ISS serving as a storage module.
- The Bishop Airlock – by Nanoracks (2020) allows for larger payload management and facilitates satellite deployment.

## Appendix B - Research Trip: SpaceCom24 and Space Coast

This appendix provides context of my research trip to Orlando and the space coast – Cape Canaveral, Florida. This trip included attendance at SpaceCom 24, The Global Spaceport Alliance Summit, a visit to the Kennedy Space Center and viewing of a SpaceX Cargo Re-Supply (CRS) mission rocket launch to the ISS.

**Global Spaceport Alliance Summit**

The trip started off with the Global Spaceport Alliance Summit. This was a private conference of space port industry leaders from across the globe. My eyes were quickly opened to the reality that spaceports operate everyday and are critical to our way of life, launching satellites and payloads that upkeep our global communications, defense, and scientific requirements.

The conversations, presentations and panels during the summit highlighted the need for an increased infrastructural network within the spaceport eco-system as the commercial space economy is expected to increase significantly. Choke-points were identified and the opportunity for designers, architects, and engineers to contribute was strongly emphasized. Payload processing facilities (PPF) are one example of the chokepoints discussed during the summit. These facilities have a range of functions as payloads vary in size and type. Payloads require manufacturing, maintenance, preparation and loading prior to launch – all of these functions and more are provided within PPFs.

The summit discussions also highlighted the differences in capabilities from spaceport to spaceport. Some spaceports launch horizontally, some vertically and each have varying payload capacities based on launch pad and rocket / spacecraft design.

During this summit I was fortunate to meet the former NASA Administrator James F. Bridenstine, Colonels in the US Space Force, and other industry professionals who all shared the gravity of spaceport infrastructure, access to space and future expectations of the space industry.



**SpaceCom24**

This conference was a 3-day event with hundreds of space industry professionals gathered in Orlando, Florida. It included panels, presentations, and key notes along with an exhibition hall filled with informative booths covering topics including space grade materials, in space manufacturing, robotics, communications technologies and architecture and engineering companies.

This is where I was able to sit in on panels including Professor Larry Toups and NASA Space Architect Krystofer Dudzinski. After these panels I was able to engage with them and many other industry professionals, asking questions related to new learnings and my thesis progress. Fortunately, they were also open to further engagement beyond the conference, and I was able to connect 1 on 1 with each of them, allowing me to learn deeper and more valuable insights into the space industry and the role of the architect in the future of space habitation and exploration.

Like the GSA Summit, SpaceCom 24 was an eye-opening experience to the fast paced, high energy and engaging space community that have worked on incredible projects and that have unbelievable aspirations for the future. This was a truly inspiring experience.

**SpaceX Cargo Re-supply Launch**

See it to believe it! I was fortunate to drive over to the space cost (Cape Canaveral) over launch hour one day to witness the NG-20 Cygnus spacecraft launch on January 30, 2024, at 12:07pm EST aboard a SpaceX Falcon 9 rocket. This spacecraft lunched from Space Launch Complex 40 at the Cape Canaveral Space Force Station. It was fascinating to experience this launch, knowing that this cargo resupply was rendezvousing with the International Space Station in just a few short hours. Again, this experience helped understand the realities of space flight, and made spacecraft and space stations just that more tangible to conceive of.





**Kennedy Space Center**

After SpaceCom 24, I was able to stay in Cape Canaveral and visit the Kennedy Space Center. This was a museum like learning experience where I was immersed in space history, technology, and culture. The following is a photographic documentation of my experience at the KSC.




Blue Origin - Facility




NASA – Vehicle Assembly Building (VAB). This is where rockets and spacecraft are assembled prior to launch, including Apollo and Artemis lunar missions.



Rocket Garden – Exhibit displaying the history of rockets. This was an early glimpse at physical rockets close-up.





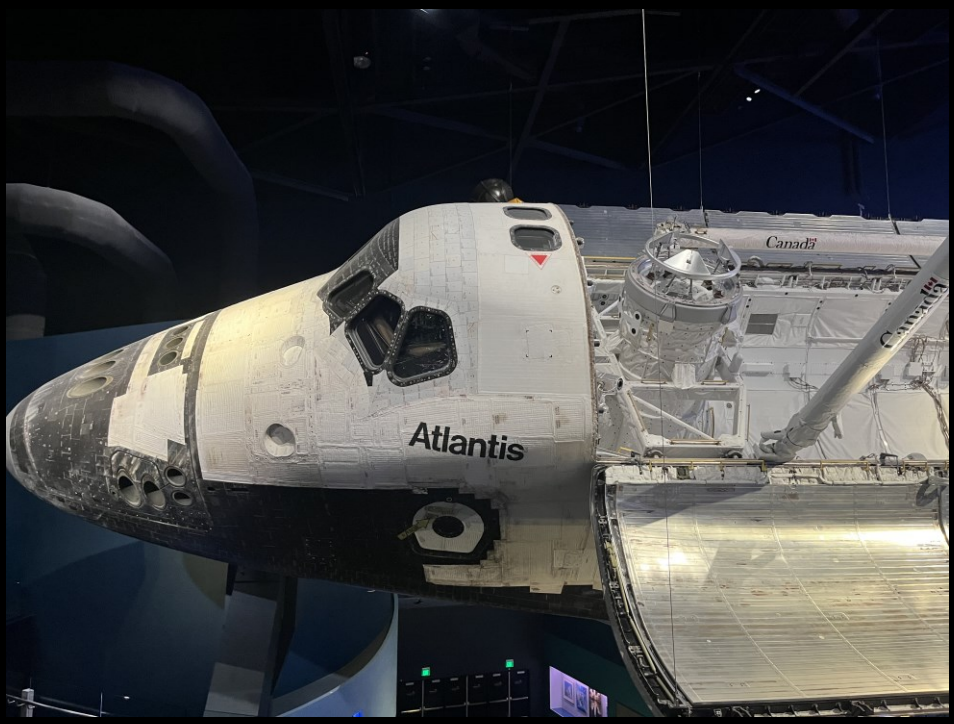

The Saturn V rocket first used during the Apollo Missions

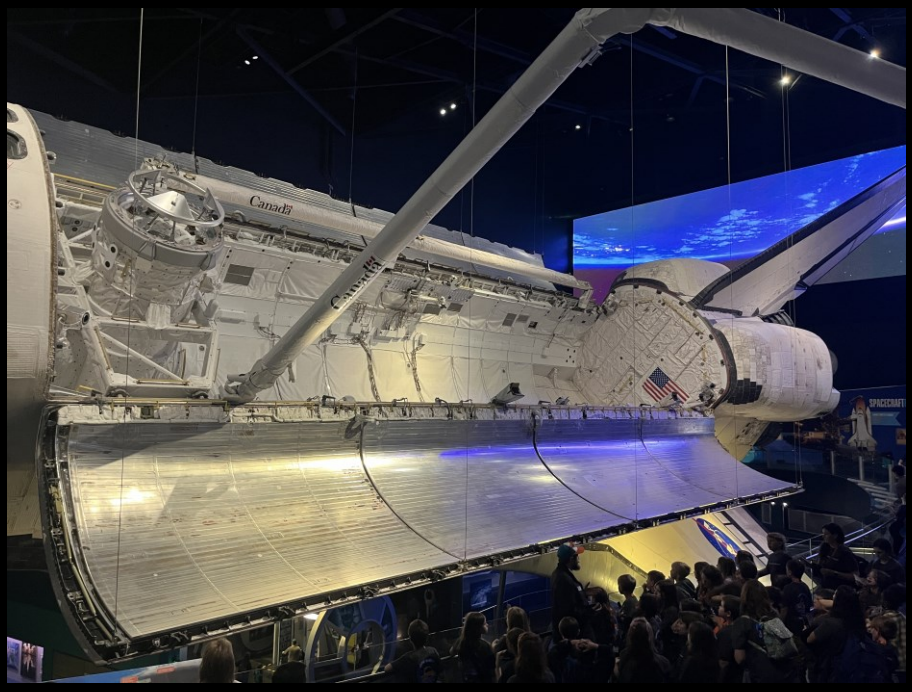



Saturn V Rocket Engines. The detail and intricacies on each engine were incredible. This solidified in my mind that I was certainly impressed with engines, but they are not a field of study and practice that interests me deeply. Leave it to the engineers!

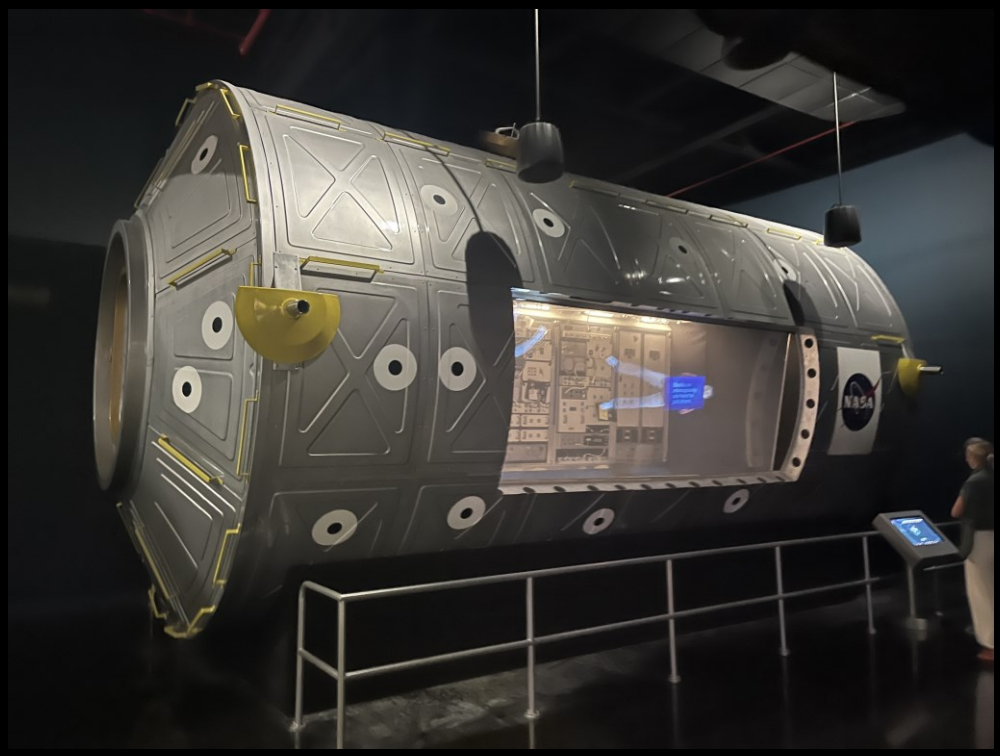


Solid rocket boosters (SRBs) and an external tank act as a gateway to the Atlantis Space Shuttle Exhibit



The Atlantis Space Shuttle was the last space shuttle to fly, its landing marked the end of the space shuttle program in 2011.







Space Station Module models gave me a first sense of scale for these space architectures, as well as insights into interior volumes and sleep quarters.





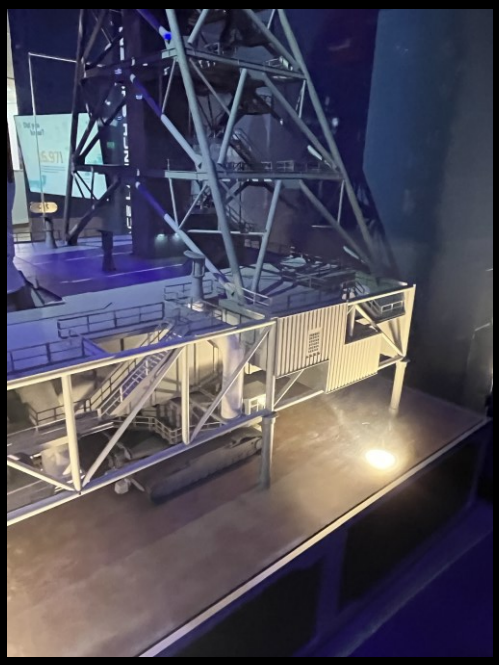

Habitation spaces within modules were displayed to get a sense of day-to-day life on the ISS





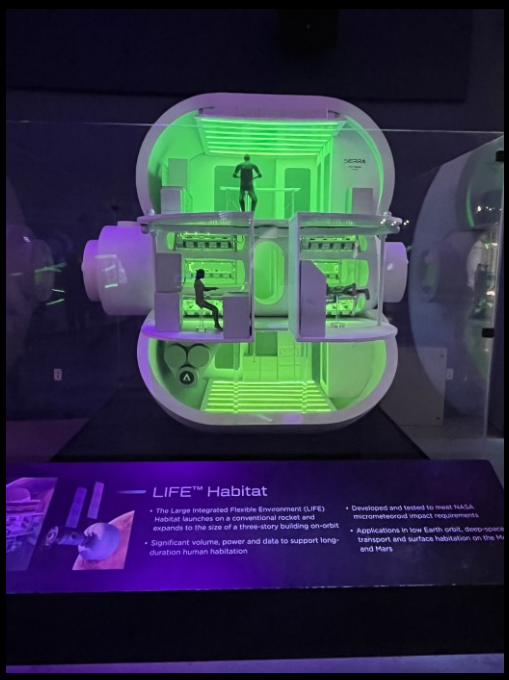

A Lockheed Martin habitat model gave a sense of laboratory orientation and equipment placement in a microgravity environment. This module size is more like the existing ISS modules, approximately 4m in diameter, providing limited internal configuration flexibility.

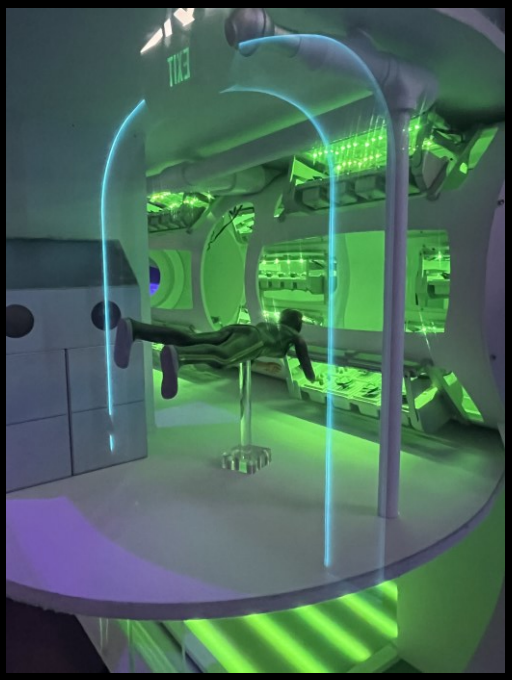


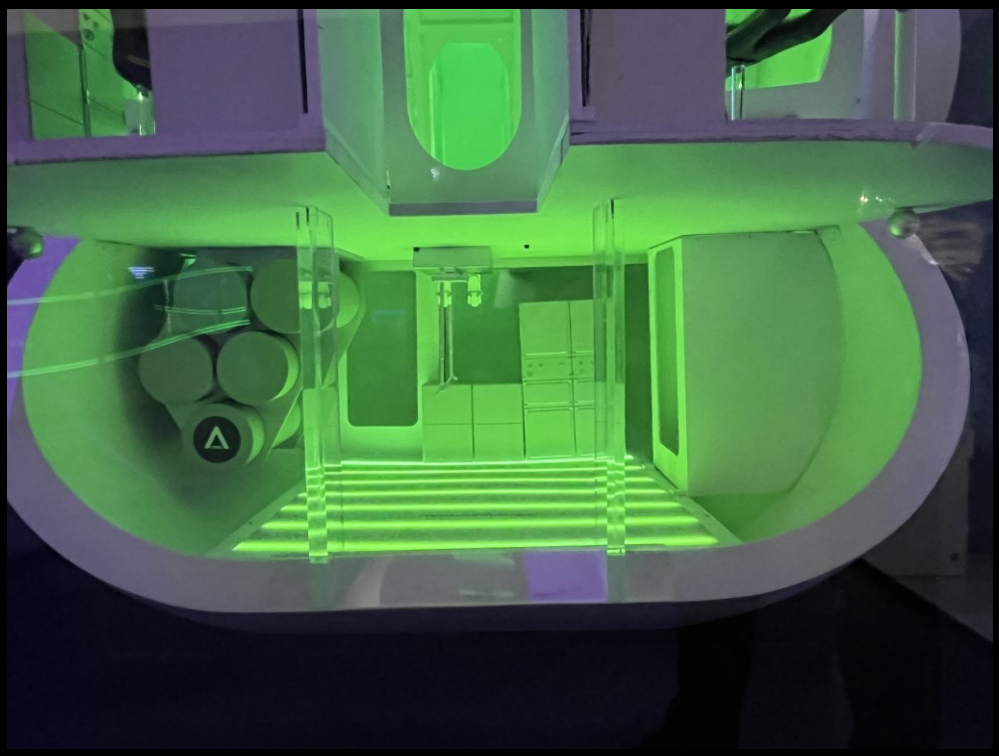





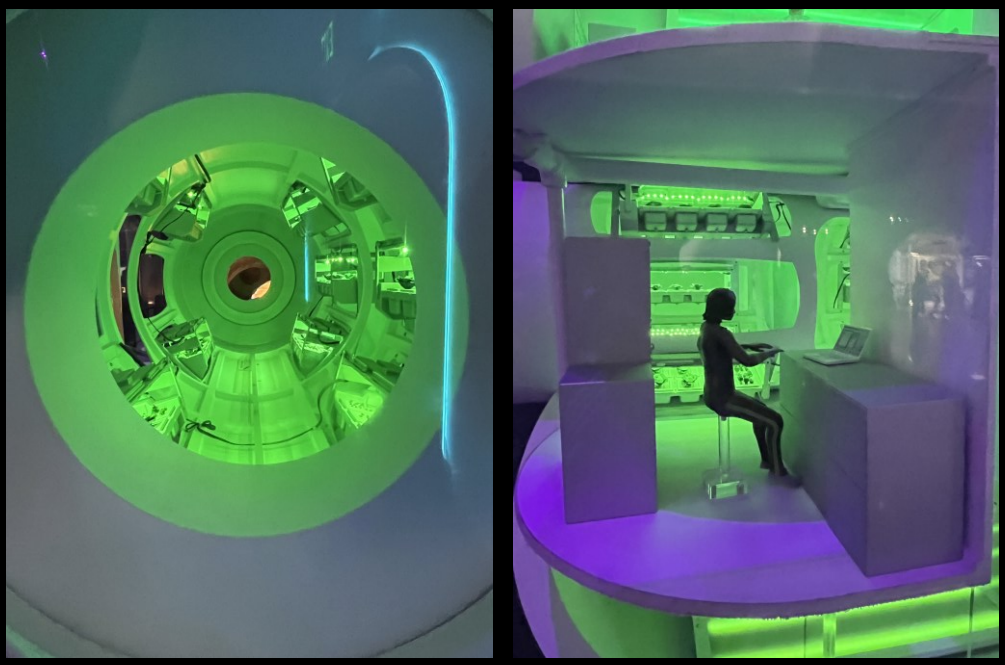
Detailed and intricate models of launch platforms, spacecraft and aircraft allowed for close up view of ground infrastructures and architecture. It was nice to see that the aerospace industry takes advantage of modelling tools as a form of communication and visualization.



Sierra Space's LIFE Habitat models. Exploring orientation and stacking techniques within an inflatable module concept.






Additional photographs of the LIFE Habitat inflatable module and interior spaces.

# Appendix C - Space Radiation

This appendix describes some of the historical a basic space radiation concepts and facts that I researched during the thesis process. Understanding how conventions evolved and are used to explain space related design provided a good basis for me to move forward with other design considerations.

To better interpret information related to radiation in space provided in the following tables, it may be helpful to define a few terms, unit conventions, and advances in radiation measurement. As different systems and conventions are used in radiation mathematics, we will use the international system of units (SI), in which units are measured in meters, kilograms, and joules.

- **Grays**: 1 gray (1 Gy) is equal to 1 joule of energy delivered to 1 kg of matter (1 J/kg). This unit of energy delivery is applicable to all matter absorbing energy – whether it is water, a rock or biological tissue. Grays were named after British physicist Louis Harrold Gray (1905-1965) and are a fundamental unit in the evolution of understanding radiation absorption calculations.

$$1\ Gy = 1\ J/Kg$$

- **1 Radiation Absorbed Dose**: 1 RAD is equal to 100 ergs of energy absorbed by 1 gram of matter (100 ergs/g). Like a gray, a RAD is a measurement of radiation absorption regardless of the absorbing matter. It should be noted, however, that the use of RAD in present day is only prominent in the United States as most other countries have moved on from the centimeter, gram, seconds (CGS) unit system onto the meters, kilogram, seconds (MKS) unit system.

$$1\ RAD = 100\ erg/g$$

- **Ergs** are a unit of measurement in the CGS system where an erg represents the work done when a force of one dyne (a CGS unit of force acting on a mass of 1 gram, increasing its velocity by on centimeter per second, every second along the direction that it acts) is acting over a distance of one centimeter and is equivalent to $1x10^{-7}$ joules.
- The conversion of RAD to gray, or centi-gray (1 cGy = 0.01Gy)

$$100\ RADs = 1\ gray, \qquad 1\ RAD = 1cGy$$

RAD and grays do well to measure energy absorption for all matter, but for life harboring space architecture, it is important to understand how radiation intensity damages **biological matter** after absorption. Ie. Human beings, other biological lifeforms, and cancer research samples on board the space station.

- **Roentgen Equivalent Man:** There are many forms of radiation in all environments, and a new system, the Roentgen Equivalent Man (named after Wilhelm Rontgen 1845-1923, discovered the X-ray), **REM**, was created to help classify the varying degrees of biological damage caused by different sources.
- The equation required to understand the leap from the old convention (RAD / Gy) to the new one is shown below in which Q represents the **quality factor** for different forms of radiation harmful to living organisms.

$$REM = Q\ x\ RAD, \qquad REM = Q\ x\ 0.01\ Gy$$

The chart below describes typical types of radiation and the quality factor (Q) for each, for use in calculating biological damage of radiation doses. Once the type of radiation exposure is selected, use the respective value of Q in the calculation to find the REM or Sievert (Sv).

| Radiation | Quality Factor (Q) |
|---|---|
| X-rays, gamma-rays, or electrons | 1 |
| Neutrons (all energies) | 10 |
| High-energy protons | 10 |
| Alpha-particles and ions | 20 |

*Table 6 - Radiation type quality factors for REM calculation – Cr. NASA Radiation Math (2011)*

- **Sievert (Sv)** is an SI unit representing radiation absorption resulting in biological health risk of ionizing radiation, describing the biological damage for varying radiation sources, using the same quality factor as the REM calculations. The sievert represents a large dose of radiation, so typical radiation exposures are represented in millisieverts (1 mSv = 1/1,000 Sv) or microsieverts (1 µSv = 1/1,000,000 Sv)

$$Sv = Q\ x\ 1Gy, \qquad Sv = Q\ x\ 100\ RAD, \qquad 1\ Sv = 100\ REM$$

- Overall, American scientists still prefer to use RAD and REM while scientists in Europe and other nations prefer gray and sieverts for calculations.
- The table below, from *spacecraft systems design & operations*[90], helps understand the levels of radiation dosage in REM (QxRAD) that cause an increasingly dangerous level of biological damage to humans.
- A space station in LEO should expect an approximate background radiation exposure to humans of about 1.2 milliSieverts per day (0.12 REM), not accounting for high radiation events, where astronauts must retreat to radiation bunkers designed to keep them safe.

| Radiation Effects on Humans | |
|---|---|
| **Effect on Humans** | **Dosage (REM)** |
| Blood count changes in population | 15-20 |
| Vomiting “effective threshold”* | 100 |
| Mortality “effective threshold”* | 150 |
| $LD_{50}$** with minimal supportive care | 320-360 |
| $LD_{50}$** with full supportive medical treatment required | 480-540 |
| | |
| *Lowest dosage affecting at least 1 member of the exposed population | |
| *$LD_{50}$ is a lethal dosage in 50% of the exposed population | |

*Table 7 - Radiation Effects on Humans – Cr. Spacecraft Systems Design & Operations (2004)*

[90] Peters, James F., *Spacecraft Systems Design and Operations*, Kendall / Hunt Publishing Company, 2004

# Appendix D – AC$^3$ROSS Project: Conceptual Design

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# AC$^3$ROSS PROJECT

ASCLEPIUS[91] CLIMATE CHANGE AND CANCER RESEARCH ORBITAL SPACE STATION



A CONCEPTUAL DESIGN EXERCISE SUPPORTING THE MASTERS OF ARCHITECTURE THESIS:

ARCH 2001: A SPACE ODYSSEY – THE ARCHITECTURE OF MICRO-GRAVITY ENVIRONMENTS.

[91] Asclepius is a Greco-Roman god of medicine. His staff, displayed in the logo above represents the knowledge of medicine descending down to humans, and the rebirth ascending the staff in the form of a snake. Image credit: Rod of Asclepius.svg https://en.m.wikipedia.org (2024)

## Introduction

The AC$^3$ROSS project attempts to tackle two of the world's most pressing existential crises. Climate change and the global cancer epidemic are two of humanity's greatest threats, jeopardizing our existence on planet Earth. The acronym "AC$^3$ROSS" stands for the *Asclepius Climate Change and Cancer (C$^3$) Research Orbital Space Station*. In addition to the design of the overall station, hereafter referred to as the "SS Asclepius," this investigation includes the schematic design of expandable laboratory and habitation modules, two of the components that comprise the space station,

The design of SS Asclepius – and its constituent components – was intended as a vehicle through which to test my understanding of facts and concepts outlined in the "primer" portion of the thesis (Chapters 2 – 4). The design of the space station was conceived as a use-case for the planning and visualization of a research facility in Low Earth Orbit (LEO). This appendix describes the development of the program, the design of the modules comprising the station, and the anticipated launch and construction sequencing.

As noted above, the SS Asclepius's will be "sited" in LEO at an orbital inclination matching that of the ISS, namely 56.1 degrees. This orbit will enable the station to orbit the Earth 16 times per day and have visual access to over 90% of Earth's inhabited areas. This orbit, as described in the primer, has many benefits for Earth observation, and is relatively easy to access from major ground infrastructure such as spaceports.

*Figure 47 - Rendering - SS Asclepius with approaching spacecraft*



## Early Concepts and Layouts

Having researched the fundamentals of space station programming and design, the layouts and exterior shell concepts illustrated below represent a first attempt at "design as research." These initial designs opened up opportunities for discussion and further research of topics for the final phase of the research and design. This process allowed for an informed building up of the space station program that would result in the SS Asclepius, as presented at the thesis defense. The programmatic functions were integrated into a refined set of modules, further developed in the design of the SS Asclepius.

The adjacency diagram below was created early in the process in response to what I had learned about the built-up systems requirements for the safe operation of a habitable space station in LEO. This provided a foundational guideline for the early concept layouts that followed.

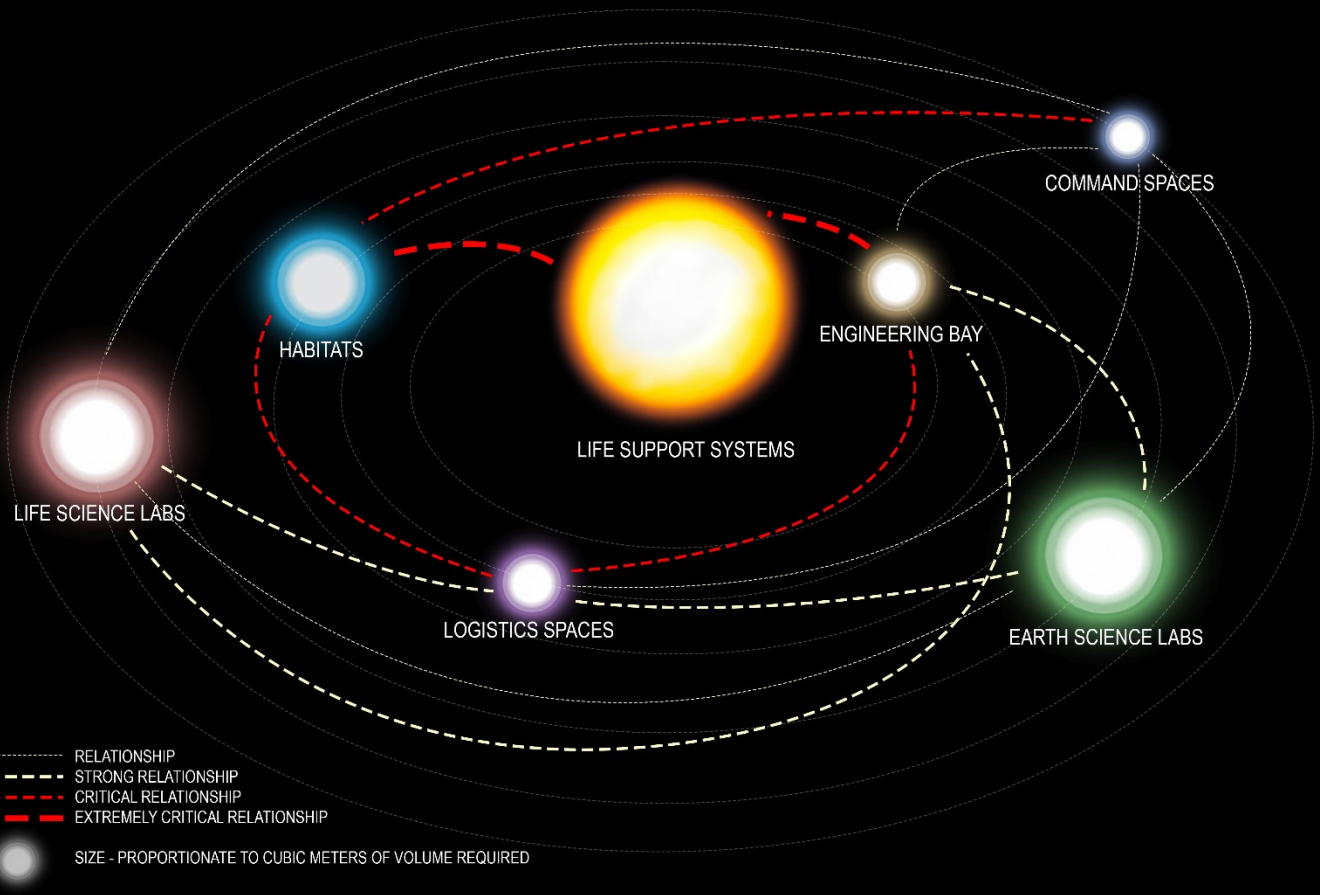



*Figure 48 - Conceptual Adjacency Diagram*

**Habitats**

Dwelling and shared living spaces (show in blue) were located near the belly of the space station, to provide additional protection from radiation, and to allow for easy access to most areas of the space station without disrupting scientific or station operation workflows. These functions were integrated into the Expandable Habitation Modules (E-HM) and Personal Adaptive Dormitory (PAD) Modules in the final design.

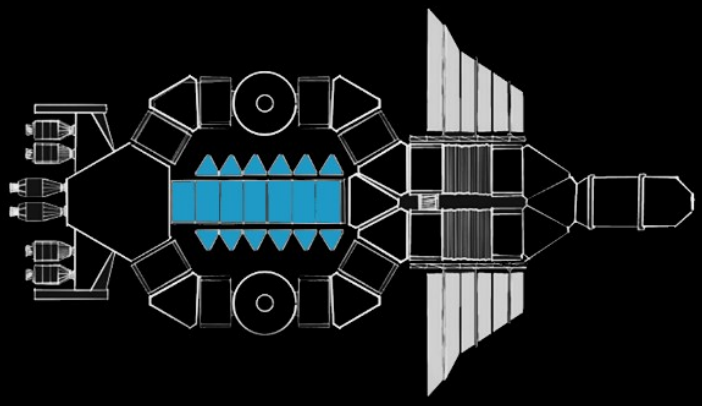


*Figure 49 - Early Program Concept - Habitation Areas*

**Laboratories**

Two lab wings, shown in green and pink below, would accommodate research on climate change and cancer, one wing for each. These wings would connect both to the engineering bays and to the loading /logistics areas, located at the bow and aft of the space station, respectively.

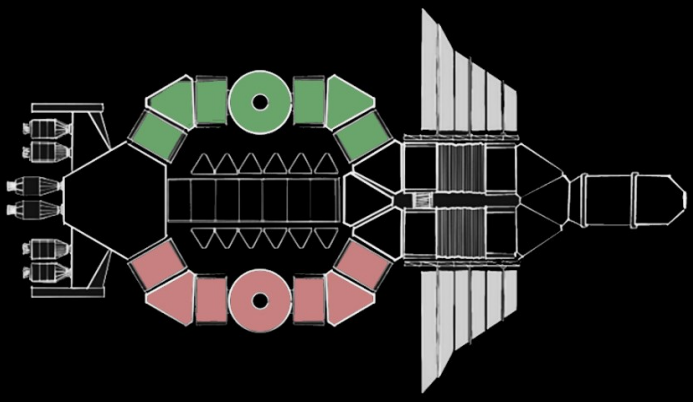


*Figure 50 - Early Program Concept - Science Laboratories*

**Logistics, Engineering Bays, and Command Center**

Logistics modules (Lavender) were conceived as the key connective elements between functional modules and would include strategically placed docking ports to provide flexibility, connectivity, and the ability to build up the station's functional components. Logistics modules would have significant storage capacity and provide functionality for space craft docking and EVA support.

Engineering Bays (Yellow) would house the key life-support and station-support systems, including ventilation, air filtration, water processing and power generation, storage, and distribution systems. Command Center modules (Blue) would house the key communications and station-monitoring systems to support the station.

These functionalities were implemented within the Expandable Engineering Bay Modules (E-EBM) and Logistics Integration Nexus Connector (LINC) Modules in the final SS Asclepius design.

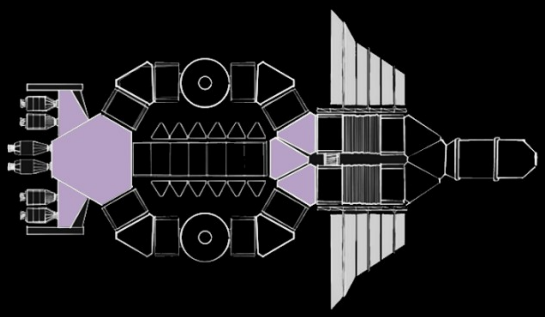


*Figure 51 - Early Program Concept - Logistics*

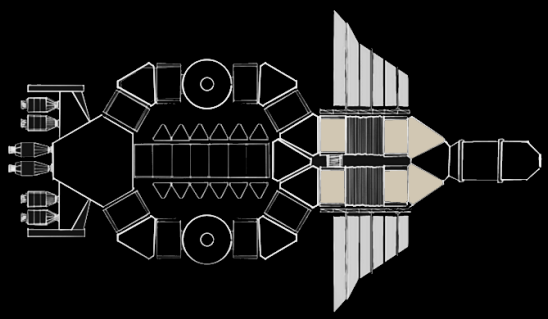


*Figure 52 - Early Program Concept - Engineering Bays*

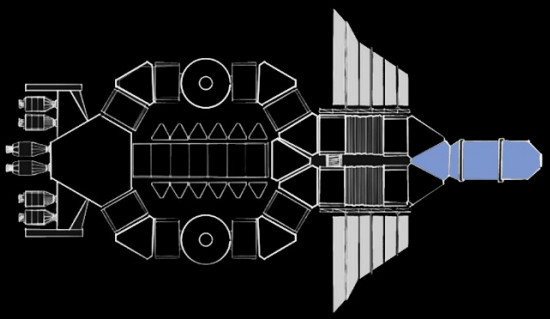


*Figure 53 - Early Program Concept – Command*

**Massing**

As the conceptual layouts evolved, the overall mass of the space station took shape, comprised as it was of a series of modules, docking areas, and solar arrays. Being able to reflect on the assembly enabled me to determine where elements might benefit from additional research and design attention. As it evolved, the so-called front of the station became increasingly distinguishable from the rear, suggesting the need for further research on nomenclature as it relates to positioning and naming conventions in outer space. In this case, a forward or “bow” and a back end or “aft” of the ship started to present itself. With this, starboard, port, zenith and nadir directions began to orient the design process in relation to the Earth.

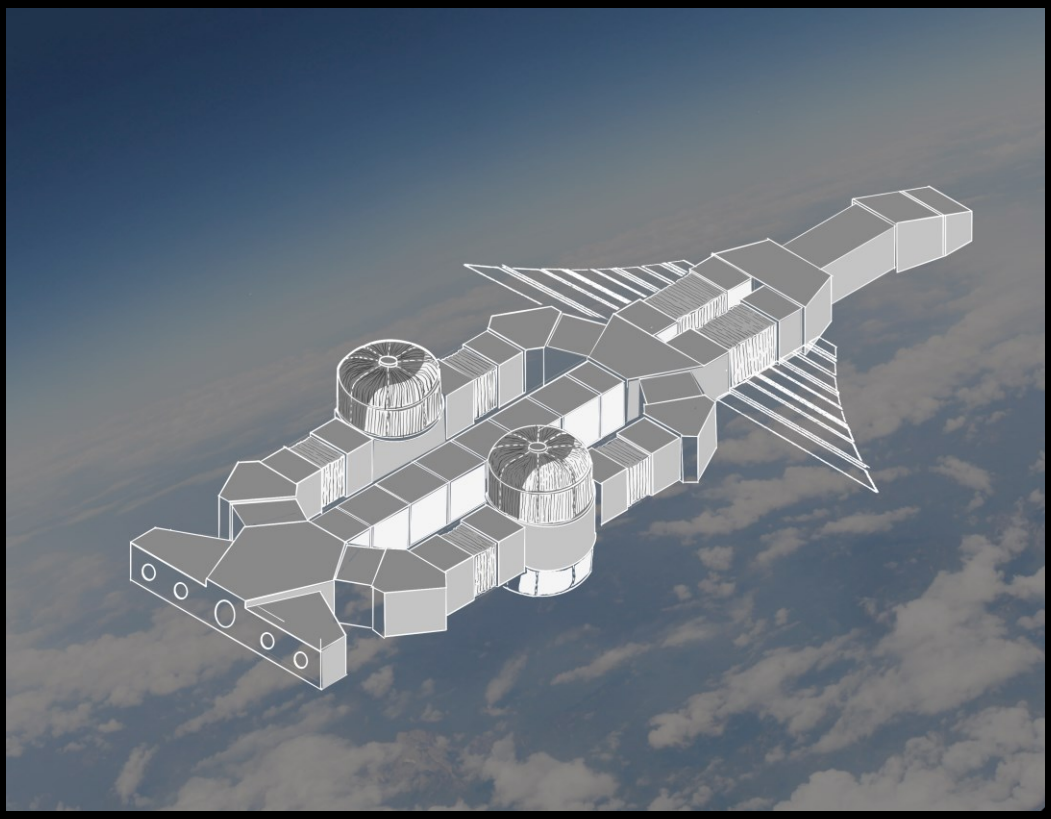


*Figure 54 - Space Station Modular Massing Diagram*

**Structural and Environmental Requirements**

The design of the skeletal superstructure of the space station played a key role in protecting the core modular components in a functional and resilient manner in the orbital environment, extending the life expectancy of the station. As noted in the chapter on *Design Parameters*, space structures must be designed to withstand environmental and operational stresses.

An exo-skeletal design was deployed, using in-space truss manufacturing and assembly opportunities to provide a substrate for enhanced shielding and built out “exterior” covered areas for docking, EVAs, and external storage.

- The design includes radiation shielding, including internal protected areas of refuge for the crew during radiation storms.
- MMOD protection is crucial as damage from space-borne projectiles can be costly and can compromise critical systems within the space station. The design concepts include cladding, module wall construction and exoskeleton design to mitigate such damage.

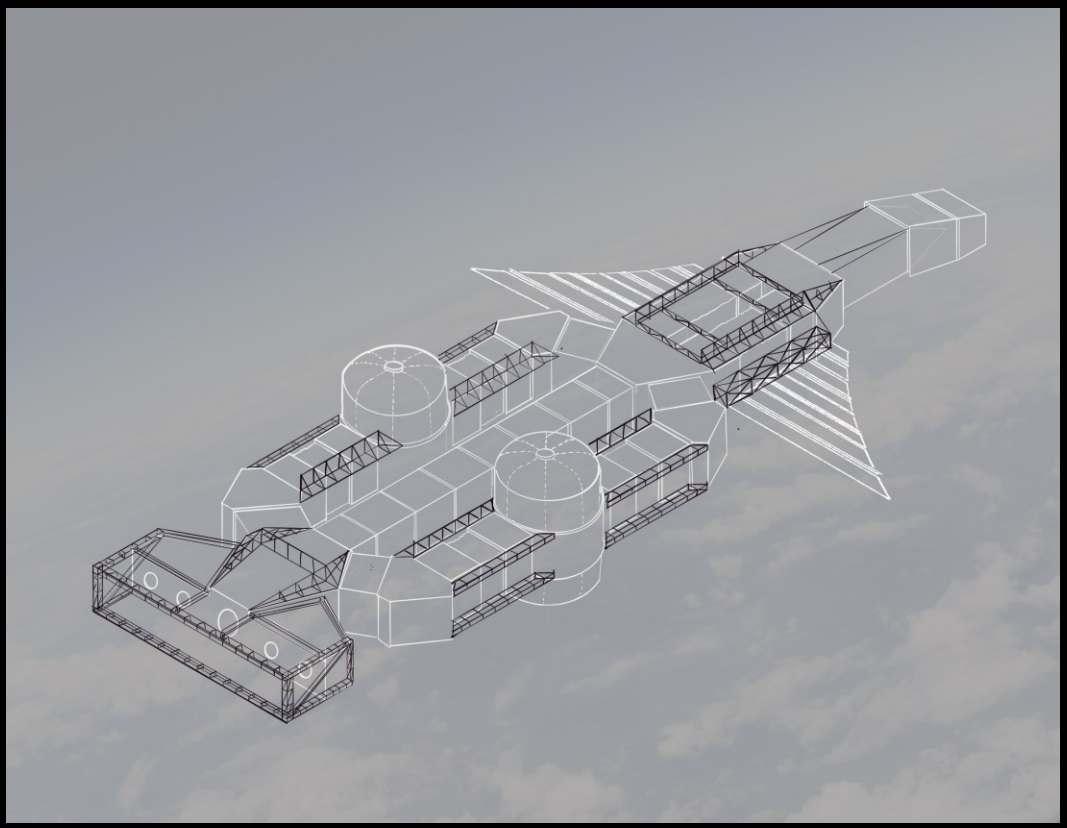


*Figure 55 - Exo-Skeleton Structural Concepts - rigidity and substrate for exterior shell*

**Exterior Cladding and Design**

The exo-skeletal trusses support an exterior skin or cladding that provides additional radiation and MMOD shielding. The gap between this cladding and the exterior walls of the station modules could accommodate additional storage and distribution space for connectivity within the station. At the very least, this second skin acts as a barrier, expanding the life of the space station and providing additional power generation opportunities.

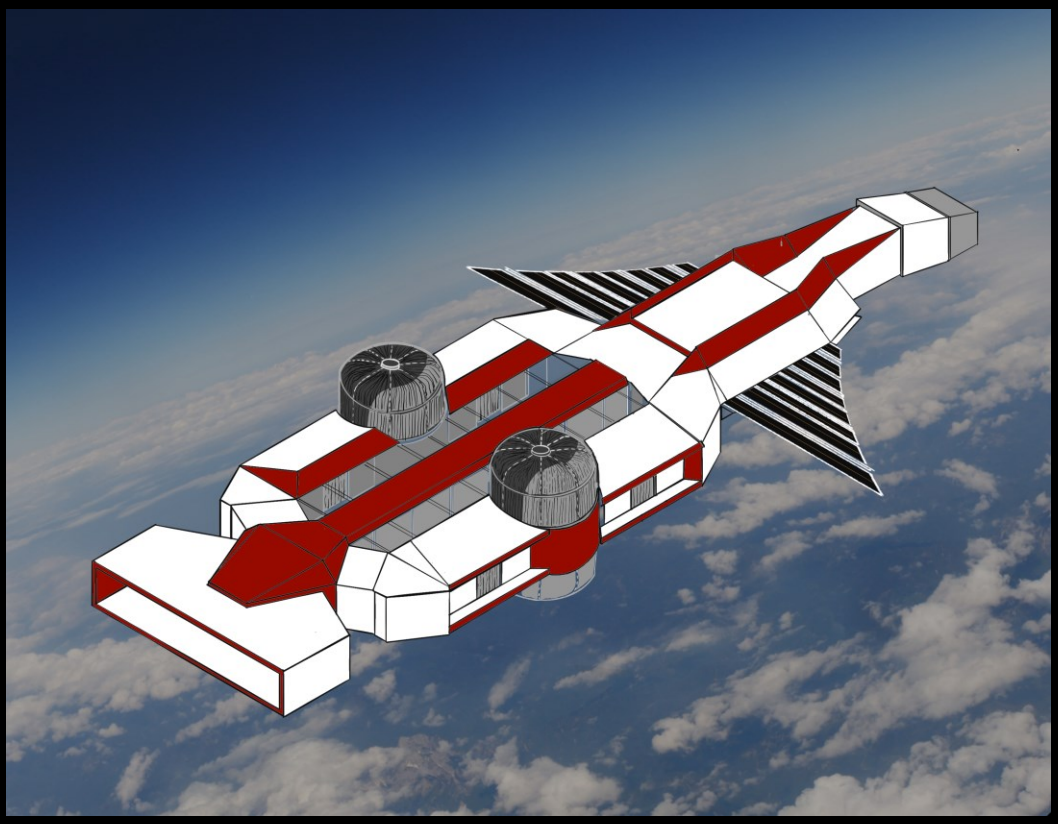


*Figure 56 - Space Station, Conceptual Visualization - Exterior Space Station Cladding*

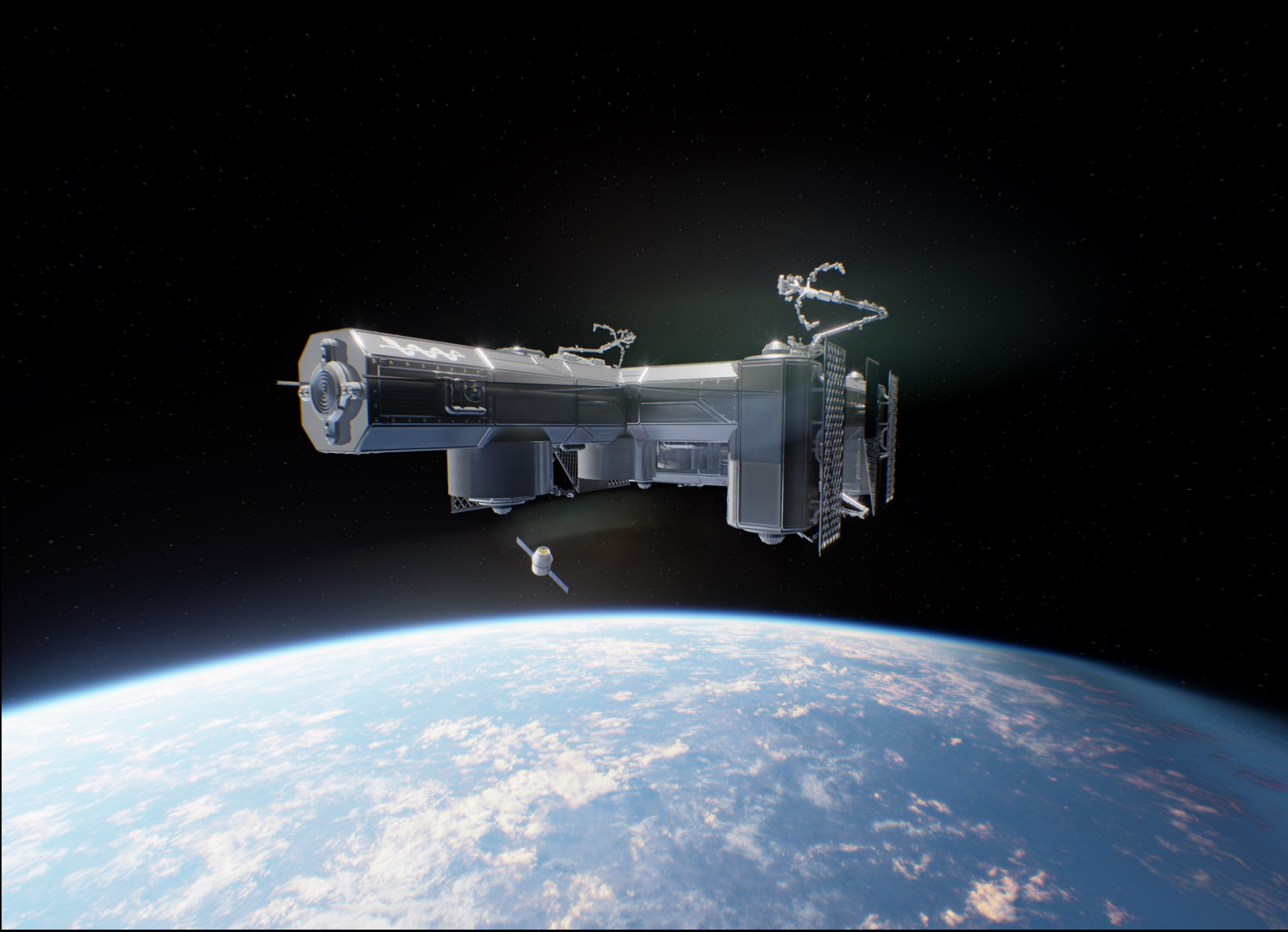
## SS Asclepius Program

The following design, which built upon the concepts described in the previous section, was presented for the final ARCH 2001 Thesis defense.

The design of the Asclepius space station comprises several modules, each of which contains specific types of spaces for various programmatic functions. These modules are described in further detail using diagrams, architectural graphics, and visualizations. This section will describe the overall form, construction, components, and functions of the modules.

The laboratory and habitation modules have developed further than the others as they are the primary components of the case studies that this thesis explores. Secondary spaces and the modules that support them -- such as engineering bays and connector modules – have been conceived of as a kit of parts, and have been developed at a schematic level to demonstrate an understanding of the assembly sequence and requirements for the station as a whole.

*Figure 57 - SS Asclepius - Forward side, Command Center viewing window.*



One of the perceptual hurdles in the planning and programming of space architecture is the distinction between planar (e.g., area) and volumetric thinking. While conventional architects consider volumetrics when planning spaces, an emphasis is put on the floor area. Volume in terrestrial architecture changes the experience and scale of a space – think of a bedroom vs. an elevator shaft. In space architecture the larger volume can be programmed, and the overall area is calculated as the sum of the areas of each of the surfaces that enclose the space. Based on the modular design of space station components, the figure below was created to get a sense of the overall volume of the SS Asclepius.

- Spaces on the SS Asclepius are a series of pressurized modules serving different operational functions.
- Station pressurized volumes are based on expected crew size of 24-32 (on 3 shifts), equipment requirements and international low earth orbit laboratory design guidelines (LEOLab-DGs)[92], facilitated by the modular design system.
- The total pressurized volume results in 7,716 $m^3$, approximately 241-321$m^3$ per crew member. an increase of 40-53% of volume per crew member compared to I.S.S. metrics of 900$m^3$ / 7-person crew (128$m^3$/p).

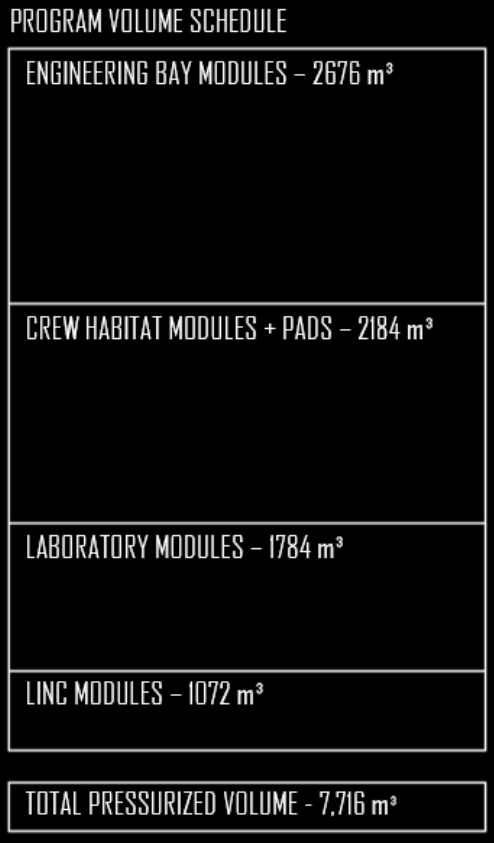



*Figure 58 - SS Asclepius Program*

[92] This is a conceived future design guideline specifically for laboratories in outer space. This can expand to different types of science and risk levels. Precedents for design guidelines like this would be the NIH Design Requirements Manual (DRM) and the Canadian Biosafety Standards (CBS)

The program component diagram below illustrates the kit of parts used to build up SS Asclepius. Each module type has different functions and is designed to take into consideration the launch and construction sequence process, described later in this document. There are three basic module variants: the *Expanding Module*, used for E-EMB, E-CHM and E-LM functions, the *Personal Adaptive Dormitory* (PAD) modules, and the *Logistics Integration Connector* (LINC) modules.

The expanding laboratory module (E-LM) and the PAD module were explored in greater detail in this design. The interiors of expandable modules can be configured in many ways to accommodate the required internal sub-structure, program, equipment, and systems required to operate the SS Asclepius.

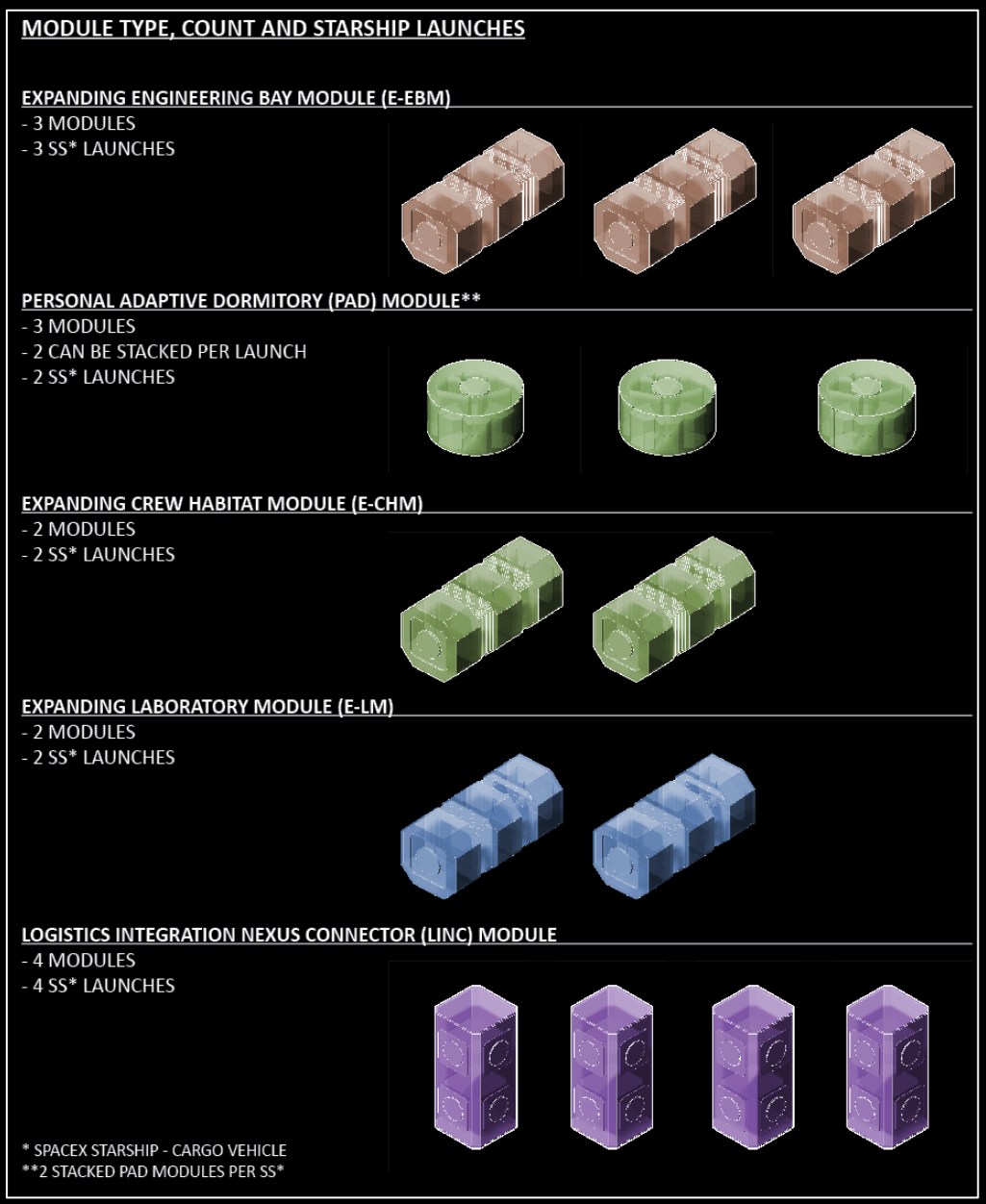



*Figure 59 - Module Types, Count, and Launches*

## Modular Design

Using a modular approach, the design of the space station is comprised of a kit of functional parts that can be assembled in outer space. As described in the *launch logistics* section of this document, space station modules are designed to fit within the cargo holds of existing space craft. In the case of SS Asclepius, all modules are designed to fit and be transported by the SpaceX Starship.

The space station modules are effectively the architecture of the project – akin to individual buildings within a small campus. Habitable structures in the vacuum of space must be capable of accommodating internal pressure loads of 0.6-1.0 atmospheres without leaking. (60.795 Kpa - 101.325Kpa - Standard atmospheric unit x 101.325 = Kpas). This is equivalent to approximately 8.8 to 14.7 PSI.

The diagram below shows early concept sketches of the space station modules, exploring expandability, docking, and module connectivity.

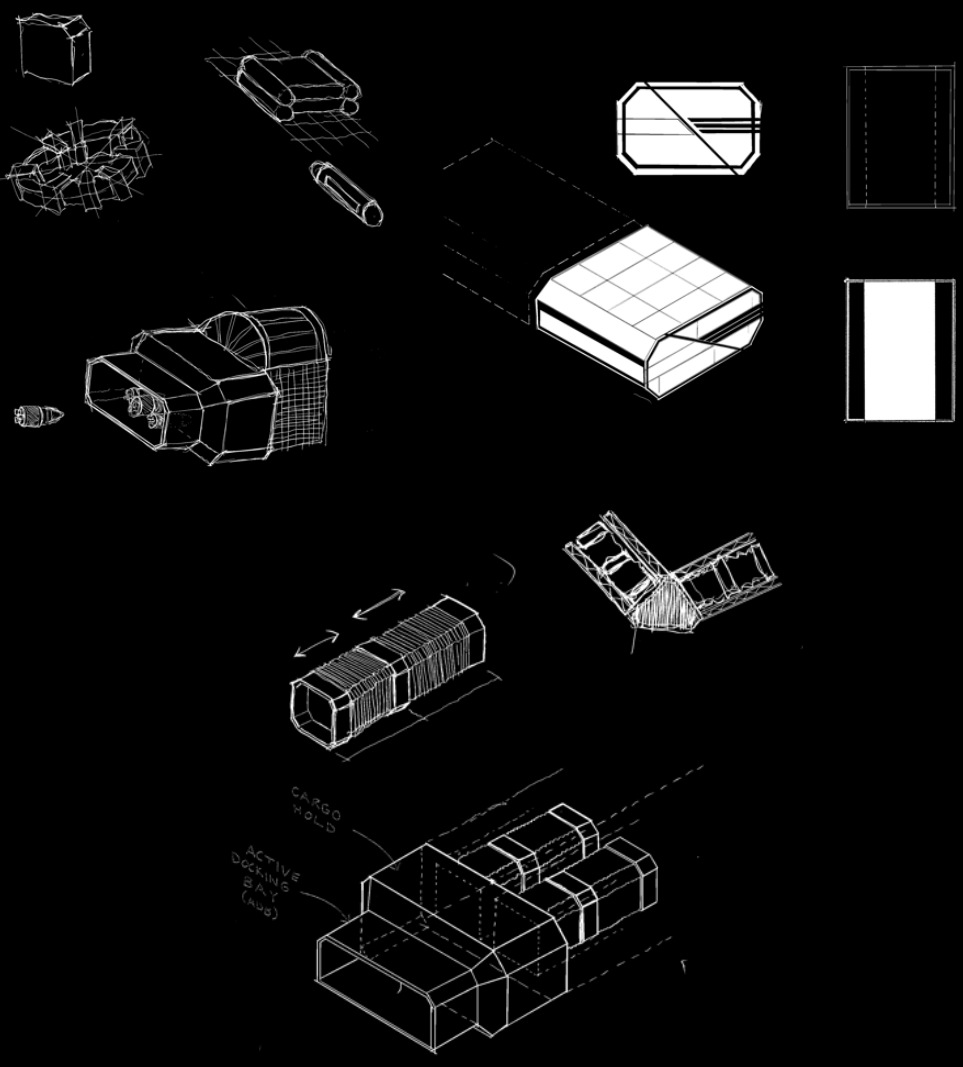



*Figure 60 - Module Design Concepts - Sketches by Ramon Renderos Soto*

## Design Requirement Considerations

### Material Selection

Although material science continues to advance, a significant amount of research and innovation have been brought to bear on the materials deployed in space. The chart below lays out key characteristics of the materials likely to be used for various components of the space station.

| Material | Advantages | Disadvantages | Typical Applications |
|---|---|---|---|
| **Aluminum** | • High Strength to weight<br>• Low cost<br>• Readily available<br>• Easy to machine<br>• Weldable<br>• Corrosion resistant | • Poor resistance to galling and wear<br>• High thermal coefficient<br>• Heat during welding causes reduced strength<br>• Prone to cracking | • Truss embers, skins, stringers, fittings, brackets, shells<br>• Face sheets for sandwich structures |
| **Titanium** | • High strength to weight<br>• Low thermal expansion coefficient<br>• Good temperature properties | • Difficult to machine or form<br>• Relatively expensive<br>• Availability | • Attachment fitting for advanced composites<br>• Fasteners |
| **Steel** | • High stiffness and strength<br>• Wear resistant, ductile<br>• Easy to machine and weld<br>• Low cost | • High mass density<br>• Low buckling strength vs. weight<br>• Varying corrosion resistance | • Fasteners<br>• Ball bearings<br>• Structural supports |
| **Magnesium** | • High buckling strength vs. weight<br>• Heat capacity and conductivity<br>• Easy casting | • Poor corrosion resistance<br>• High thermal coefficient<br>• Low stiffness | • Lightly loaded structures, especially those critical for buckling<br>• Light castings |
| **Beryllium** | • High stiffness vs. weight<br>• Low thermal coefficient | • Expensive and toxic<br>• Low ductility and fracture<br>• Poor cross grain properties | • Mirrors and optical support structures<br>• Precision gimbal and telescope housings<br>• Hinges, high temperature applications |
| **Heat Resistant Alloys** | • Strength at high temperatures<br>• High stiffness and strength<br>• Oxidation resistant | • High mass density<br>• Low buckling strength<br>• Relatively hard to machine | • Fastening hardware<br>• High temperature applications (nozzles heat shields) |
| **Graphite/ Epoxy Composite** | • Relatively light<br>• Tailored to have high strength stiffness, low thermal coefficient<br>• High conductivity | • Brittle (no ductility), repair<br>• Absorbs water, outgases<br>• Process development cost<br>• UV sensitive | • Truss members, antenna booms<br>• Face sheet for sandwich structure<br>• Optical benches<br>• Fuel tanks |

*Figure 61 - Comparison of Conventional Space Construction Materials – Credit: SSD&O (2004)*

### Life Support and Emergency Systems

The table below lists critical life-support systems that sustain human life and safety and operations on a functioning space station. While many of these systems are common to all space station modules, major life-support systems are assumed to be accommodated within engineering bay modules, described in further detail in the following pages.

| Structures & Mechanisms: | Thermal Control Systems: | Propulsion & Motion Control: | Guidance, Navigation & Control |
|---|---|---|---|
| ▪ Pressurized elements<br>▪ Viewports & airlocks<br>▪ Utility systems/ interfaces<br>▪ Berthing/docking mechanisms<br>▪ Trusses/ attachments<br>▪ Robotic systems/workstations | ▪ Passive insulation/ coatings<br>▪ Fluid loops/ pumps/ valves<br>▪ Electric heaters<br>▪ Heat pipes & cold plates<br>▪ Active & passive radiators<br>▪ Multiplexers/ demultiplexers | ▪ Propellant tanks<br>▪ Rocket engines<br>▪ Auxiliary/ nuclear power<br>▪ Orbital maneuvering systems<br>▪ Reaction control systems<br>▪ Hydraulic systems | ▪ Computer systems<br>▪ Celestial trackers<br>▪ Attitude sensors<br>▪ Attitude actuators<br>▪ Control moment gyros<br>▪ Antennas/ GPS |
| **Electrical Power Systems:** | **Env. Control /Life support:** | **Command & Data Handling:** | **Communications & Tracking:** |
| ▪ Photovoltaic arrays<br>▪ Radioisotope generators<br>▪ Batteries & fuel cells<br>▪ Power conditioning<br>▪ Distribution busses<br>▪ Fault detection systems | ▪ Atmosphere supply/ distribution<br>▪ Trace contaminate control<br>▪ Ventilation/temp./hum. control<br>▪ Fire detection/ suppression<br>▪ Waste collection/ treatment<br>▪ Regenerative life support | ▪ Computer networks<br>▪ Crew interface computers<br>▪ Multiplexers/ demultiplexers<br>▪ Flight control systems<br>▪ Caution & warning systems<br>▪ Telemetry systems | ▪ Data relay satellites<br>▪ Tracking/ data network<br>▪ External/ interior antennas<br>▪ Audio/ visual systems<br>▪ Data distribution<br>▪ Robotic workstations |

*Figure 62 - Life support and emergency systems – Credit: SSD&O (2004)*

## Ground Spaceport Infrastructure

Although the focus of this thesis was a research facility in an orbital environment, other opportunities exist to support research in space. Among these, facilities here on earth present rich opportunities for architectural involvement, contribution, and innovation. These include the following:

**Ground Support Facilities** serve a range of functions such as mission control, station module manufacturing, communications, payload processing, and a wide range of behind-the-scenes work that would need to be done to execute the AC$^3$ROSS project successfully.

For a detailed list of existing ground facilities maintained by various global agencies, visit: https://www.nasa.gov/international-space-station/ground-facilities/

**Spaceports** are a crucial component of the expanding commercial space economy. As of 2024 there were 35 spaceports and launch facilities around the world able launch satellite or spacecraft into sub orbit, orbit, and beyond.[93] Organizations such as the Global Spaceport Alliance (GSA) serve this spaceport ecosystem by providing a forum for the collaboration and development of this global infrastructure. For more information on the GSA, visit: https://www.globalspaceportalliance.com/

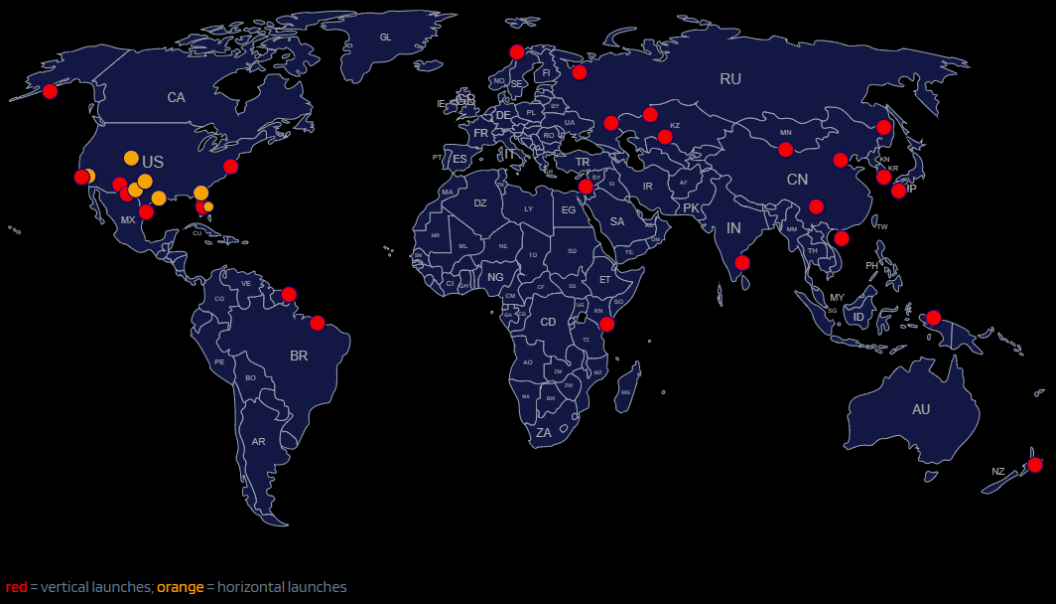



*Figure 63 - Global Spaceports & Launch Sites. Credit: go-astronomy.com (2024)*

[93] Go-Astronomy, "Spaceports & Launch Sites". 2024. https://www.go-astronomy.com/space-ports.php

**Scientific Laboratories** on Earth will continue to support whatever research work is done in outer space. Extensive study and preparation is required before experiments can be executed in a space station environment. The *Tumors in Space* project headed by Dr. Larose, for example, receives support from the Norwegian Space Agency, other EU partners and from China. To prepare for her upcoming work on the Chinese Space Station she conducted experiments in a number of international laboratories here on Earth,[94] including:

1. Organoid development at Hubrecht Organoid Technology in Utrecht, Netherlands.
2. Ground-based experiments at the European Space Research and Technology Center in Noordwijk, Netherlands.
3. Radiation biology experiments at the Belgian Nuclear Research Center, Belgium.
4. Collaboration with the International Space University and parabolic flight campaigns conducted in France.

In addition, Chinese laboratories will conduct post-experiment analysis on the tumors when they return from space.

This list serves as an example of the network required to conduct even one groundbreaking experiment in space. Work on and with this global network of laboratories presents an opportunity for future lab and space architects to contribute to the commercial space economy and to the state-of-the-art science that targets our global climate and health issues. Led by pioneers like Dr. Larose, this network will support the research that is envisioned to take place aboard the SS Asclepius.

## Launch logistics

The ARCH 2001 primer notes numerous constraints associated with design for human habitation in outer space. It also describes the history and evolution of space craft up to the most recent development, namely SpaceX's Starship, which represents a major leap in spacecraft design.

The size of our space craft is a critical constraint as it provides us the maximum dimensions and weight of elements that can be transported into space. And although cost considerations also figure heavily in the feasibility of any space-bound mission, these were not considered in any detail in the primer. Cost will vary based on the rocket and spacecraft selected, the distance of travel, and other technical issues that warrant further exploration.

**SpaceX - Starship and Super Heavy** are the space craft and rocket (collectively named "Starship") I propose would be selected for the AC$^3$ROSS project. During the period in which I have worked on this thesis, Starship has been tested three times, achieving increased success at each interval, with a fourth test flight scheduled for May of 2024. The test flight history for Starship is as follows.

Starship Integrated Flight Test 1 (IFT-1) – April 20, 2023.

Launched from Starbase in Texas, Starship successfully lifted off, reached an apogee of 39km before experiencing multiple engine outages leading to flight termination.[95]

[94] Larose, "Tumors in Space: Preparation for Spaceflight".1-14
[95] SpaceX. 2024. "Starship's First Flight Test" For more information, visit: https://www.spacex.com/launches/mission/?missionId=starship-flight-test

Starship Integrated Flight Test 2 (IFT-2) – November 18, 2023

The second flight test was able to go even further and conduct additional operations including vehicle separation (between rocket and space craft) and the initiation of second stage Raptor engines – the first time this has been executed with a vehicle of this size. Additional maneuvers were also successfully executed, enabling the Starship space craft to reach an altitude of 150km with a velocity of 24,000 km/h – the first SpaceX Starship spacecraft to reach outer space. The rocket and space craft experienced rapid unscheduled disassembly and safe command destruct respectively, bringing IFT-2 to an end.[96]

Starship Integrated Flight Test 3 (IFT-3) – March 14, 2024

Starship's third IFT accomplished additional milestones and firsts. *Super Heavy (rocket)* successfully completed its flip maneuver and boost back burn, and commenced its return to the designated splashdown point, during which engines commenced the first ever landing burn before experiencing rapid unscheduled disassembly (RUD – catastrophic explosion). Starship reached its desired orbit using its six, second-stage Raptor engines, completing its full-duration ascent burn. In orbit, Starship conducted several additional flight tests, including the opening and closing of its payload door, and a propellent transfer demonstration. Starship went on to experience its first-ever entry from space providing data on heating and vehicle control during hypersonic reentry.[97]

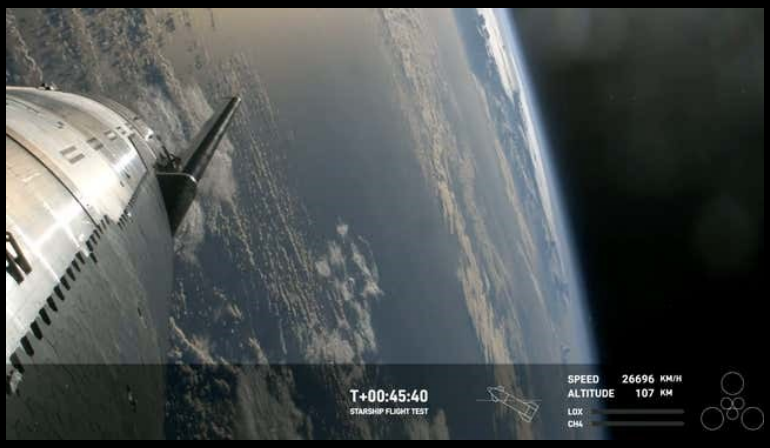



*Figure 64 - Starship IFT-3 Screenshot from orbit. Credit: SpaceX*

Starship Integrated Flight Test 4 (IFT-4) – Scheduled for May 2024

Starship IFT-4 will attempt additional Super Heavy (rocket) landing maneuvers, and Starship orbital tests, and reentry – building on the lessons learned and data collected from previous IFTs. The ultimate goal for SpaceX's Starship is for the Super Heavy rocket to return safely to Earth for re-use, and a successful deployment of payload in orbit by the starship spacecraft. SpaceX aims to "build a fully reusable transportation system capable of carrying both crew and cargo to Earth orbit, help humanity return to the Moon, and ultimately travel to Mars and beyond."[98]

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[96] SpaceX. 2024. "Starship's Second Flight Test" For more information, visit: https://www.spacex.com/launches/mission/?missionId=starship-flight-2

[97] SpaceX. 2024. "Starship's Third Flight Test" for more information, visit https://www.spacex.com/launches/mission/?missionId=starship-flight-3

[98] See footnote 97

**Starship Significance**

The purpose of highlighting the IFT chronology is to clarify that Starship is still is still very much in development, despite being used as the baseline for decisions about the size of the modules that will comprise the SS Asclepius assembly. The following diagrams highlight the significance of the Starships cargo payload capacity. Compared to the 4 m diameter (+/-) of the Space Shuttle's cargo bay, Starship boasts a 9m diameter cargo bay, and a payload capacity of 100-250 tonnes, significantly increasing the design possibilities for space station modules.

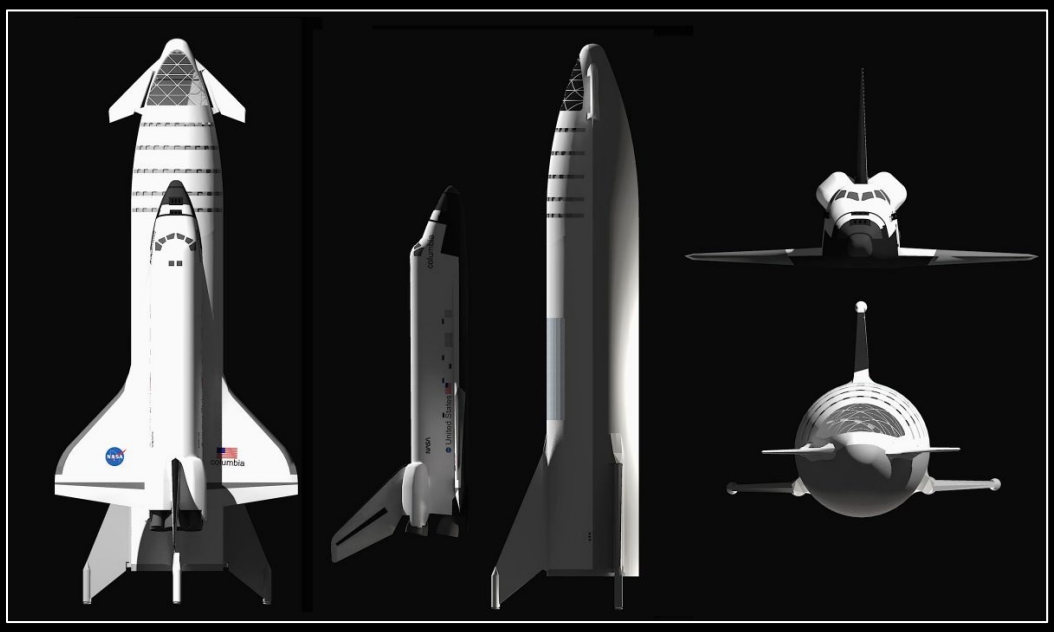


*Figure 65 - SpaceX Starship Space Shuttle Comparison. Credit: quora.com*

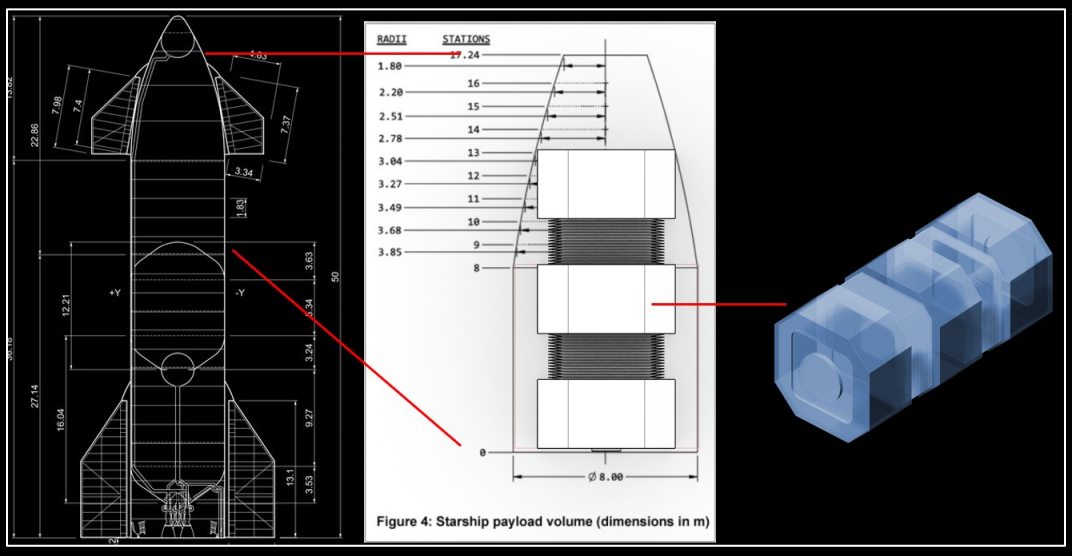



*Figure 66 - E-LM within Starship payload. Base images Cr. X, Fael097 (2021), SpaceX (2020)*



*Figure 67 - Concept SS Asclepius Module within Starship. Base image Cr. ERCX, X (2022)*

## In-orbit manufacturing

Emerging manufacturing techniques and technologies include solutions for the 3D printing of trusses and advanced robotics for maintenance and repair procedures. This suggests room for further exploration in the future. It is assumed that the SS Asclepius will take advantage of up to four *Canadarm3*s and two *DEXTRE* robots, which will be supplied by the Canadian Space Agency to assist with the assembly of the ship's modules, exterior trusses (exoskeleton) and cladding systems.

## Crew Planning

The number of occupants or users is a fundamental metric for architectural design and is an even more critical design consideration for space stations like the SS Asclepius. The SS Asclepius will be able to provide significantly more volume per crew member than the ISS thanks to the larger cargo bay of the Starship and the proposed design for the habitation modules (PADs). Studies on volume per person can lead to future explorations on quality of life in outer space. The SS Asclepius is designed to accommodate up to 24 long-duration inhabitants, with ample space for additional short-duration habitation.

## Orbital Selection Rationale

The rationale for siting the SS Asclepius in Low Earth Orbit was described earlier in Chapter 4.3 *Orbital Selection Criteria*. The motivation was ease of access to and from Earth and to enable to observe a high percentage of Earth's populated areas and overall surface. Yet the rationale for the selection of an orbital space station vs. a laboratory on a lunar or Martian surface – a question raised at the thesis defense – should be addressed. The rationale is relatively simple. As evidenced by industry participation and investment, the commercial, low-earth-orbit economy is the next frontier.

A study by Deloitte in 2022 called "The Commercialization of Low Earth Orbit" divided the LEO human space flight industry into the following categories: [99]

- infrastructure,
- services,
- on-orbit research & development,
- on-orbit manufacturing,
- space tourism and media, and
- entertainment & advertising.

The study concluded that the potential market value of this industry will be as high as $312 billion USD by 2035, and no lower than $151 billion USD in a low-end, "constrained growth" scenario. Considering that the feasibility of orbital habitation and scientific exploration are well established, and considering that my thesis privileged research on cancer and climate change to the immediate benefit of humanity, it made little sense to dedicate time and effort on designing a lunar or Martian environment for the distant future. While I understand that my decision to pursue research and design of space architecture in LEO is ambitious, I believe going any further would be tantamount to escapism.

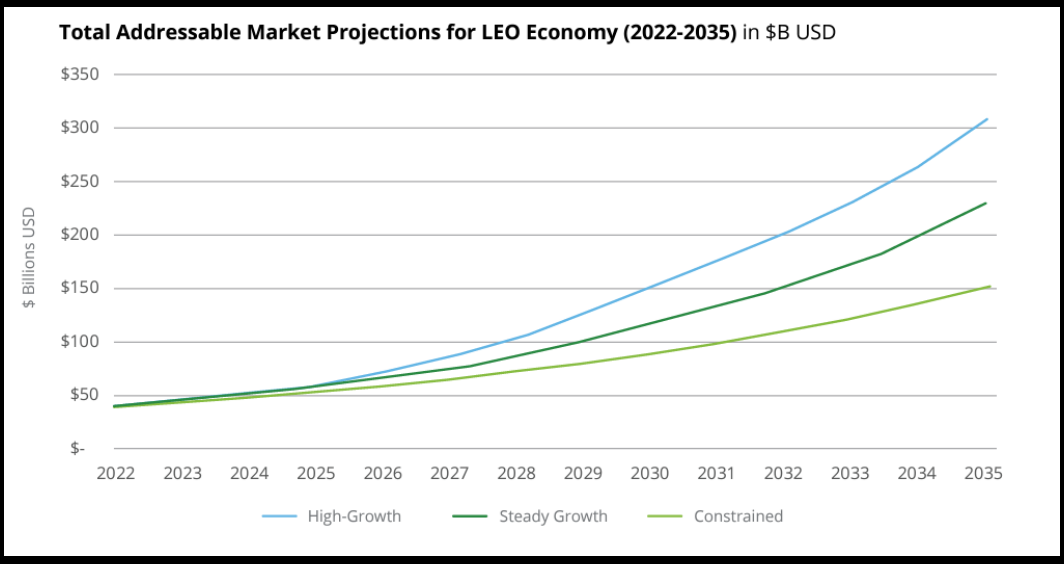



*Figure 68 - LEO Market Projections. Credit: Deloitte (2022)*

[99] Deloitte. "The Commercialization of Low Earth Orbit". 1-23

## SS Asclepius Modules

**Extendable Modules**

The majority of the modules that comprise the SS Asclepius are designed to be extendable and/or expandable once they are anchored into place. Extendable modules serve as habitats, laboratories, and engineering bays. The diagrams below provide the overall dimensions of the generic module in both its un-extended and extended states. For the purposes of this investigation, it is assumed that these modules will be transported into space using a SpaceX Starship, the dimensions of which provided working constraints.

The extendable laboratory modules are described in greater detail in the ensuing sections of this appendix.

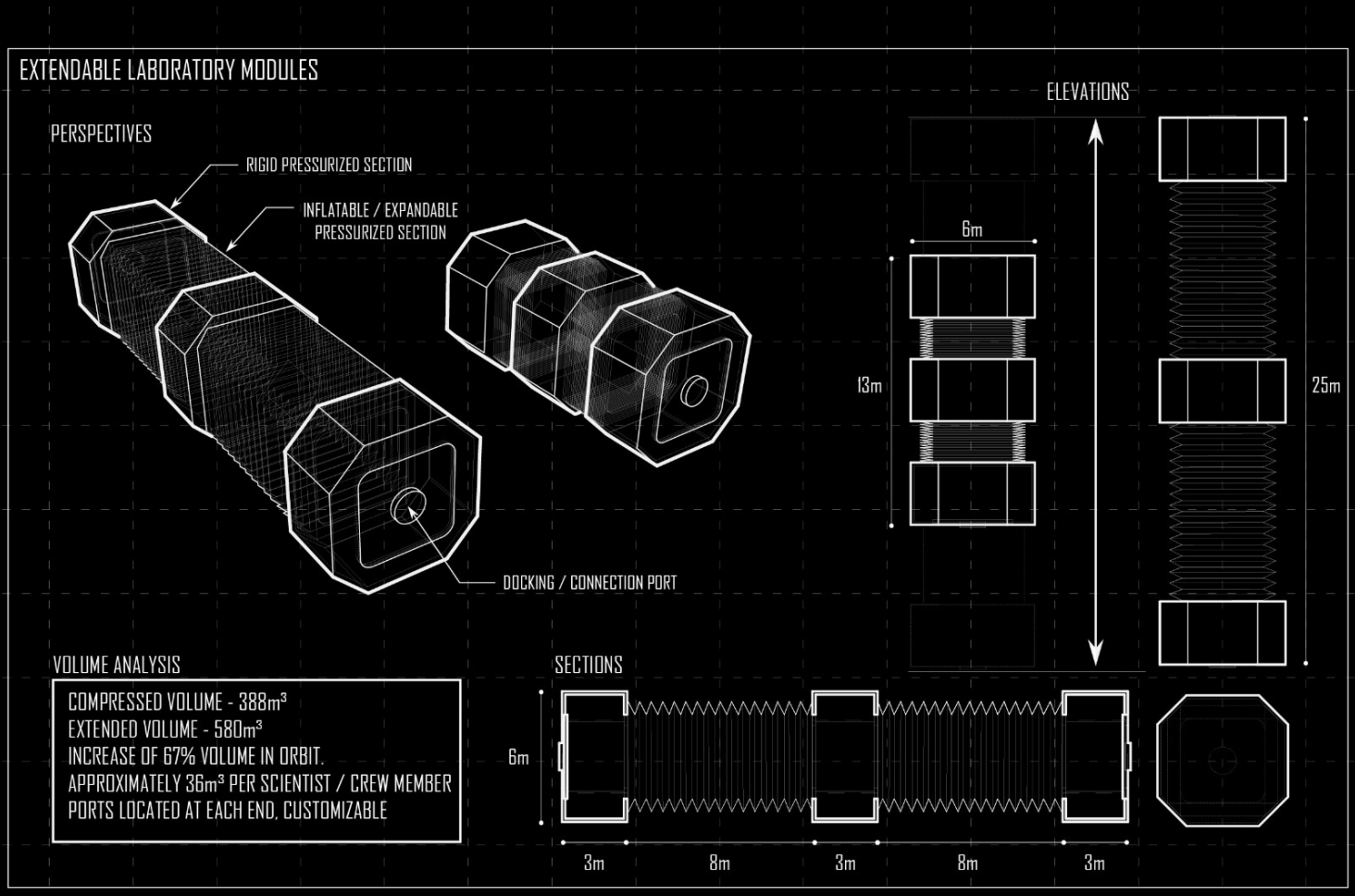



*Figure 69 - Extendable Module Concept Design*

**Extendable Laboratory Modules (E-LM)**

As the primary function of the SS Asclepius is to support research, laboratories are key to its success and must provide the required scientific infrastructure to conduct experiments in micro-gravity. These laboratories will support research in Life Sciences, notably cancer, and in the Environmental Sciences, including carbon research, earth observation, and energy-related research as it relates to climate change initiatives. Each lab must accommodate bench lab functions, equipment, and lab storage. In addition to the basic infrastructure required to conduct experiments, labs will include equipment to support specialized workflows.

As designed, the scientific laboratories are located along the port and starboard sides of the station, with direct connections to the LINC modules and to the engineering bays that support them. As envisioned, this location allows for flexibility, enhanced interior and exterior maintenance capabilities, and expandability thanks to strategically placed birthing ports located on the LINC modules. For example:

- A private or commercial environmental research module can be docked for the duration of the research project to a strategically placed port on the laboratory or adjacent LINC.
- If a containment level 3 (CL3) capability were required, a CL3 containment laboratory module could be docked onto the life sciences laboratory for the duration of a research project and/or as a permanent high containment module capability added to the facility.
- Some debate has occurred on whether samples returning from asteroids, Mars or other celestial bodies should be returned to sample return laboratories on Earth. A containment laboratory on SS Asclepius is a sample return option that protects our Earth's environment from potentially harmful substances that may be found within incoming samples. Initially, samples can be analyzed in an orbital containment laboratory before being sent down to Earth.

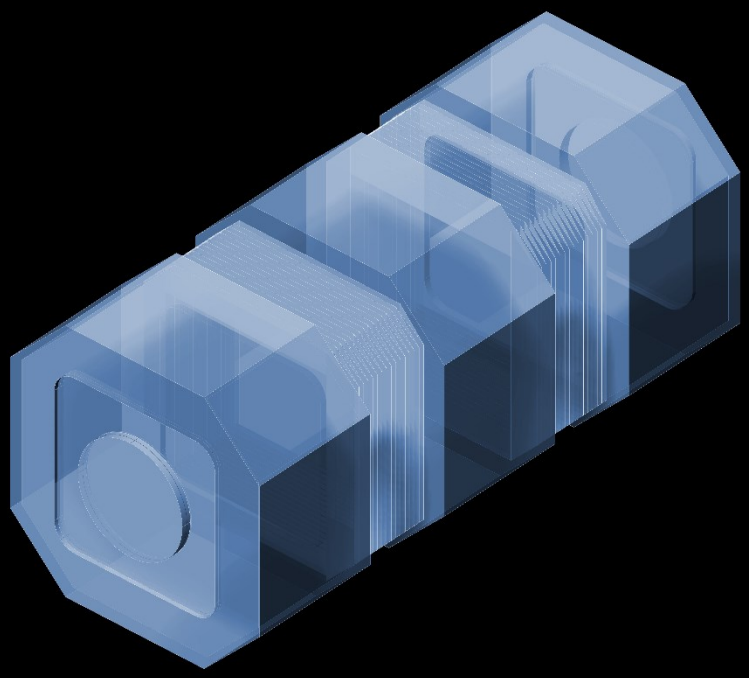


*Figure 70 - E-LM Concept, un-extended.*

**Personal Adaptive Dormitory (PAD) Modules**

Crew members aboard the I.S.S. currently sleep in small compartments or tin sleeping bags throughout the station in, using restraints to keep themselves as stationary as possible. Neither option provides much in terms of comfort, materiality, or a sense of personal space.

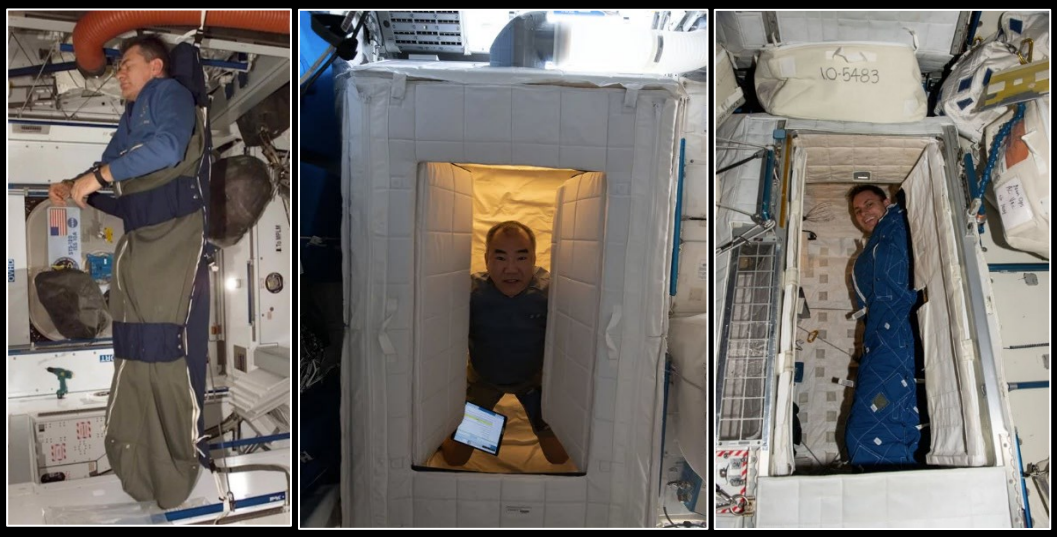


*Figure 71 - Left to Right -"Sleep among the stars", Cr. CNN (2017). JAXA Astronaut Soichi Noguchi, CSA Astronaut Josh Cassada. "Sleeping in Space". Cr. CNN (2023)*

Using the attic bedroom shown below as a precedent, design elements were introduced into the PAD modules to provide a sense of home, escape, materiality, and personalization.



*Figure 72 - Left to Right - Attic bedrooms. Alex Geerts, Jonathan Borba, Cr. Unsplash (2024), Attic Bedroom, Cr. thenorrdroom.com (2024)*

As designed, PAD modules consist of a series of pie-shaped berths that fit within a cylindrical module. The module, in turn, is sized to fit within the cargo hold of Starship, which can accommodate two modules per launch, or 16 berths. Each berth provides a private getaway for Asclepius crew members and includes an adaptable viewport or window. This viewport faces the Earth and acts as an opening through which to view the earth, a shade, or an entertainment screen for viewing or video communication with family and friends.

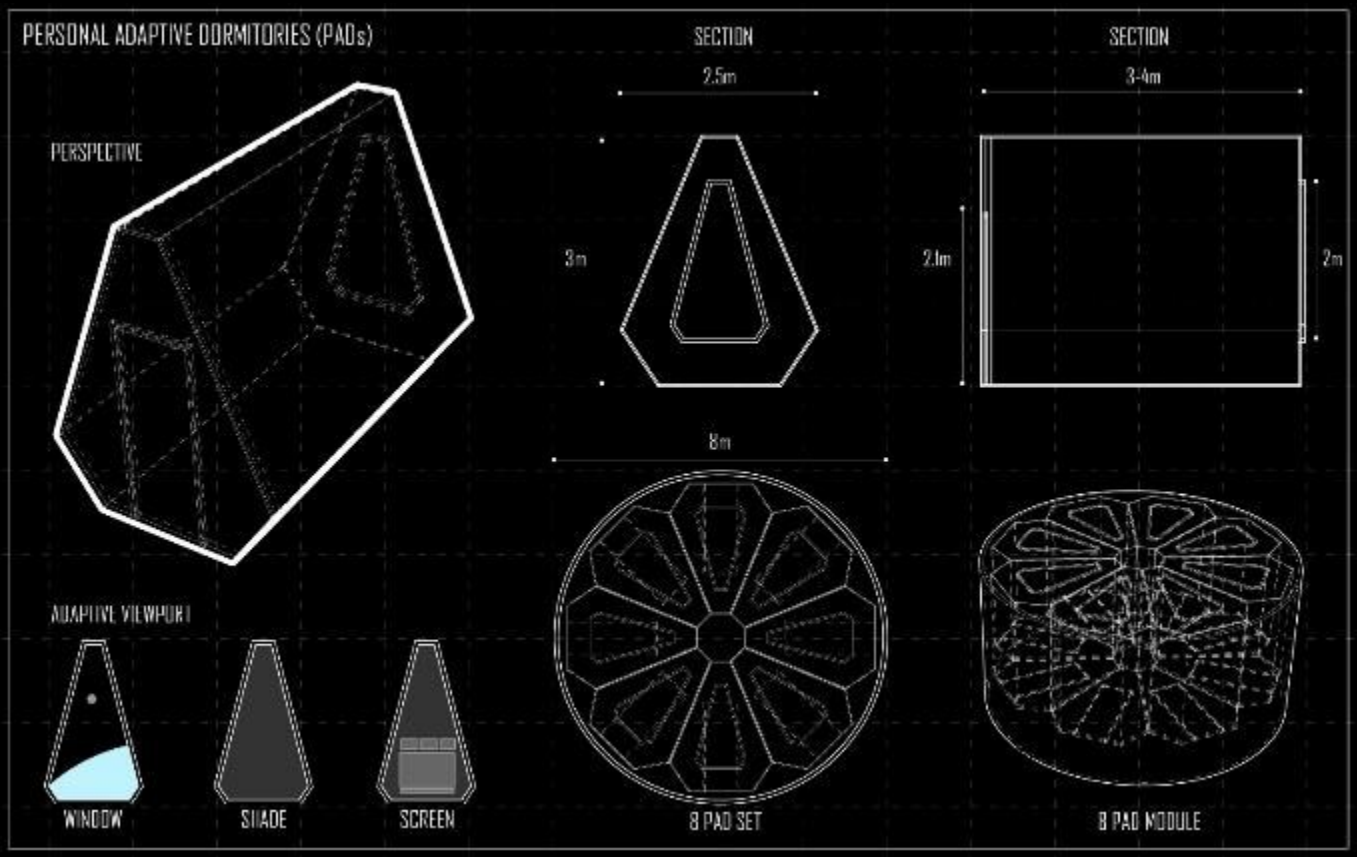



*Figure 73 - PAD Module Concept Design.*

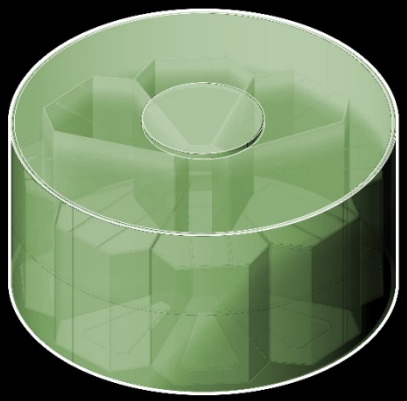


*Figure 74 - PAD Module Concept*

**Extendable Engineering Bay Modules (E-EBMs)**

E-EBMs house the critical support systems that make it possible to accommodate human life in the vacuum of outer space. The systems and equipment associated with engineering bays include:

- Energy production, storage, and distribution.
- Oxygen generation and distribution.
- Water filtration systems and distribution.
- Laboratory gas generation, storage, and distribution.
- Command center with monitoring and communications equipment allowing for on-board, inter-station and Earth communications.

The space station will have a dedicated loading dock E-EBM at the aft (rear) of the station that provides a protected environment for multiple spaceships (crew and cargo) to dock simultaneously.

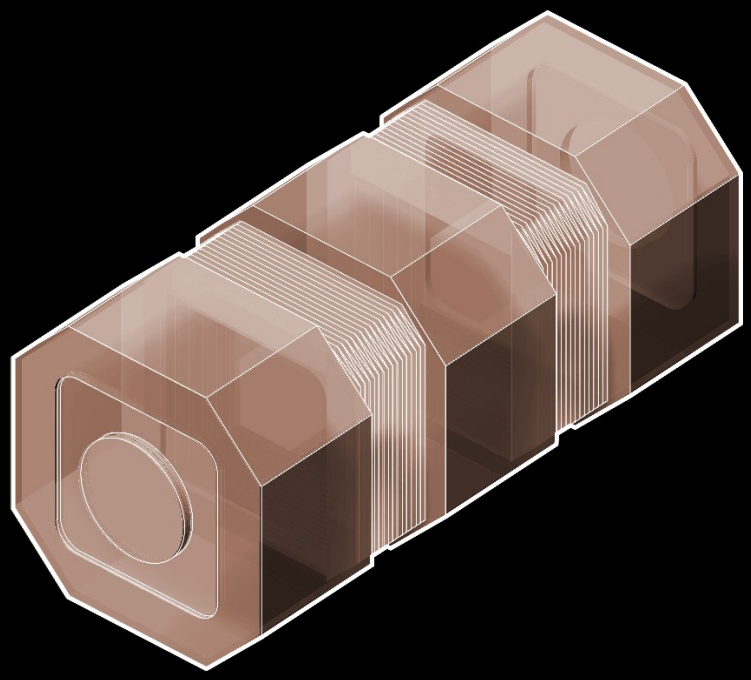


*Figure 75 - E-EBM Concept, un-extended.*

**Extendable Habitat Modules (E-HM)**

While the entire pressurized assembly of the SS Asclepius is habitable, distinctions can be made between work and leisure spaces – each with its own subset of spaces. Habitat spaces promote rest, and relaxation, and support interactions between crew members and visitors to the space station. Habitat modules have flexible internal configurations that provide:

- connectors for PAD modules
- additional temporary sleeping quarters for visiting astronauts
- a gymnasium
- an entertainment / recreation room
- crew mess areas where crew prepare and share meals and conversation.

The habitat modules are located at the core of the station, adjacent to and easily accessible from major functional work areas including the logistics spaces within engineering bays and science laboratories. This central location also provides enhanced radiation shielding during cosmic radiation storms, during which crew must retreat to specially designed shielded areas.

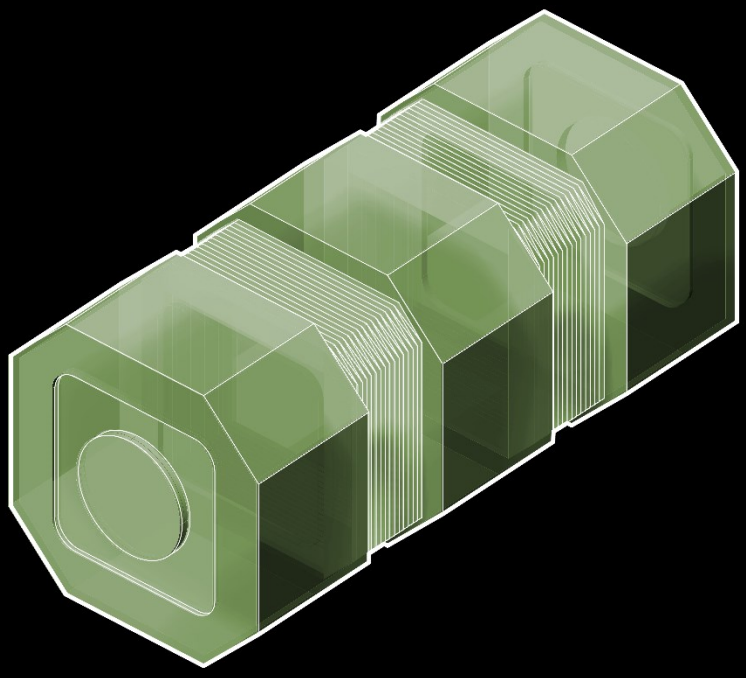


*Figure 76 - E-HM Concept, un-extended.*

**Logistics Integration Nexus Connector (LINC) Modules**

While logistics areas are not typically inhabited for long periods of time, they act as the connective elements for the SS Asclepius. In addition, some LINC modules provide thresholds between the interior and exterior of the space station. Located at four key quadrants of the space station, the LINC modules accommodate the functions listed below:

- up to 10 docking ports for receiving space station modules. This provides space station connectivity and expandability.
- loading Dock(s) for space craft.
- access to cargo holds.
- airlocks.
- suit rooms for donning and doffing procedures supporting EVAs
- space station storage.
- internal configurations are flexible to allow for additional work and collaborative areas.

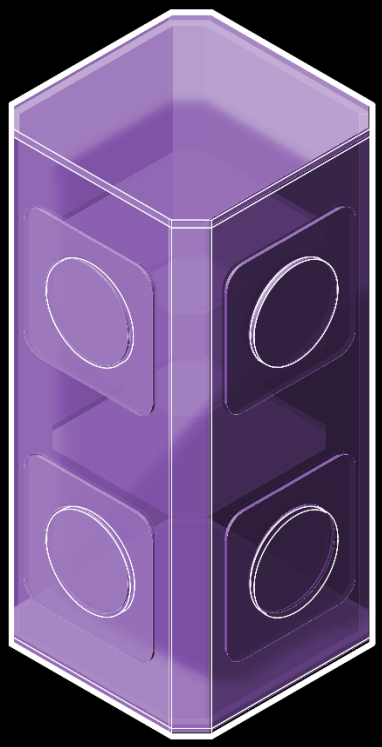


*Figure 77 - LINC Module Concept.*

## SS Asclepius Laboratories

The parameters around which laboratories on Earth are designed include design drivers such as risk level, personnel count, linear bench space, and equipment requirements. These drivers influence design decisions such as orientation, adjacencies, and circulation, along with mechanical, electrical, and plumbing layouts to support operations. For the purpose of this thesis, elements such as linear bench space, equipment, and circulation were studied as they relate to the extendable laboratory modules and to the micro-gravity environments through which scientists and facility engineers on the SS Asclepius must navigate.

**Terrestrial Laboratories**

**Linear bench space** is typically calculated in feet or meters and provides an idea of the available workspace for laboratory users as well as space for bench-top equipment. Detailed equipment lists are typically reviewed early in the design process to ensure that adequate bench space is provided.

While **large equipment** can sometimes be bench-top mounted, elements such as fridges, freezers, and incubators are typically floor-mounted. Large equipment must be considered early in laboratory design to ensure that adequate floor and bench space is provided. The amount of large equipment in laboratories can affect circulation, power, and heat loads.

**Circulation** is an important driver for scientific workflows and collaboration efforts. Laboratories typically include circulation that connects laboratories together, but also include more refined strategies to accommodate circulation within the labs to ensure safe and ergonomic workflows.



*Figure 78 - Terrestrial Laboratory – Credit: labnetinternational.com (2019)*



*Figure 79 - Lab with Large Equipment – Credit: medium.com (2023)*



*Figure 80 - Lab with Large Benchtop Equipment. Credit: labnetinternational.com (2019)*

**Lab Fundamentals**

Taking the fundamentals of linear bench space (green), benchtop and large equipment (red) and circulation within a lab space (blue arrows), the diagrams below illustrate how a typical double bay laboratory would be configured here on Earth. The overall width of this laboratory is about six meters, the same width as the proposed expandable modules on the SS Asclepius.

Free bench space is typically designed along the perimeter, with larger equipment located in the middle and sometimes the back end of the laboratory. Circulation space between benches and equipment should be wide enough to accommodate back-to-back work and safe circulation through the space. The challenge, however, is to translate these principles into a micro-gravity environment where lab users are not bound to the floor by gravity. What happens to the orientation of the laboratory (up/down) and its users? Where should equipment go? And how might circulation be re-envisioned?

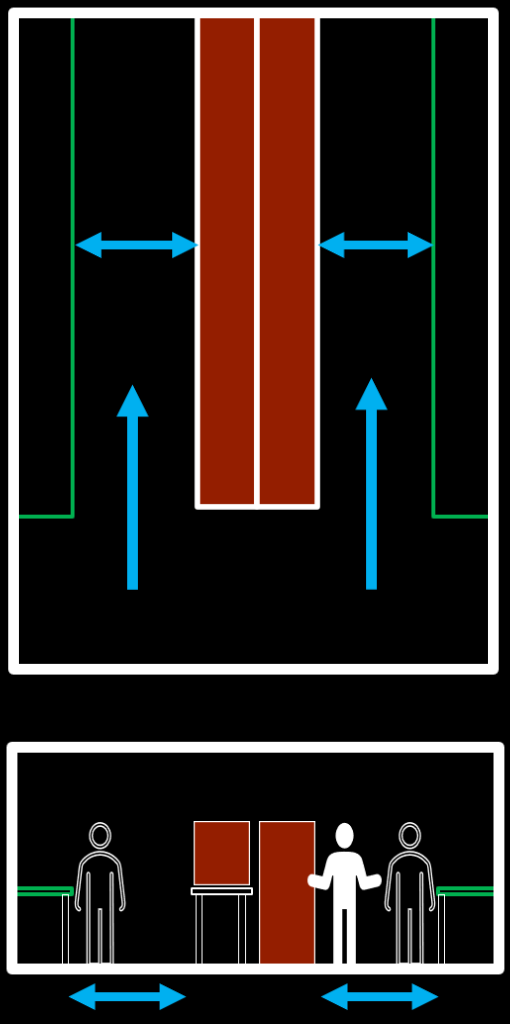


*Figure 81 - Terrestrial Laboratory Plan and Section.*

**Orbital Laboratories**

Understanding that space stations are primarily intended to support scientific research, the thesis looked at how laboratories in current stations contend with the challenges of bench space, equipment, and circulation. At a glance, it is clear to see that some of the fundamentals of space designed to support scientific work on earth also appear in space.

Below, Astronaut Christina Koch works on a scientific workbench, with small equipment and tools. As seen, there is some overhead storage of laboratory tools and instruments to support the experiments.



*Figure 82 - Christina Koch, Nasa Astronaut. Credit: NASA (2019)*

In the next photograph, an astronaut is seen in front of a scientific glovebox similar to those found in containment laboratories here on Earth. It allows the user to manipulate samples or conduct experiments within an enclosed environment This protects the experiment, the user, and the surrounding environment.



*Figure 83 - Astronaut at Glovebox – Credit: electronicsweekly.com (2020)*

In both International and Chinese Space Stations we can see that there is an implied top / bottom and left / right orientation driven by the placement of equipment along the “side walls.” The space in which researchers work doubles as a corridor for scientists to travel through and is wide enough to enable them to work back-to-back. The overall amount of space available is constrained by the size of the module, which is typically about four meters in diameter, due to the size constraints of the cargo bays of the spacecraft in which they are transported into space. The design constraints will change as the size of cargo bays increase. Indeed, larger, and expandable or inflatable module designs are being explored and tested today.

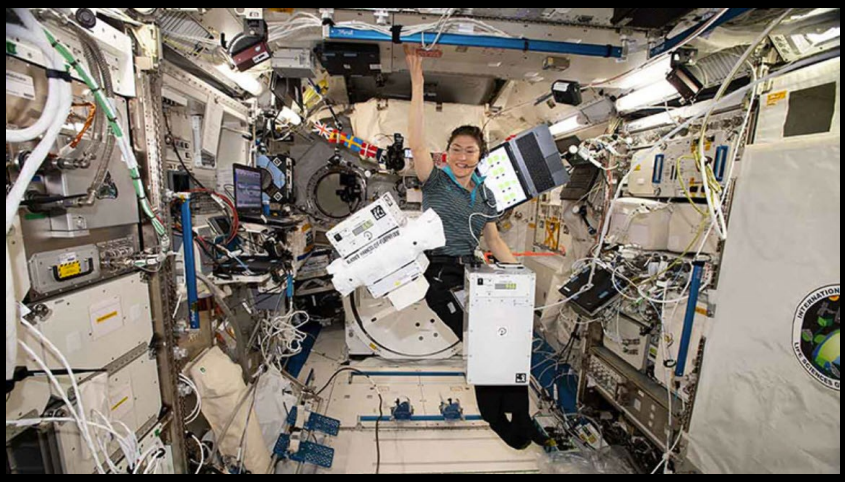


*Figure 84 - ISS National Laboratory – Credit: plasticstoday.com (2024)*



*Figure 85 - Tianhe Module, CSS. Taikonaut Wang Yaping. Cr. global.chinadaily.com.cn (2021)*

**Extendable Lab Module (E-LM) Orientation**

In the micro-gravity environment of LEO, the internal orientation of spaces is subject to design of interior planes (wall, floor, ceiling surfaces). Given the larger diameter of the module, (6 m as compared to the current standard of 4 m), new configurations are possible. Essentially, like in the ISS and CSS examples, up is what we make it, and thus the extendable modules can be visualized as either long, corridor-like spaces or tall, shaft-like spaces divided into multiple levels.

Let us start by rotating the module 90 degrees so that it is “upright”, regardless of its directional orientation in relation to Earth.

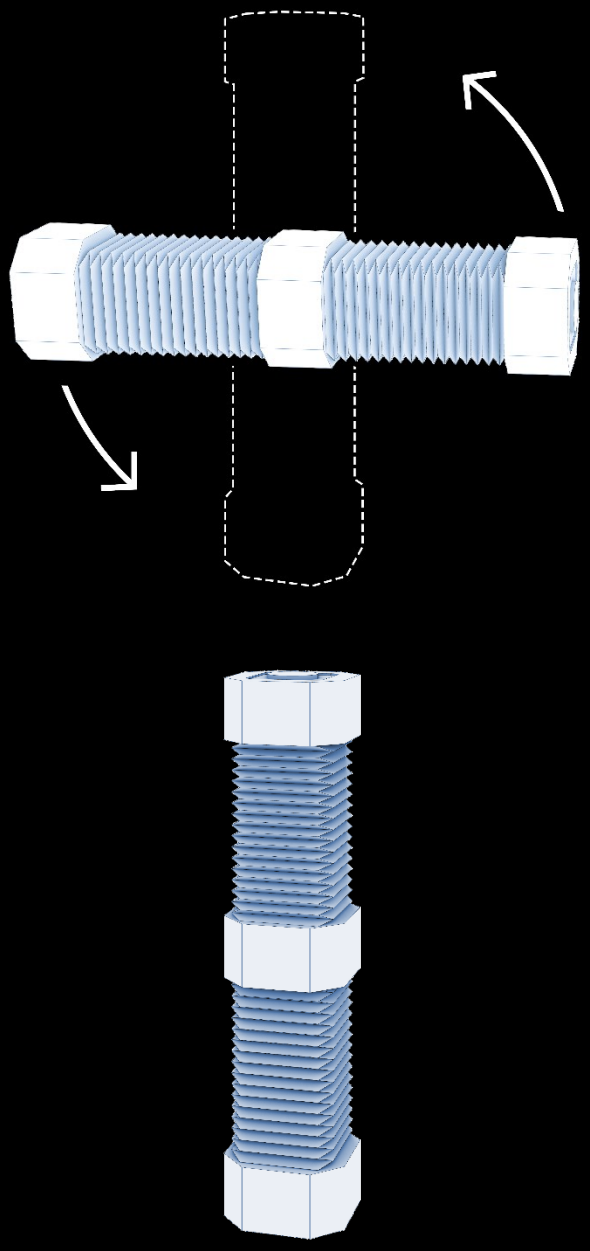


*Figure 86 - Capsule rotated upright.*

**Laboratory Levels and Circulation**

In its “vertical” orientation, the E-LM can be subdivided into multiple levels like stories in a building. This is illustrated in the sections below. The E-LM now serves as a lab tower with up to 9 levels of laboratories, providing operational flexibility and a variety of laboratory configurations depending on research needs.

Laboratory circulation is now vertical in relation to the laboratories. Hatch openings and a ladder are used to safely move between laboratories. As with an elevator, users ascend and descend from level to level as indicated by the blue arrow on the drawing to the right.

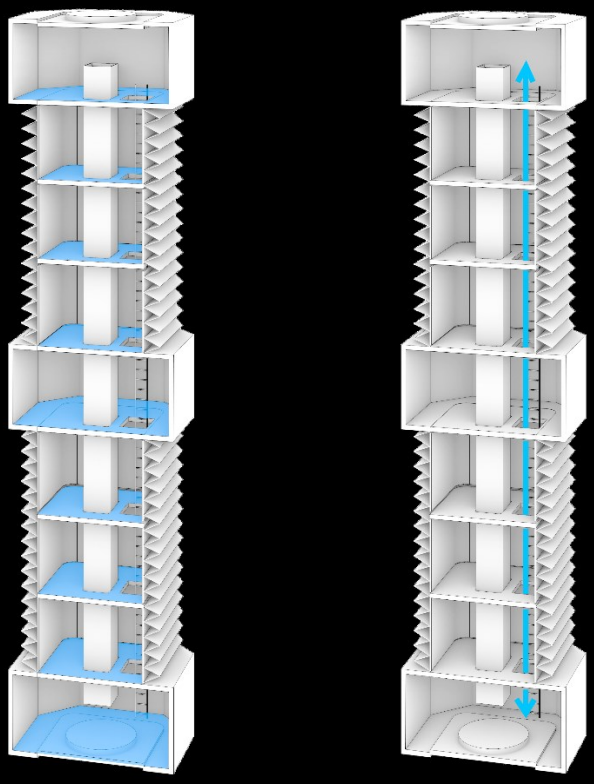


*Figure 87 - E-LM Lab Levels and circulation*

**Equipment and Bench Space**

A continuous Sample Transfer Core (STC) runs through the center of the E-LM (shown in red), allowing for safe management and transfer of biological samples and specimens. Automation capabilities enable samples to be passed between labs transferred directly into gloveboxes.

Dr. Tricia Larose described the issues and changes associated with the manual transfer of samples and specimens between equipment and gloveboxes. This STC will automate storage and transfer of samples, augmenting safety, and scientific integrity.

Wall area (shown in green) can now be fully optimized for linear bench space, storage and equipment as needed. Since scientists are not bound to the floor planes, they are able to use the full height of the lab.

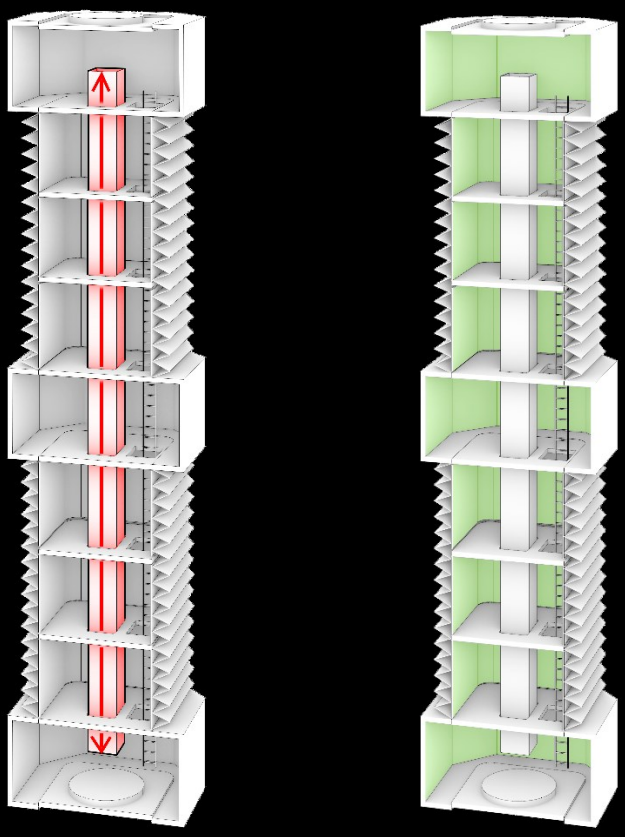


*Figure 88 - E-LM Sample Transfer Core and Wall Space Utilization*

**Floorplans**

The standard layout at the rigid sections of the E-LM is shown below. The floorplates are approximately 6m x 6m wide, and organized like standard double-wide laboratory modules here on Earth. The plan includes a framing system along the perimeter to receive equipment and laboratory work benches.

Highlighted in blue is the circulation corridor/shaft. Astronauts enter the space through the floors and ceilings of the laboratory through the open hatches shown. A ladder is provided for stability and safety, also reinforcing the up/down travel directions in relation to the layout of the laboratories.

The topic of "Circulation Driven Orientation (CDO)" – i.e., aligning the direction of circulation with the general orientation of researchers in the lab – warrants further study.

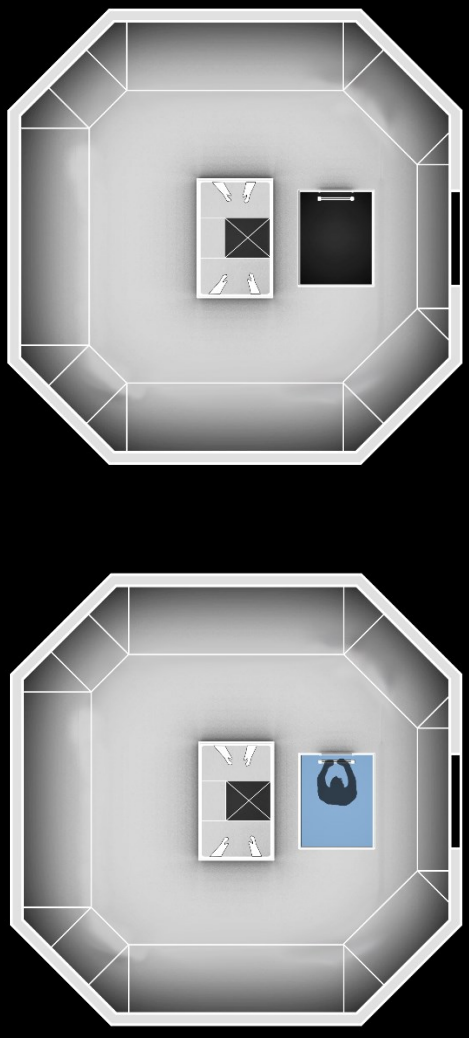


*Figure 89 - E-LM Concept Plan Diagrams: Standard and Circulation*

Reporting stations, typically comprised of a work surface and a monitor, are typically included inside or direclty adjacent to laboratories. Researchers use these stations to document experiments and conduct required administrative tasks related to day-to-day scientific operations. A reporting station is shown in yellow adjacent to a proposed window, looking down to Earth on the Nadir side of the space station.

Bench spaces along the perimeter of the lab are highlighted in green. As seen in Figure 88, the full height of the wall can be used in micro-gravity. This expands the capacity of each meter of linear bench space provided. As configured, each lab can accommodate about 15 meters of linear bench space. Overhead space can be used for equipment storage or for additional (tiered) bench space. Undercounter space can be used for equipment or storage.

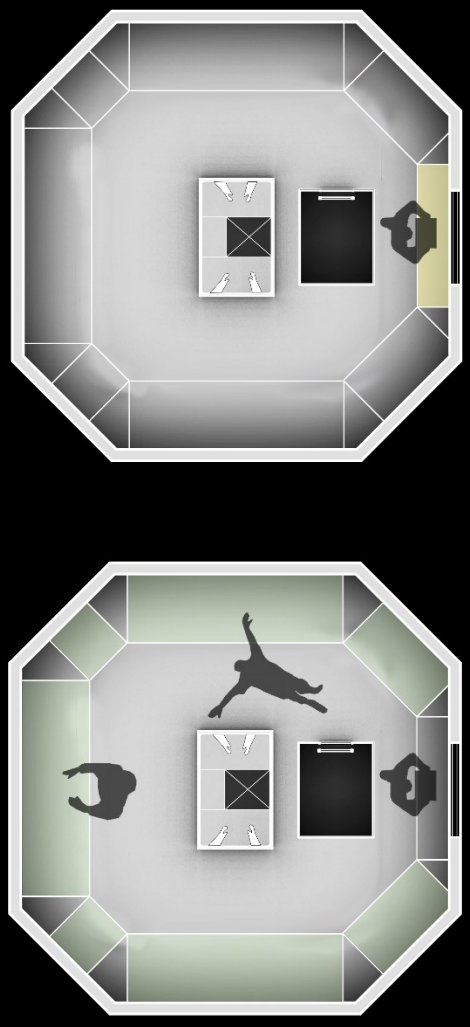


*Figure 90 - E-LM Concept Plan Diagrams: Reporting and Bench Space*

The Sample Transfer Core (STC) shown at the center of the floorplate (in dark red) provides a central hub at each lab level. This automated core connects labs and is surrounded by a frame that houses gloveboxes, pass-thru chambers, and other equipment as needed. This is envisioned as a smart dumbwaiter that transports biological samples between level and would incorporate advanced robotics to accurately pick-up and place sample boxes in the desired locations.

The red areas along the perimeter can accommodate standardized cancer or climate research equipment racks that fit the module dimensions (height and width). While the locations shown allow for bench and storage space in between, diagrams on the following page show alternatives.

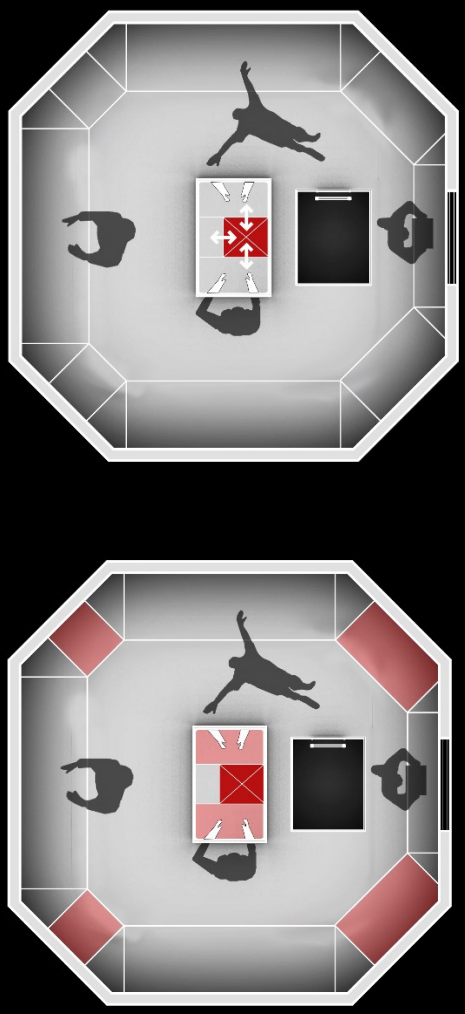


*Figure 91 - E-LM Concept Plan Diagrams: Sample Transfer Core and Equipment Racks*

The two configurations shown below show the versatility and flexibility built into the laboratory planning strategy. The all-red, all-equipment option suggests that some laboratory levels can act as equipment rooms or instrument rooms, like those commonly found within laboratories on Earth. These rooms are dedicated to larger, louder, and higher-heat-producing equipment. Aggregating larger or higher amounts of equipment in one space or on one level allows for more efficient systems distribution (power, plumbing, lab gases), HVAC (Heating Ventilation and Cooling) design to mitigate heat loads, and enhanced user comfort and ergonomics. Grouping heat and noise generating equipment in dedicated spaces frees up space in user-occupied laboratories.

The second diagram below shows a balanced or hybrid laboratory that has a mix of laboratory bench space and equipment racks as well as the standard STC and reporting station.

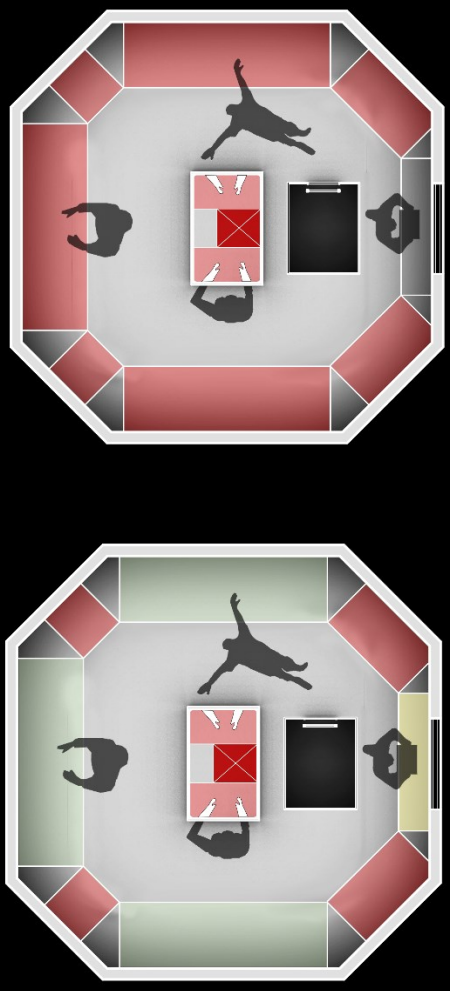


*Figure 92 - E-LM Concept Plan Diagrams: Equipment and Standard Labs*

## SS Asclepius Launch and Assembly Sequence

Among the topics I researched was the sequence by which space stations are assembled. The well-documented assembly of the ISS provided an excellent case study to understand the chronological sequence in which modules were added to provide critical capabilities and additional functionality.

The table below lists out a launch and orbital assembly sequence for SS Asclepius. It includes an 8-phase launch schedule beginning with critical modules and orbital construction components. As the structure is fleshed out, additional modules are added to achieve the target configuration. Diagrams on the following pages illustrate this launch and assembly sequence.

| LAUNCH # | CARGO | DETAILS |
|---|---|---|
| 1A | E-EBM-1, CANADARM3 | Systems: Comms, Life Support, Power storage. |
| 1B | Solar Sail/Arrays, SCP**, PAD-1 | Power generation, crew PADs pressurized. +8 Crew |
| 2A | E-CHM-1 | Birthed onto E-EBM1, Extended + Pressurized. (E+P) |
| 2B | E-CHM-2 | Birthed onto E-CHM-1, (E+P) |
| 2C | SCP-1** | Exo-skeleton + Envelope constructed in tandem. |
| 3A | HANGAR-1 BAY SCP-2** | Hangar-1 build provides port side framework. |
| 3B | HANGAR-2 BAY SCP-3** | Hangar-2 build provides starboard side framework. |
| 4A | LINC-1 | Provides E-LM-1 connectivity, storage + Docks. (P) |
| 4B | E-LM-1 | Birthed onto LINC-1, (E+P), Cancer research begins. |
| 4C | SCP-4** | Exo-skeleton + Envelope constructed in tandem. |
| 5 | PAD x 2 | Station capabilitiy: +16 crew. |
| 6A | LINC-2 | Provides E-LM-2 connectivity, storage + Docks. (P) |
| 6B | E-LM-2 | Birthed to LINC-2, (E+P), Climate research begins. |
| 7A | LINC-3 | Birthed to E-LM-1, connectivity for E-EBM-1/2. |
| 7B | LINC-4 | Birthed to E-LM-2, connectivity for E-EBM-1/2. |
| 7C | SCP-5** | Exo-skeleton + Envelope constructed in tandem. |
| 8A | E-EBM-2 | Birthed to E-CHM-1, (E+P), Birthed to LINC-1+2. |
| 8B | E-EBM-3 | Birthed to E-EBM-2, Dock + EVA capability en-hanced. |
| 8C | SCP-6** | |

* Each module provides altitude control thrusters, synchonized together as station construction advances.
** Station Construction Payloads (SCPs) - Include required robotics, supplies, exo-skeleton structural members and envelope cladding panel systems for station assembly by robotic arms and crew EVAs.

*Figure 93 - SS Asclepius, Launch and Assembly Sequence*

**Illustrated Launch and Assembly Sequence**

**Launch Phase 1** includes an E-EBM, solar sails, and a PAD module. This provides the initial energy generation, life support systems, and dormitory space for a crew of eight facility engineers who will conduct and supervise subsequent assembly phases. In addition, a station construction payload (SCP) is launched, and exo-skeleton and second skin cladding construction begins.

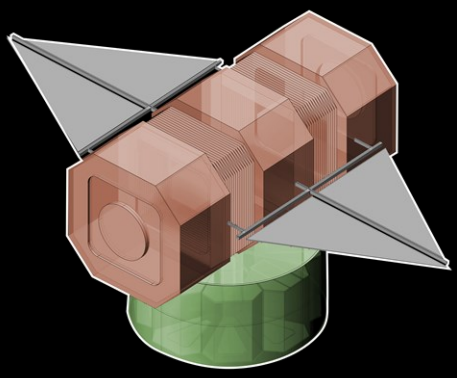


*Figure 94 - Launch Phase 1*

**Launch Phase 2** adds two E-HM modules and additional solar sails. Once modules are connected, the station can be extended, adequately pressurized, and solar sails can be expanded for maximum solar energy capture and storage.

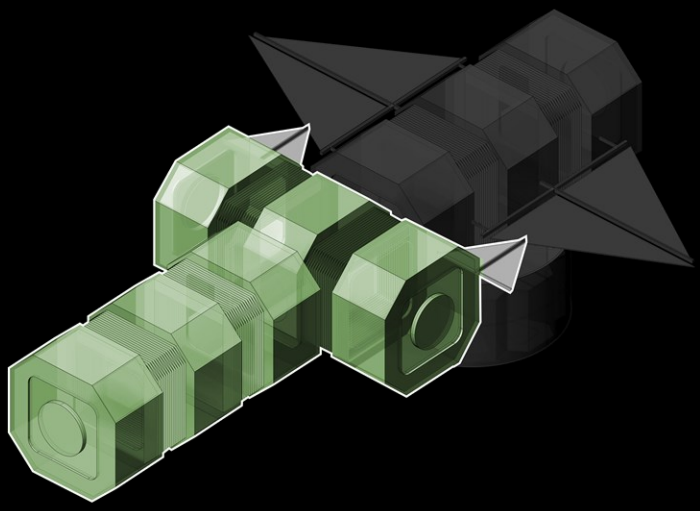


*Figure 95 - Launch Phase 2*

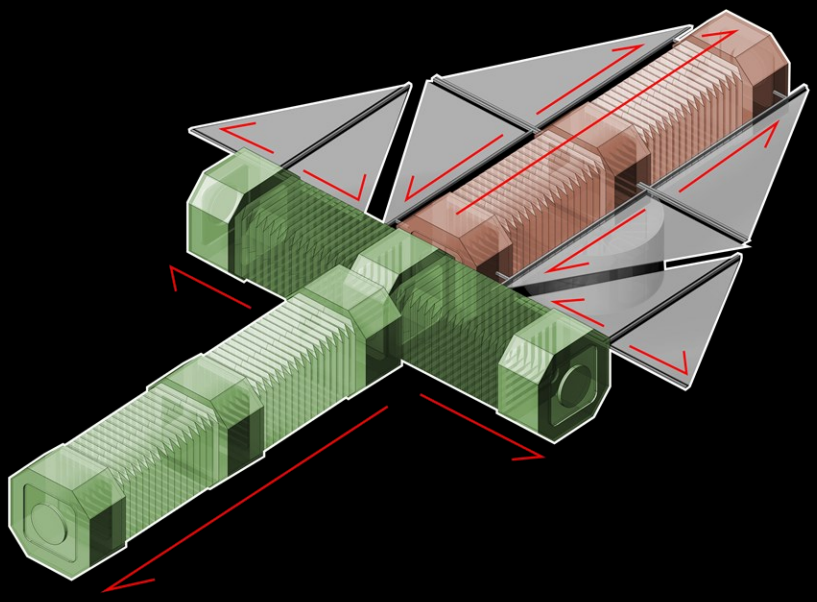


*Figure 96 - Launch Phase 2, extended*

**Launch Phase 3** includes construction components for two exterior hangars/cargo bays that fit within the station's exo-skeleton. These bays provide sheltered exterior environments in which crew members can conduct routine maintenance and repairs, and in which equipment and supplies can be stored. Hangar doors located on the Zenith and Nadir (top and bottom) of the hangars act as shortcuts to move "through" the space station without the need to travel around the entire exterior perimeter.

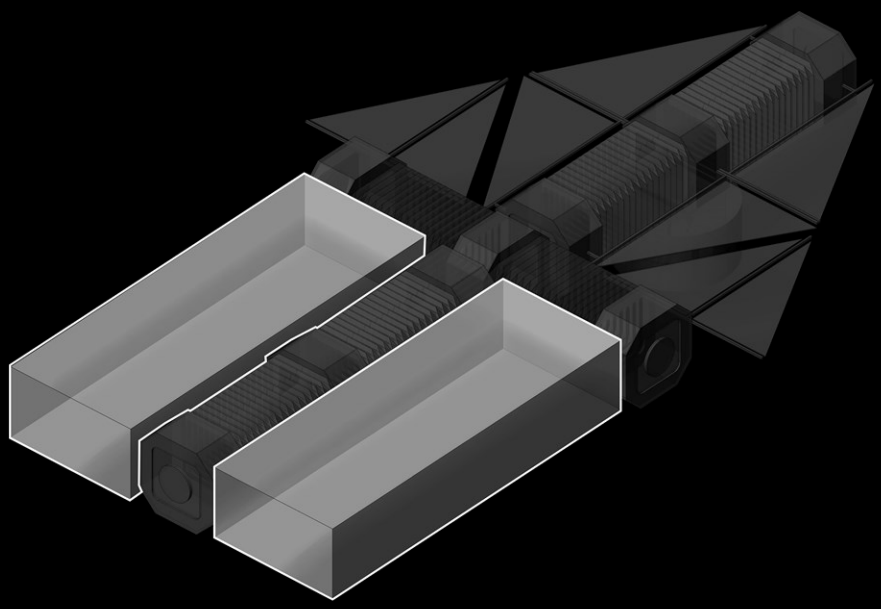


*Figure 97 - Launch Phase 3 - Hangers / Cargo Bays*

**Launch Phase 4** provides the stations first LINC module and E-LM. Once docked, the laboratory is extended and pressurized.

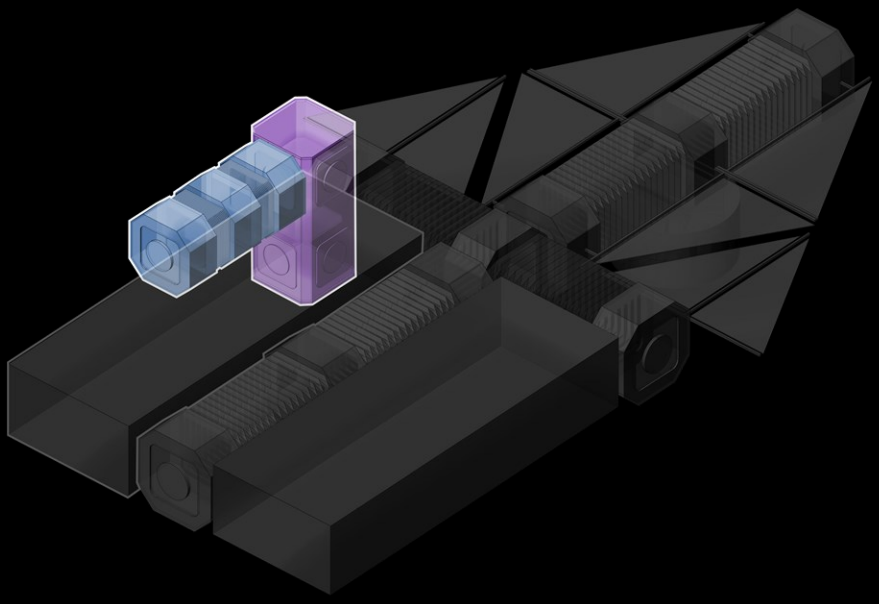


*Figure 98 - Launch Phase 4 - LINC Module + E-LM*

Viewing the space station from below for clarity, **Launch Phase 5** adds two additional PAD modules to the Nadir face (underside) of the station, increasing the full-time crew count to 24. The PAD modules are intended for long duration crew members to have an enhanced experience. Short term or visiting crew will have adequate dormitories within the E-HMs. (image showing station from below for clarity)

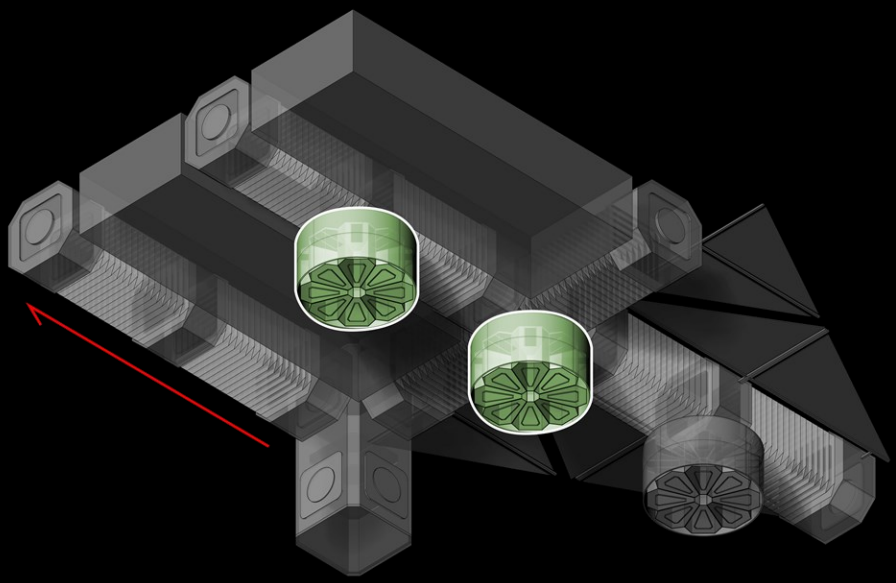


*Figure 99 - Launch Phase 5 - PAD Modules*

**Launch Phase 6**, similar to launch phase 4 provides a LINC module and an E-LM. Once docked, the E-LM is extended and pressurized.

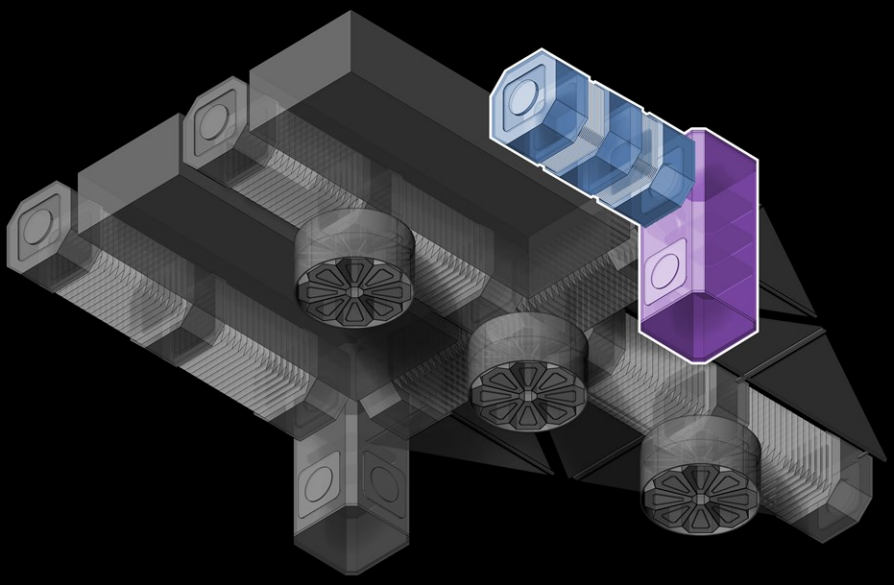


*Figure 100 - Launch Phase 6 - LINC Module + E-LM*

**Launch Phase 7** adds two additional LINC modules to the aft side of the station. These modules are docked to the ends of the two E-LMs and provide future connectivity for the final assembly phase.

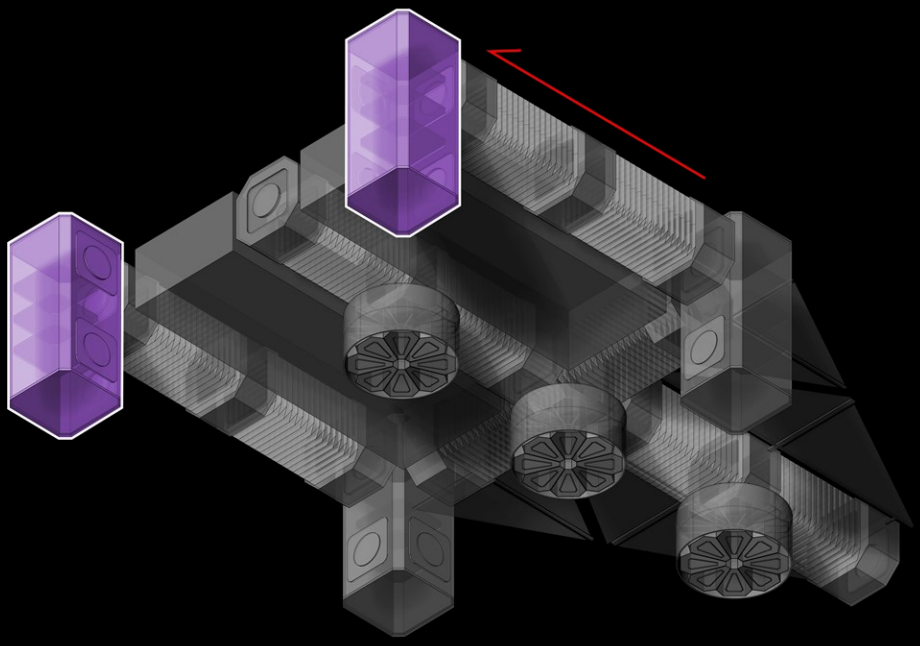


*Figure 101 - Launch Phase 7 - LINC Modules*

**Launch Phase 8** is the final assembly phase of SS Asclepius, adding two additional E-EBM that serve as the primary logistics spaces for storage, workshops, EVA airlocks, and as a loading dock area for incoming crewed or cargo resupply spacecraft.

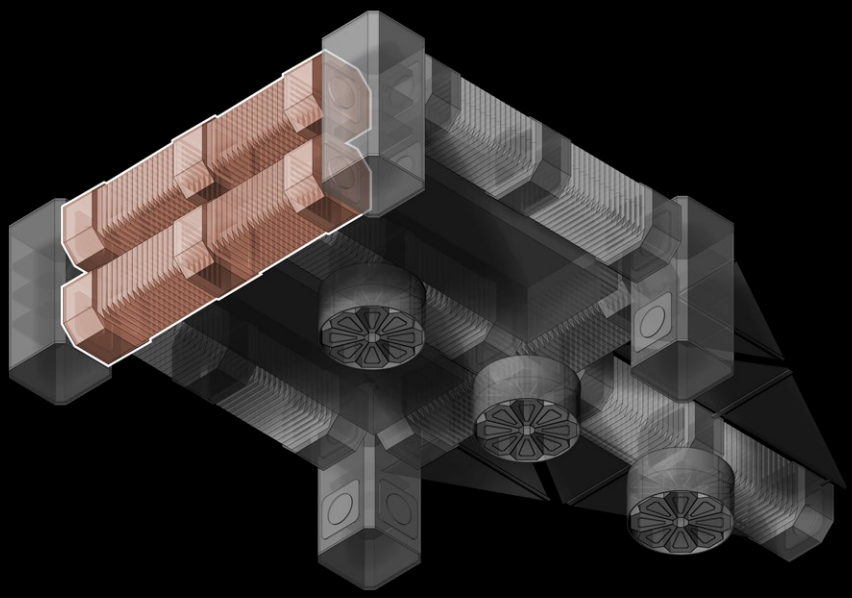


*Figure 102 - Launch Phase 8 - Final E-EBMs*

**Finalized SS Asclepius.** Although this completes the vision for the station, LINC modules and other strategically planned docks provide growth and expansion capabilities on the station. At the very least, two additional laboratory modules can be added below the two initial E-LMs, doubling the scientific experimental volume on the station. Additional link modules can be connected to the existing ones to enable additional modules to be added to the top of the station, i.e., the Zenith side that faces away from earth.

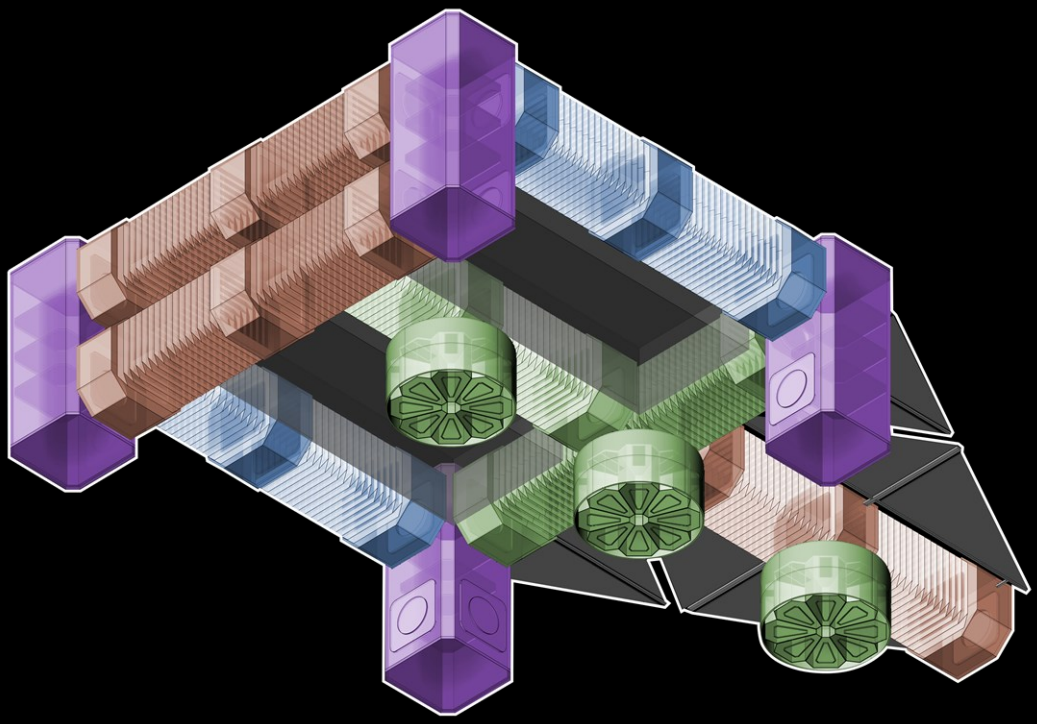


*Figure 103 - Full Program*

## Conclusion

The AC$^3$ROSS Project encompasses the conceptual design of the SS Asclepius, the design concepts for the modular components that comprise it, and the launch and assembly sequence required to assemble it. Fundamentals of terrestrial laboratory design were applied to the layouts of laboratories operating in LEO. The laboratory modules aboard the SS Asclepius are substantially larger than those in the ISS, taking advantage of the increased cargo capacity of SpaceX's Starship – the largest and most advanced space craft designed to date. Innovative designs for berths (PADs) took their inspiration from traditional attic bedrooms and are intended to enhance the quality of life for the researchers aboard. The larger purpose of the station is to support research to address two of the most significant crises facing humanity, namely Climate Change and Cancer.

Further design and research are inevitable and future students are encouraged to take some of the topics explored in this design even further. Topics for additional investigation include:

- Material studies into exo-skeleton and second skin cladding for space stations.
- Automation for use externally on the space station or internally within laboratories such as the Sample Transfer Core.
- Circulation Driven Design (CDO) – how is circulation and orientation connected or disconnected in microgravity? What is the importance of circulation on Earth and how can that translate or change in outer space?
- Habitation – creating ergonomic and sustainable habitats in low Earth orbit. What are the psychological effects of living and working in outer space – how can architectural design enhance quality of life for astronauts?
- Classical Order defines many architectures on Earth. What does a future "order" look like for astrotectonics which are not bound by gravity?

*Figure 104 - Rendering, space eye view of SS Asclepius*



The final thesis defense concluded with the following transmission from the future…

“TRANSMISSION – DATE: APRIL 24, 2100, 0900h

ANNOUNCEMENT FROM THE AGENCY FOR RESEARCH AND INFRASTRUCTURE OF EXTRA-TERRRESTRIAL LABORATORIES - ARIEL

ARIEL IS PROUD TO CELEBRATE THE 70TH YEAR OF SCIENTIFIC OPERATIONS ABOARD THE SS ASCLEPIUS, INAUGURATED IN THE YEAR 2030.

THANKS TO CANCER RESEARCH BREAKTHROUGHS IN BOTH SCREENING AND TREATMENT TECHNOLOGIES, HUMANS WERE ABLE TO AVOID THE ANTICIPATED 77% INCREASE IN CANCER RATES PREDICTED BY MID-CENTURY (2050, WHO). CANCER RATES HAVE DROPPED TO BELOW 20% OF 2030 NUMBERS.

HUMANITY WAS NOT ABLE TO KEEP GLOBAL WARMING BELOW 1.5 DEGREES, AS TARGETED. HOWEVER, ADVANCED CARBON, COMBUSTION AND BOTANY RESEARCH ALONG WITH DETAILED MONITORING AND MEASURING OVER THE LAST 7 DECADES ABOARD THE SS ASCLEPIUS HAS ALLOWED US TO SLOW THE GLOBAL TEMPERATURE RISE AT 1.8 DEGREES WITH HOPES OF REVERSING THE EFFECTS OF CLIMATE CHANGE OVER THE NEXT CENTURY.

AS THE EARTH HEALS, OUR LUNAR COLONY EXPANDS AND MARS-LABS BEGINS OPERATIONS, ARIEL’S NETWORK AIMS AT CONTINUING SCIENTIFIC EXCELLENCE IN SUPPORT OF HUMANITY’S GOALS OF SUSTAINABLE USE OF OUR HOME SOLAR SYSTEM…”

**AD ASTRA**

*Figure 105 - Rendering - Zenith View of SS Asclepius*

